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## Key Points:

- The dissolution of calcite leads to a decrease of the magnitude of the streaming potential coupling coefficient
- The pore water conductivity is dominated by the calcite dissolution in carbonates
- Calcite dissolution rates calculated from streaming potential measurements are consistent with PHREEQC simulations

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Influence of CO<sub>2</sub> on the Electrical Conductivity and Streaming Potential of Carbonate RocksA. Cherubini<sup>1,2</sup> , B. Garcia<sup>1</sup>, A. Cerepi<sup>2</sup>, and A. Revil<sup>3</sup> 

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**Abstract** Minimally intrusive geophysical methods are required to monitor CO<sub>2</sub> leakages from underground storage reservoirs. We investigate the impact of gaseous CO<sub>2</sub> on both electrical conductivity and electrokinetic properties of two limestones during their drainage. These data are contrasted with measurements performed on one clay-free sandstone. The initial NaCl brine concentrations before drainage (from 8.5 to 17.1 mMol/L) correspond to the limit between freshwater and slightly brackish water. These values are representative to saturated brine formations inside which CO<sub>2</sub> can be stored. Using these water salinities, the surface conductivity of the samples represents less than 5% of the overall electrical conductivity. A CO<sub>2</sub> release leads to an increase of the electrical conductivity of the rock during drainage in limestones and no change in sandstone. This increase in the electrical conductivity is due to the dissolution of calcite with the concomitant release of Ca<sup>2+</sup> and HCO<sub>3</sub><sup>-</sup> in the pore water. It is not due to the CO<sub>2</sub> dissociation in the pore water in the pore pressure range 0–0.5 MPa and at a temperature of  $T = 20$  °C. The measurements of the streaming potential show a substantial decrease of the streaming potential coupling coefficient and zeta potential magnitudes after a CO<sub>2</sub> release in carbonates. This observation is explained by the increase of the ionic strength of the pore water in the course of the experiment. This change can be used, in turn, to determine calcite dissolution rates from the measurement of the electrokinetic properties.

## 1. Introduction

Geological CO<sub>2</sub> sequestration is considered as a viable way of reducing CO<sub>2</sub> concentration into the atmosphere (Benson & Cole, 2008; Moore et al., 2004). In order to host CO<sub>2</sub>, underground locations have been proposed such as depleted oil and gas reservoirs (Jenkins et al., 2012), saturated brine formations (Benson & Cole, 2008), and coal seams (Shi & Durucan, 2005). The injected CO<sub>2</sub> is generally trapped below low-permeability and high capillary entry pressure seals corresponding for instance to clay-rich anticlines (Liu et al., 2011). CO<sub>2</sub> dissolves partly into the pore water and interacts with the surrounding minerals promoting dissolution/precipitation reactions (Bachu et al., 1994). Two well-known geophysical galvanometric methods can be used to monitor fluid flow in shallow or deep formations, namely, the self-potential (SP) method and the electrical conductivity/resistivity tomography methods (Le Roux et al., 2013; Zhou et al., 2012).

The electrical conductivity method is an active geophysical method. A low-frequency current is injected in the ground and the resulting field is recorded in order to reconstruct the conductivity/resistivity distribution of the subsurface. This method is already used in the field to identify CO<sub>2</sub> leakages (Le Roux et al., 2013; Zhou et al., 2012). Recent studies showed the ability of electrical conductivity monitoring to detect variations caused by CO<sub>2</sub> flowing in porous media (Giese et al., 2009; Kiessling et al., 2010; Schmidt-Hattenberger et al., 2011; Strazisar et al., 2009; Würdemann et al., 2010; Zhou et al., 2012). Strazisar et al. (2009) showed, for instance, the increase of the size of a conductivity anomaly observed after the injection of CO<sub>2</sub> in a vertical injection well. They suggested that the dissolution of CO<sub>2</sub> in the pore water is responsible for such a conductivity increase. Along the same line, Lewicki et al. (2007) argued that only a small amount of CO<sub>2</sub> is dissolved in the pore water system during its release, leading only to a slight pore water conductivity increase. This increase is due to the CO<sub>2</sub> dissociation in pore water, leading to an increase of the ionic strength of the solution. Recently, some authors focused on the carbonate/CO<sub>2</sub> interactions to explain the observations mentioned above (e.g., Cohen et al., 2013; Garcia et al., 2013; Loisy et al., 2013). The conductivity of the pore water decreases when the water saturation decreases. However, there is also two other effects impacting

the conductivity of the rock mixture, namely, (a) the  $\text{CO}_2$  dissociation in water and (b) the carbonate dissolution. These two additional effects can counterbalance the saturation effect, leading to an increase of the conductivity of the mixture (Le Roux et al., 2013).

The self-potential method is a passive galvanometric technique consisting in recording the electrical field associated with in situ sources of electrical current in the subsurface. Such current source densities can be, for instance, associated with the flow of the liquid water phase in a porous material. Such component is said to be of electrokinetic nature (Aubert & Y  n   Atanga, 1996; Darnet et al., 2003; Jougnot et al., 2012; Naudet et al., 2003; Naudet et al., 2004). The resulting electrical current density is called the streaming current. The self-potential method is a well-recognized method in geophysics used to identify fluid flow characteristics such as preferential flow paths (Corwin & Hoover, 1979; Revil et al., 2004) or transport properties including permeability and the specific storage coefficient (Cerepi et al., 2017; Cherubini et al., 2018; Jackson, 2010; Perrier & Morat, 2000; Revil et al., 2007; Soueid Ahmed et al., 2016). Streaming potential signals are due to the presence of an electrical diffuse layer on the surface of the minerals (Dukhin & Derjaguin, 1974; Overbeek, 1952). The drag of the excess of charge of the diffuse layer by the flow of the pore water generates the streaming current. This current serves as a source to generate electromagnetic signals, which can be measured remotely.

Our goal in this study is to develop experiments showing how electrical conductivity and electrokinetic properties are influenced by a release of  $\text{CO}_2$  in carbonate rocks. We use two limestones known to be reactive in presence of  $\text{CO}_2$ . These data will be contrasted with similar experiments realized with a sandstone core sample. We use a clean silica-based Fontainebleau sandstone for these additional experiments. Regarding the potential influence of gaseous  $\text{CO}_2$  on the electrical conductivity, we want to test the two different assumptions proposed in the literature and briefly discussed above, that is, (i) the dissociation of  $\text{CO}_2$  in the pore water can control the pore water conductivity (see Dreybrodt et al., 1997; Lide, 2003) and (ii) calcite dissolution can change the ionic strength of the pore water. We propose a model allowing us to quantify calcite dissolution rates with the self-potential method, using the streaming potential coupling coefficient as a tool to define these rates.

## 2. Theory

### 2.1. Electrical Conductivity

The electrical conductivity  $\sigma$  of rocks is the sum of two contributions, (i) one contribution through the bulk pore space and dependent on the conductivity of the pore water  $\sigma_w$ , plus (ii) a second contribution called surface conductivity and written below as  $\sigma_s$ . This second contribution is dispersive (i.e., frequency dependent) and related to the conduction in the electrical double layer (Stern and Gouy-Chapman diffuse layers) coating the surface of the minerals. We use the following conductivity equation defining the surface conductivity  $\sigma_s$ ,

$$\sigma = \frac{1}{F} \sigma_w + \sigma_s, \quad (1)$$

where  $F$  denotes the intrinsic formation factor (dimensionless) and is connected to the porosity  $\phi$  (dimensionless) by the first Archie's law (Archie, 1942),

$$F = \phi^{-m}. \quad (2)$$

In equation (2),  $m$  is called the cementation or porosity exponent.

According to Waxman and Smits (1968), the effect of saturation  $S_w$  on electrical conductivity is given by

$$\sigma = S_w^n \frac{1}{F} \sigma_w + S_w^{n-1} \sigma_s, \quad (3)$$

where  $n$  (-) represents the saturation or second Archie's exponent. The resistivity index is defined as the ratio of the conductivity at full saturation ( $\sigma_0$ ) and the conductivity at a given saturation ( $\sigma$ ). The resistivity index  $RI$  (-) is therefore given by (Revil et al., 2014)

$$RI = S_w^{-n} \left( \frac{\sigma_w + F\sigma_s}{\sigma_w + F\sigma_s/S_w} \right). \quad (4)$$

When surface conductivity can be neglected compared to bulk water conductivity, the resistivity index can be written as

$$\lim_{\sigma_s \rightarrow 0} RI = S_w^{-n}, \quad (5)$$

and under the same assumption  $F$  can be approximated as the ratio of the pore water conductivity ( $\sigma_w$ ) to the conductivity of the sample at full saturation ( $\sigma_0$ ) as

$$\lim_{\sigma_s \rightarrow 0} F = \frac{\sigma_w}{\sigma_0}, \quad (6)$$

In the case of a gaseous  $\text{CO}_2$  injection, the presence of  $\text{CO}_2$  impacts the saturation  $S_w$ . Since  $\text{CO}_2$  is not an immiscible gas, it will also impact the conductivity of the pore water solution through the dissolution of the carbonic acid in the pore water and the potential dissolution of the mineral into water. In this case, Le Roux et al. (2013) formulated the resistivity index ( $RI_{\text{CO}_2}$ ) in terms of the pore water conductivity before  $\text{CO}_2$  release ( $\sigma_w$ ) and pore water conductivity at  $\text{CO}_2$  saturation ( $\sigma_{w-\text{CO}_2}$ ) assuming that  $m$ ,  $n$ , and  $\phi$  are constant under  $\text{CO}_2$  or  $\text{N}_2$  conditions,

$$\lim_{\sigma_s \rightarrow 0} RI_{\text{CO}_2} = \frac{\sigma_w}{\sigma_{w-\text{CO}_2}} S_w^{-n}. \quad (7)$$

## 2.2. Streaming Potential

There are two types of electrical current in porous media in which pore water flows. The first current is the classical conduction current and is due to the fluxes of charges under the effect of the electrical field. The second current is a source current associated with the drag of the electrical charges by the flow of the pore water itself. This advective current is called the streaming current. The total current density is the sum of the two current densities. For the case of a rock sample crossed by fluid flow and acting as an open system, the total current density is equal to zero and that the streaming current density counterbalances the conduction current density. In this case, the streaming potential  $\Delta\psi$  (V) is given by (e.g., Revil et al., 2007)

$$\Delta\psi = \frac{\epsilon_w \zeta}{\eta_w F \left( \frac{\sigma_w}{F} + \sigma_s \right)} \Delta p. \quad (8)$$

where  $\epsilon_w$  (F/m) denotes the dielectric permittivity of the liquid pore water,  $\zeta$  (V) the zeta potential (the potential of the diffuse layer at the shear plane close to the mineral surface where the velocity of the pore water is zero),  $\eta_w$  (Pa s) the dynamic viscosity of the liquid water phase,  $\psi$  (V) the electrical potential of electrokinetic nature and measured on the end faces of the rock sample, and  $p$  (Pa) the pore fluid pressure. If the surface conductivity is neglected, the streaming potential reduces to

$$\lim_{\sigma_s \rightarrow 0} \Delta\psi = \frac{\epsilon_w \zeta}{\eta_w \sigma_w} \Delta p. \quad (9)$$

The streaming potential coupling coefficient  $C(S_w)$  (V/Pa) is given by (Helmholtz, 1879),

$$C(S_w) = \left( \frac{\Delta\psi}{\Delta p} \right)_{J=0}. \quad (10)$$

The zeta potential is related to the streaming potential coupling coefficient (e.g., Alroudhani et al., 2016; Guichet et al., 2006; Li et al., 2016; Revil et al., 1999) using the intrinsic formation factor,  $F$ , as

**Table 1**  
Petrophysical Characteristics of the Limestones (L1 and L2) and the Sandstone (S1) Samples Studied

| Sample | $\phi_w$ | $k$<br>(mD) | F    | m    | n     |        | $N_w$ | $\sigma_s$ (S/m)                  |
|--------|----------|-------------|------|------|-------|--------|-------|-----------------------------------|
|        |          |             |      |      | $N_2$ | $CO_2$ |       |                                   |
| L1     | 0.28     | 80          | 16.7 | 2.21 | 2.47  | 2.12   | 4.5   | $7.0 \times 10^{-4}$              |
| L2     | 0.27     | 100         | 16.7 | 2.15 | 2.30  | 2.62   | 4.5   | $7.0 \times 10^{-4}$              |
| S1     | 0.13     | 130         | 51.6 | 1.93 | 2.87  | 3.04   | 4.6   | $1.1 \times 10^{-4}$ <sup>a</sup> |

Note. The parameters  $\phi$  and  $k$  are respectively the porosity and the permeability.  $F$  is the formation factor,  $m$ ,  $n$ , and  $N_w$  are respectively the cementation, saturation, and Corey exponents;  $\sigma_s$  is the surface conductivity.

<sup>a</sup>Data calculated from the formula of Revil et al. (2014) for the Fontainebleau sandstones  $\log \sigma_s = -3.11 - \log(F\phi)$ .

$$C_{\text{sat}} = \frac{\varepsilon_w \zeta}{\eta_w F \sigma}, \quad (11)$$

and can be reduced to the well-known Helmholtz-Smoluchowski equation if we neglect the surface conductivity (e.g., Hunter, 1981),

$$\lim_{\sigma_s \rightarrow 0} C_{\text{sat}} = \frac{\varepsilon_w \zeta}{\eta_w \sigma_w}. \quad (12)$$

According to Linde et al. (2007) and Revil et al. (2007), the streaming potential coupling coefficient can also be related to the dynamic excess of charge dragged by the electrolyte  $\hat{Q}_V$  (given by a volume average of the local current density given by the local charge density of the diffuse layer times the local fluid flow velocity) and the permeability  $k$ , as

$$C_{\text{sat}} = -\frac{k \hat{Q}_V}{\eta_w \sigma}. \quad (13)$$

The relative streaming potential coupling coefficient is given by the ratio of the coupling coefficient at saturation  $S_w$  to the value of the coupling coefficient at full saturation (Revil & Cerepi, 2004),

$$C_r(S_w) = \frac{C(S_w)}{C_{\text{sat}}}. \quad (14)$$

Revil et al. (2007) proposed a relationship tested in Cerepi et al. (2017) and Cherubini et al. (2018) with nitrogen between the relative streaming potential coupling coefficient  $C_r$  (-) and the saturation  $S_w$  (-) using the Corey exponent ( $N_w$ ) defined using relative permeability curves according to Brooks and Corey (1964), and the saturation Archie's exponent ( $n$ ) as

$$C_r(S_w) = \frac{1}{S_w^{n+1}} (S_e)^{N_w}, \quad (15)$$

with

$$S_e = \left( \frac{S_w - S_{wi}}{1 - S_{wi}} \right), \quad (16)$$

where  $S_{wi}$  and  $S_e$  denote the irreducible water saturation and the effective saturation, respectively.

### 3. Experimental Methodology

#### 3.1. Core Samples

Three core samples were investigated at full and partial brine saturation. They include two algal rhodolith packstones (L1 and L2, also called Estailades limestones) from Provence (southeast of France) and one clay-free sandstone (S1). This clay-free sandstone corresponds to a Fontainebleau sandstone from the Paris basin. This pure silica sandstone is considered much less reactive (if not reactive at all) with respect to  $CO_2$ . The three core samples have a length of 80 mm and a cross-section diameter of 39 mm. They were drilled parallel to the stratification using a drilling tool made with tungsten carbide to avoid fracturing the samples. They were dried in an oven (60 °C) before each experiment and then saturated with a degassed brine using a vacuum pump (5 Pa). We checked that the drilling operation does not create any fractures using a micro scanner (CT-Scan). Samples L1 and L2 were drilled in the same block, few centimeters from each other.

The petrophysical properties of the core samples are reported in Table 1. The connected porosity is calculated using the dry mass difference method ( $\phi_w = V_V/V_T$ , where  $V_V$  ( $m^3$ ) and  $V_T$  ( $m^3$ ) are the volume of the connected voids in the sample and the total volume of the sample, respectively). The quantity  $V_V$  is

calculated from the mass difference between saturated and dry samples, using a pore water density equal to  $1.0 \text{ g/cm}^3$ . We obtained  $\phi_w = 0.28$ ,  $\phi_w = 0.27$ , and  $\phi_w = 0.13$  for the samples L1, L2, and S1, respectively. For the sample L1, we also calculated the porosity using a scanner, considering all the voids of the sample (connected and disconnected voids). We obtain  $\phi_{\text{scan}} = 0.29$  that is slightly higher than the value calculated using the dry mass difference method ( $\phi_w = 0.28$ ). This slight difference could be explained by a minor presence of disconnected porosity ( $\phi_{\text{disconnected}} \approx 0.01$ ). The permeability ( $k$ , in square meters) at brine saturation is determined by measuring the pressure difference measured for each brine rate, using the Darcy's law, when steady state is reached. The saturation Archie's exponent ( $n$ ) is calculated using equation (5) using conductivity data at different saturations. The cementation exponent ( $m$ ) is calculated using the first Archie's law (equation (2)) using the porosity and formation factor data. Finally, the Corey exponent ( $N_w$ ) is determined by fitting relative permeability data with the Brooks and Corey (1964) model.

### 3.2. Apparatus and Experimental Procedure

The experimental setup used to measure the streaming potential coupling coefficient and the conductivity of the samples is shown in Figure 1. The cylindrical core samples are wrapped in a rubber sleeve and confined within a core holder with a confining pressure equal to 3 MPa. We proceeded to some tests of fluid flow with a scanner to control fluid paths in real time. With the confinement pressure of 3 MPa, the pore fluid flows through the core sample and is not expected to flow through the sample-sleeve interface. The confining pressure is regulated by a pump to avoid pressure fluctuations due to temperature variations and possible leakages and is constant throughout the experiment. The stainless steel body of the core holder does not come into contact with the sample or fluids (except the oil confinement) and is grounded.

The electrical measurements (streaming potential and electrical conductivity experiments) as well as the relative permeability measurements are performed using a steady-state flooding technique. Such a system simultaneously allows injecting both a NaCl brine prepared by mixing pure dehydrated NaCl with deionized water (through the two inlets shown in Figure 1) with flow rates between 0.1 and 20 ml/min and gas (one inlet in the center of the spiral injection, Figure 1) in the sample (rocks are water wet). For each drainage experiment on the samples L1, L2, and S1, the gas partial pressure ( $N_2$  or  $CO_2$ ) is in the range 0.1–0.4 MPa. Brine and gas are therefore mixed when entering the rock sample. The steady state is reached when the pressure difference ( $\Delta p$ ) between the inlet and the outlet of the sample is stable, after few minutes. The spirals are also used as electrodes to measure the conductivity of the core sample and the streaming potential coupling coefficient. These electrodes are made of hastelloy, a material highly resistant to corrosion.

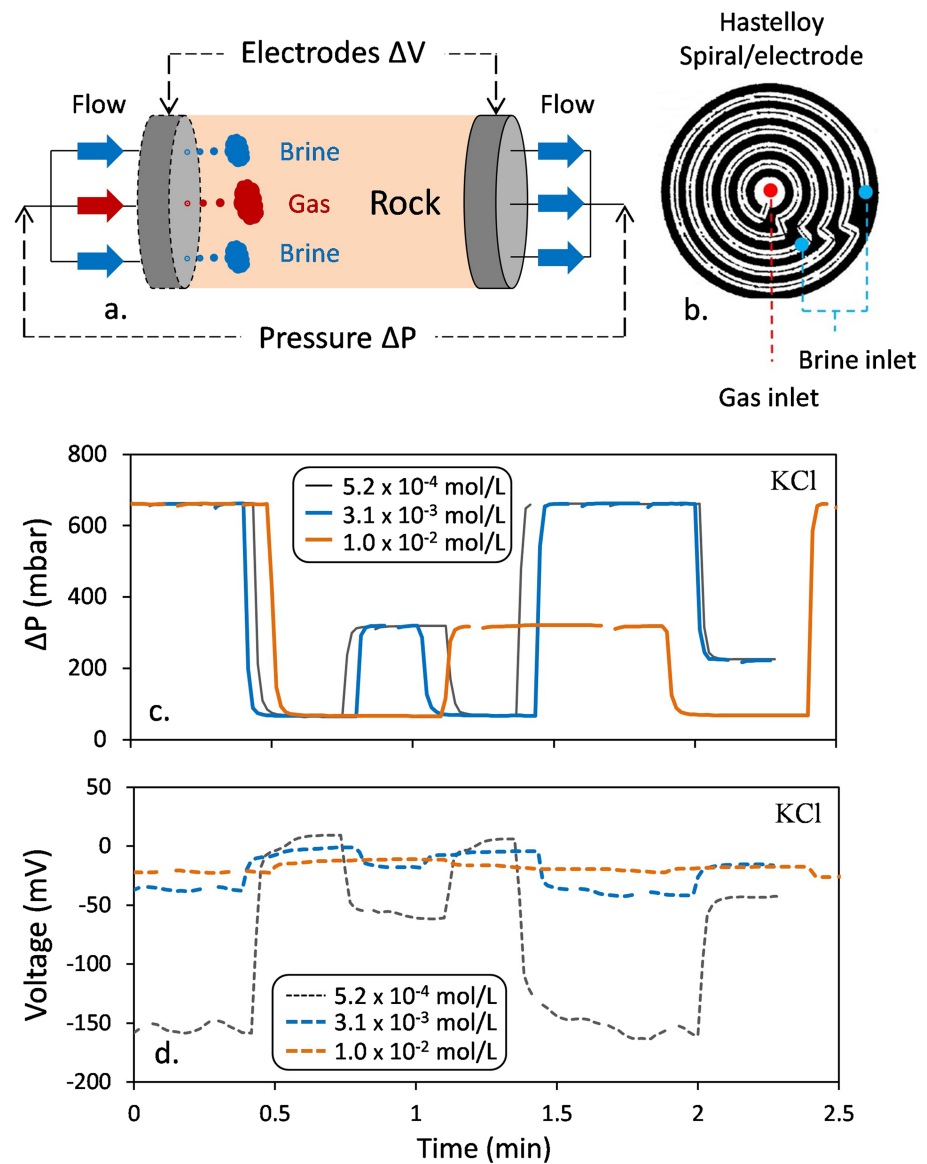
One pump is used to inject brine in the core at constant flow rate, while a gas flow regulator (Bronkorst F-201-CV) regulates the flow of gas. The brine/gas flow rate ratio is decreasing during the drainage phase. The pressure difference measured for each brine-gas ratio permits to measure relative permeabilities using Darcy's law when steady state is reached and when the water saturation in the sample is stable. At each saturation, the electrical conductivity of the sample and the streaming potential coupling coefficient are measured. Concerning electrical conductivity measurements, the voltage is self-regulated by the impedance meter in the range 0–3 V, with a phase error less than 1 mrad. Measurements are performed with a Solartron 1265 at 1 kHz, using a four-electrode arrangement, separating the current and voltage electrodes. Current electrodes are 8 cm apart, while voltage electrodes are separated by 2.5 cm.

As we calculate the pore water conductivity using the model of Le Roux et al. (2013), it is necessary to verify if the porosity remains the same, before and after a  $CO_2$  injection. If not, Le Roux et al. (2013) model corresponding to equation (7) is not valid in our case. The sample L1 was investigated using the computerized tomography method (Figure 2). The X-ray absorption was calculated in the full saturated sample before  $CO_2$  release (red line, Figure 2a) and after drainage, when the sample is saturated again (blue line, Figure 2a).

## 4. Results and Discussion

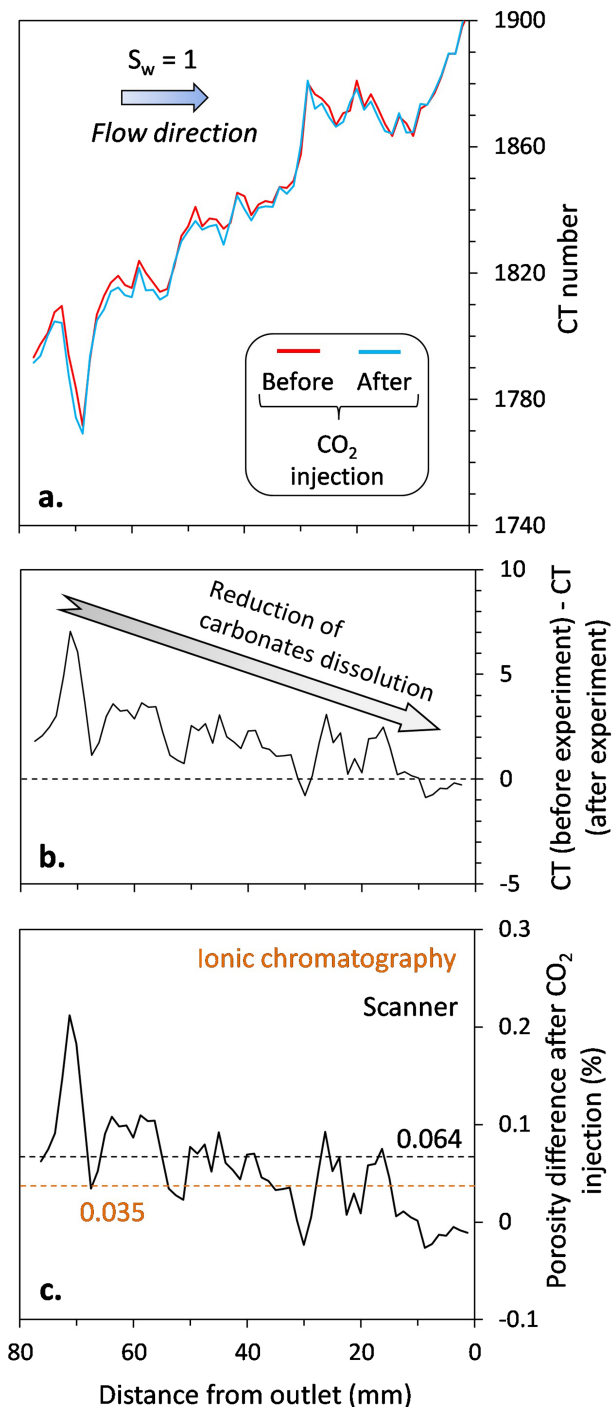
### 4.1. $CO_2$ Effects on Porosity During Drainage

A drainage experiment using gaseous  $CO_2$  was leaded on the sample L2, at ambient temperature (20 °C). The pore water pressure before drainage is around 0.15 MPa (the water flow rate is equal to 10 ml/min), and  $CO_2$



**Figure 1.** Sketch of the experimental setup used to measure the streaming potential coupling coefficient and electrical conductivity of the porous plug. (a) Positioning of the electrodes (gray circles). (b) Sketch of the nonpolarizable Hastelloy electrodes used to measure the streaming potential and the sample electrical conductivity at 1 kHz. The electrode has the same diameter as the cylindrical core sample wrapped in the sleeve. At the same time, the shape of the electrodes allow the simultaneous injection of both the gas and brine in the core sample. (c) The streaming potential coupling coefficient is calculated using the ratio of the pressure difference between the inlet and the outlet of the sample the voltage difference. (d) Streaming voltage difference across the core sample. We use a KCl brine at three different salinities ( $5.2 \times 10^{-4}$ ;  $3.1 \times 10^{-3}$ ;  $1.0 \times 10^{-2}$  Mol/L) on the sample L1.

partial pressures (in MPa) are in the range  $0.1 < p_{\text{CO}_2} < 0.4$  during drainage. When the irreducible water saturation is reached, the sample is saturated again using the same brine. The CT number difference (which is a normalized value of the X-ray absorption coefficient of a pixel) has a positive value along the core (Figure 2b), meaning that the density of the sample-brine system was higher before drainage. We also note that the effective dissolution is higher close to the inlet and is almost equal to zero near the outlet. The system brine- $\text{CO}_2$ -dissolved ions tend toward chemical equilibrium as the  $\text{CO}_2$  flows in the sample. This observation is consistent with the studies by both Grigg et al. (2003) and Luquot and Gouze (2009).



**Figure 2.** Computed tomography scan response of the sample L1 under  $N_2$  and  $CO_2$  conditions. (a) CT-scan along the core sample L1 before drainage by  $CO_2$  (red line) and after  $CO_2$  injection (blue line). These test are performed at full brine saturation ( $[NaCl] = 0.017$  Mol/L). (b) Difference between the CT numbers showing higher values in the vicinity of the inlet. Note that the difference tends to zero at the outlet of the sample. This implies in turn an increase of porosity in the vicinity of the inlet. (c) Generated porosity after the drainage phase using  $CO_2$ .

With the scanner laboratory team, we first establish a CT number baseline using the sample saturated by deionized water. Then, we remove the sample of the sleeve to dry it in an oven (at  $60^\circ C$ ). We replace the dry sample in the sleeve and establish another baseline of the CT number. In each case (saturated or dry state), the sleeve is in the stainless steel body to consider the X-ray attenuation due to the body composition. Knowing the CT numbers of saturated and dry samples, and the densities of the water and the samples, we are able to define a calibration curve allowing us to convert the CT number to porosity (Figure 2c). The result shows that the increase of porosity during the drainage process is negligible. Indeed, the average of absolute porosity variation along the core is only equal to  $+0.06\%$ . In comparison, the dissolved material balance calculated using ionic chromatography gives an absolute porosity variation equal to  $+0.035\%$ , that is in the same range. We also noted that the dissolution is higher close to the inlet (porosity increase of  $+0.2\%$ ) than to the outlet of the sample (no porosity variation). However, we could not compare these results with respect to the porosity calculated using the dry mass difference before and after experiment. Indeed, we observe a disaggregation of the sample during the removal process from the sleeve.

Other authors found porosity variations much higher than our values (Auffray et al., 2016), between  $+5\%$  and  $+15\%$  for gaseous  $CO_2$  partial pressures between 3 and 9 MPa. The solubility of  $CO_2$  is pressure, temperature, and ionic strength dependent. According to the Henry law in a  $CO_2/H_2O$  system, the solubility of  $CO_2$  in water decreases when the temperature increases up to  $150^\circ C$  under atmospheric pressure (Enick & Klara, 1990) and increases for temperatures higher than  $150^\circ C$ . On the other hand, the aqueous solubility of  $CO_2$  is greater at elevated pressure (Rosenbauer et al., 2005), which may explain the high dissolution rate of calcite in the study of Auffray et al. (2016).

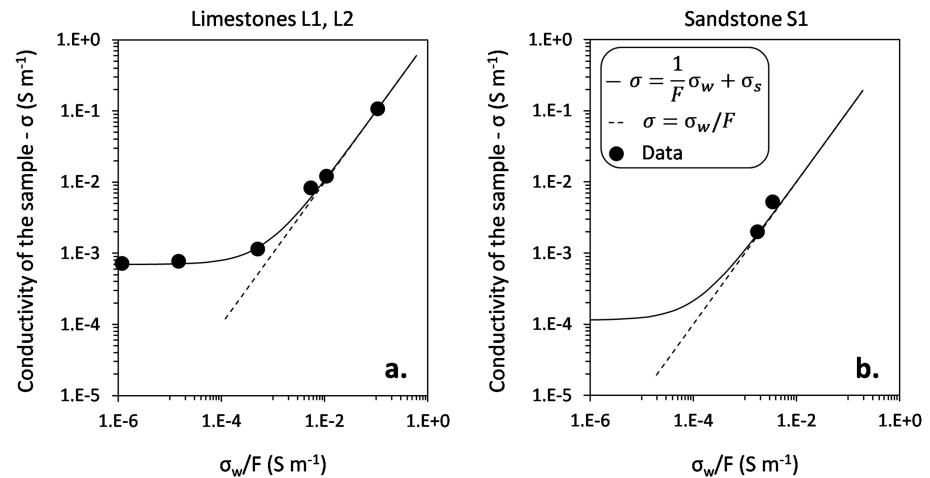
## 4.2. Influence of $CO_2$ on the Conductivity

### 4.2.1. Surface Conductivity

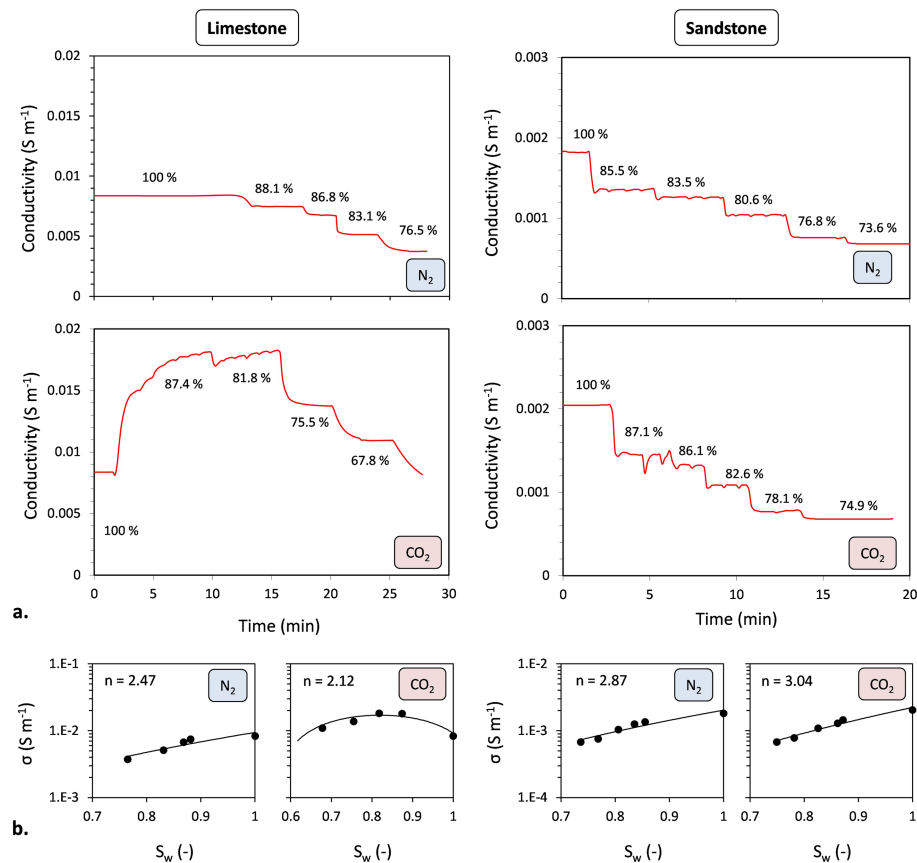
In order to determine if the surface conductivity can be neglected, we performed electrical conductivity measurements on the three core samples at different pore water conductivities (Figure 3). We see for all samples the high-salinity asymptotic behavior (equation (6)) for which the pore water conductivity dominates (from  $\sigma_w/F = 1.4 \times 10^{-2}$  S/m for the limestones L1 and L2 and  $\sigma_w/F = 2.2 \times 10^{-3}$  S/m for the sandstone S1) and the low-salinity behavior for which surface conductivity dominates. Then, the surface conductivity for the limestones is equal to  $\sigma_s = 7.0 \times 10^{-4}$  S/m. For the Fontainebleau sandstone, the surface conductivity has been calculated following the formula  $\log \sigma_s = -3.11 - \log(F\phi)$  given in Revil et al. (2014) and validated for a large data set of Fontainebleau sandstones, using the formation factor and the porosity. The surface conductivity is equal to  $\sigma_s = 1.1 \times 10^{-4}$  S/m. In each case,  $\sigma_s \ll \sigma_w/F$  and the surface conductivity can be neglected with respect to the pore water conductivity. The intrinsic formation factor is equal to 11.9 for the two limestones and 51.6 for the sandstone.

### 4.2.2. Second Archie's Exponent in a Brine-Gas System

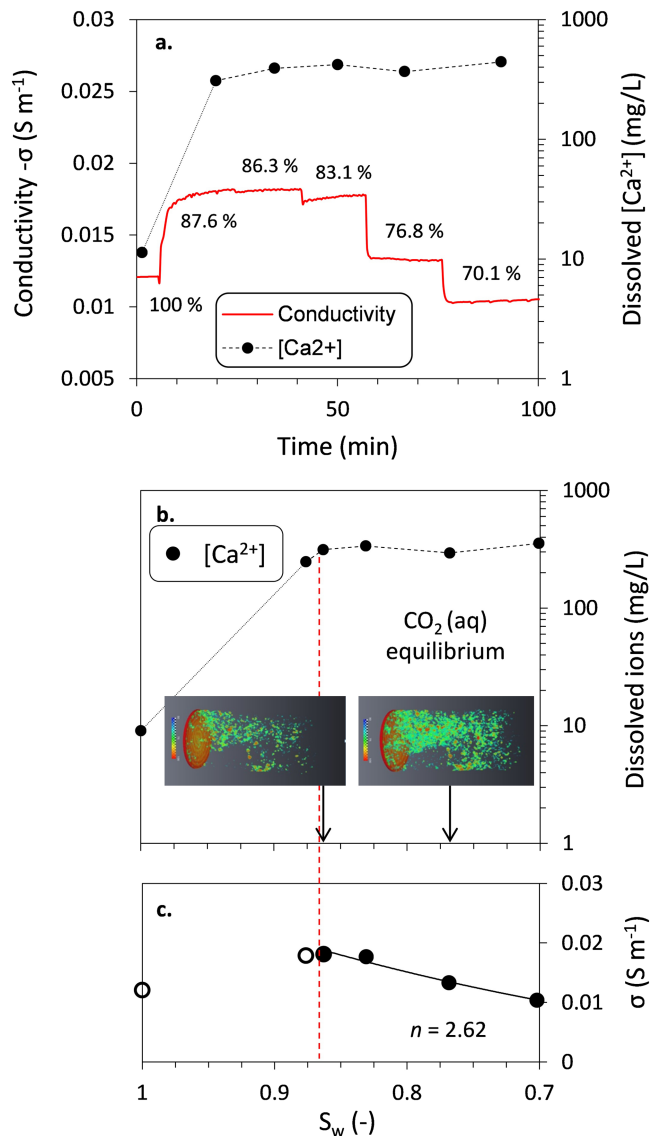
The electrical conductivity of two samples (one pure limestone, L1, and one sandstone, S1) was measured during a drainage process. We used these two types of rocks with the goal of determining the influence of  $CO_2$  on the rock conductivity in presence of carbonates. In contrast,



**Figure 3.** Conductivity of the samples versus pore water conductivity  $\sigma_w$  at full saturation. The data (black circles) are fitted with the conductivity model (plain line) of Waxman and Smits (1968). The surface conductivity  $\sigma_s$  is equal to (a)  $7.0 \times 10^{-4}$  S/m in the limestones L1 and L2 and (b)  $1.1 \times 10^{-4}$  S/m in the sandstone S1. For the sandstone,  $\sigma_s$  is calculated using the formula  $\log \sigma_s = -3.11 - \log(F\phi)$  from Revil et al. (2014) due to the lack of data at low salinities.



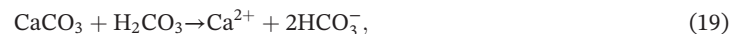
**Figure 4.** Conductivity during drainage under  $N_2$  and  $CO_2$  conditions (a) in the limestone L1 (left column) and the sandstone S1 (right column). (b) When  $N_2$  is injected, the conductivity follows equation (5) in both rocks (surface conductivity can be neglected), while it is not the case in the presence of  $CO_2$  for the limestone (the pore fluid conductivity has to be corrected). The brine is composed of 8.55 mMol/L NaCl ( $\sigma_w = 1.0 \times 10^{-1}$  S/m at  $20^\circ\text{C}$ ) in each case.



**Figure 5.** Influence of calcite dissolution on the conductivity during drainage under  $CO_2$  conditions. (a) Conductivity of the sample L2 during the drainage phase, with a brine concentration equal to 17.1 mMol/L before  $CO_2$  injection, associated with (b) calcite dissolution into the pore water for each saturation step. Two tomograms show the gas phase (green bubbles) for two water saturation steps (0.863 and 0.768).  $CO_2$  equilibrium is reached at the first drainage step ( $S_w = 0.876$ ), allowing to calculate (c) the saturation Archie's exponent ( $n = 2.62$ ).

#### 4.2.3. Calcite Dissolution in a Brine- $CO_2$ System

A drainage experiment in carbonate using  $CO_2$  has been performed with sample L2 with an initial NaCl concentration equal to 0.017 Mol/L. From equation (17), the formation of  $H_2CO_3$  and  $H^+$  leads to chemical reactions responsible for the dissolution of calcite (Talman et al., 1990)

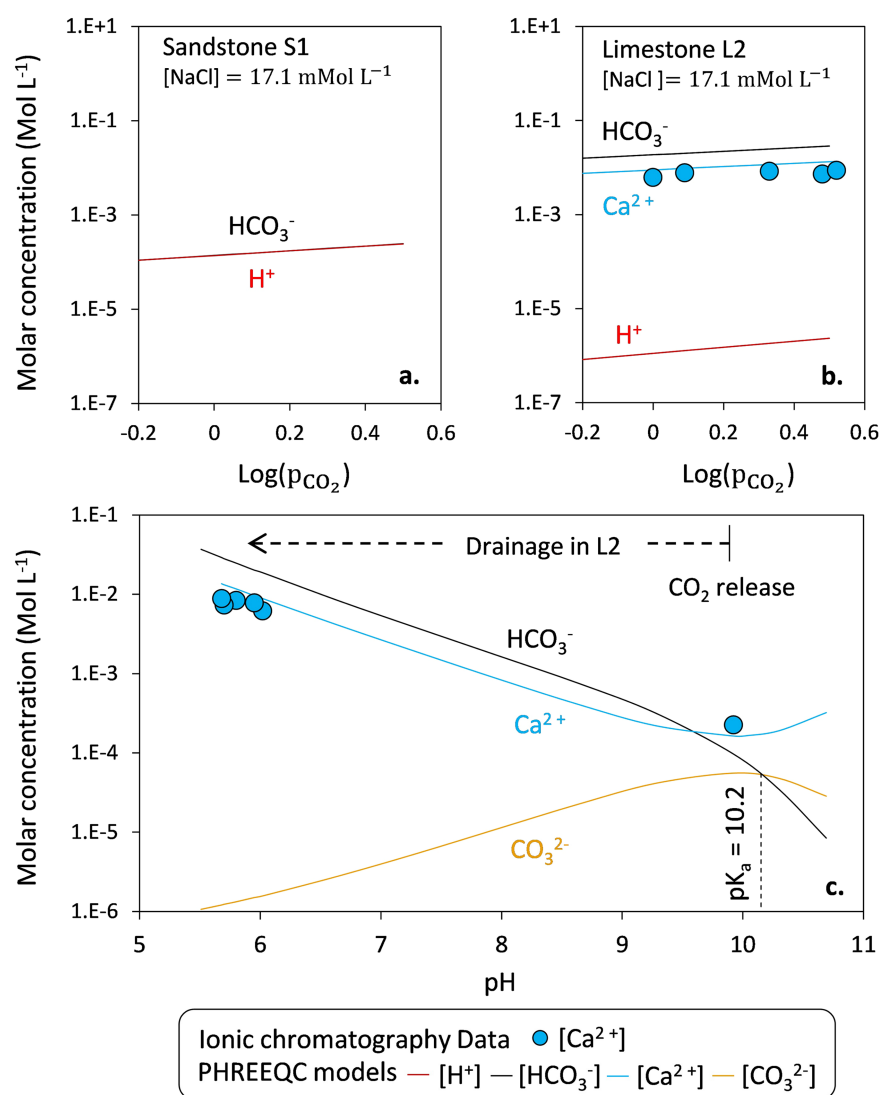


silica materials are not expected to react in presence of  $CO_2$ . In the general case, if the gas does not react with the sample, the electrical conductivity of the core sample decreases when the water saturation decreases too. Both drainage experiments were first performed using nitrogen acting as neutral gas (Figure 4a, upper line). Then, these drainage experiments were performed again using carbon dioxide (Figure 4a, lower line). In all these experiments, the salinity of the pore water is equal to 8.55 mMol/L of NaCl ( $\sigma_w = 1.0 \times 10^{-1}$  S/m at 20 °C) in each case. As expected, the rock conductivity and the water saturation are related to each other by a power law (equation (5)) under nitrogen conditions (Archie, 1942). Equation (5) is indeed valid in this case since as shown in section 4.2, the surface conductivity can be safely neglected for the salinity used in our experiment. We obtain  $n = 2.47$  and  $n = 2.87$  for the core samples L1 and S1, respectively (Figure 4b). Under carbon dioxide conditions,  $n = 3.04$  for S1, while  $n$  cannot be directly estimated for L1 without pore water conductivity corrections, due to the conductivity increase during the first part of drainage (between  $S_w = 1$  and  $S_w = 0.818$ ). The same phenomenon was previously observed in Le Roux et al. (2013). A significant increase of conductivity is observed in this case when carbon dioxide is injected in carbonate rocks (Figure 4a, lower left corner) due to the elevation of the ionic strength of the electrolyte (between 0 and 15 min of injection). The unexpected peaks (Figure 4a, lower right corner) during the first step of the  $CO_2$  release (at  $S_w = 0.871$ ) could be due to the heterogeneity of the mixture close to the electrodes area. The gaseous  $CO_2$  does not seem to have finished migrating by gravity to the top of the sample.

This conductivity increase can be associated with a change in the pore water conductivity related to  $CO_2$  dissolution (Lindsay, 1979; Plummer et al., 1979; Rasmuson et al., 1990). The dissolution of  $CO_2$  in the pore water can be written as



These ionic species (bicarbonate and carbonate) influence the pore water conductivity in a given set of conditions and, thereby, the electrical rock conductivity. This reaction holds also in sandstones. However, in the same experimental conditions (temperature and pore pressure), the electrical rock conductivity during drainage has the same trend in the sandstone S1 using  $N_2$  or  $CO_2$ . From these observations, we expect that the dissociation of  $CO_2$  in pore water cannot be the mechanism controlling the electrical rock conductivity change in carbonates. Next, we investigate the influence of the dissolution of calcite regarding the observed change in the electrical conductivity.

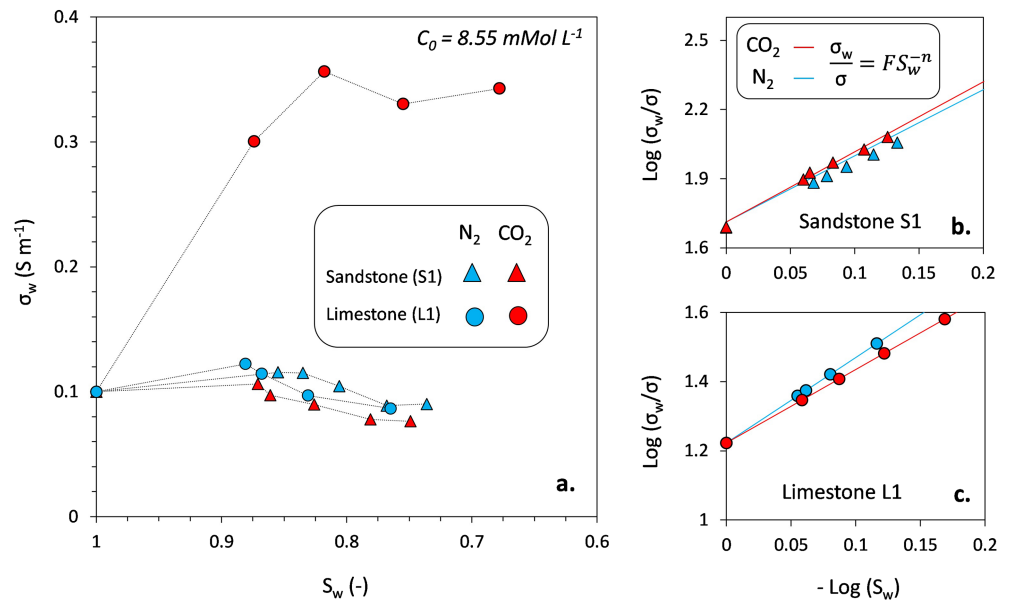


**Figure 6.** Simulated Molar concentration (using PHREEQC) of dissolved ions versus CO<sub>2</sub> partial pressure during drainage. (a) Simulation for sandstone S1, following the reaction  $\text{CO}_2 + \text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{HCO}_3^-$ . (b) Simulation for limestone L2 following the global reaction  $\text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{Ca}^{2+} + 2\text{HCO}_3^-$ . (c) The model fits very well with data of dissolved Ca<sup>2+</sup> (blue circles) measured using ionic chromatography and predicts a drastic decrease of CO<sub>3</sub><sup>2-</sup> during drainage and an increase of Ca<sup>2+</sup> and HCO<sub>3</sub><sup>-</sup> due to the decrease of the pH.

Although H<sup>+</sup>, H<sub>2</sub>CO<sub>3</sub>, and H<sub>2</sub>O reaction with calcite occur simultaneously throughout (far from equilibrium, as well as at equilibrium), the calcite dissolution is dominated by reaction with single species as a function of pH and CO<sub>2</sub> partial pressure (Plummer et al., 1979). The reaction of calcite with water is not considered in our dissolution rate, because it occurs when  $p_{\text{CO}_2} \leq 0.01$  MPa (Plummer et al., 1979), which is significantly lower than the CO<sub>2</sub> partial pressure in our system. Moreover, we consider that there is no longer CO<sub>2</sub> in the brine (liquid or supercritical) when the system reaches the equilibrium due to temperature (20 °C) and pressure (<0.5 MPa) conditions. Then, dissolution of calcite is determined by the overall reaction



We test now the assumption that this change is responsible for the observed conductivity change. To check our assumption, we analyzed the changes of the concentration of Ca<sup>2+</sup> (the main cations in the pore water) by ionic chromatography at different saturations (Figure 5). The initial calcium



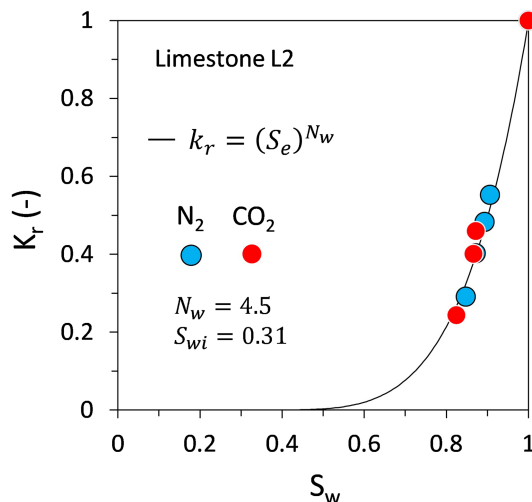
**Figure 7.** Pore water conductivity during the drainage (a) under  $\text{CO}_2$  (red symbols) and  $\text{N}_2$  (blue symbols) conditions, calculated using equation (7), for the limestone L1 (circles) and the sandstone S1 (triangles). The initial NaCl brine concentration is 8.55 mMol/L. (b, c) Rock conductivity dependence of pore water conductivity for the sandstone S1 (upper graph) and the limestone L1 (lower graph). Models (blue and red lines) are calculated using equation (5).

content (at  $S_w = 1$ ) is about 10 mg/L (when the brine is equilibrated with the surface of the rock), whereas its final concentration (during drainage) is close to 350 mg/L. This is 40 times higher than the initial concentration. The  $\text{Ca}^{2+}$  equilibrium in the pore water is reached at the end of the second desaturation step ( $S_w \approx 0.863$ ). At the same time, the electrical rock conductivity reaches its higher value during this drainage step (Figure 5a). When the  $\text{Ca}^{2+}$  equilibrium is reached, the conductivity decreases with decreasing the saturation following the second Archie's law (Figure 5c). This result is

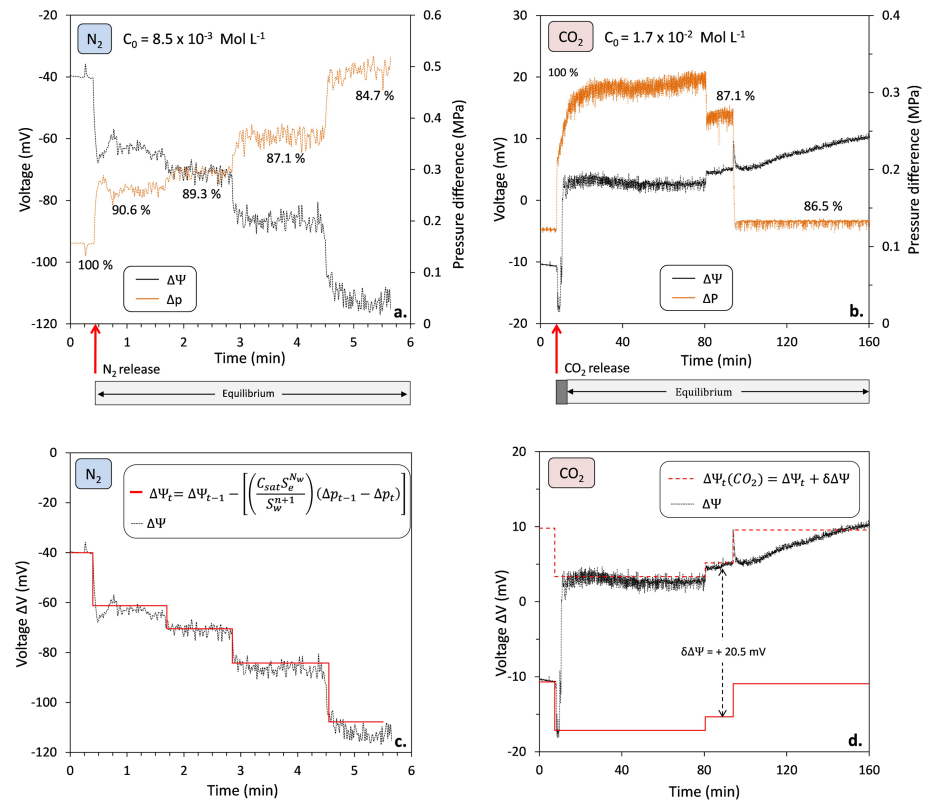
not however consistent with the studies of Dreybrodt et al. (1997) and Lide (2003), which argue that conductivity modifications in pore water are mostly related to  $\text{CO}_2$  dissociation in water than calcite dissolution due to the high value of gas dissolution kinetic of  $\text{CO}_2$  for the first ones, and due to the higher value of  $\text{H}^+$  ionic conductivity than  $\text{Ca}^{2+}$  ionic conductivity for the second one.

In order to go one step further in our analysis, we performed a numerical simulation test with the software package PHREEQC (Parkhurst & Appelo, 1999). We use this software to estimate the evolution of the overall dissolution reactions in the sandstone S1 and the limestones L2, using experimental  $\text{CO}_2$  partial pressures and brine NaCl concentration (Figure 6). As expected from equation (17), the  $\text{CO}_2$  dissolution in pore water (equation (17)) is the only reason of the presence of  $\text{HCO}_3^-$  in the pore fluid (Figure 6a). Note that the negligible concentration of  $\text{CO}_3^{2-}$  (equation (17)) is explained by the low value of the pH during drainage (Figure 6c). The presence of  $\text{HCO}_3^-$  and  $\text{H}^+$  have a low impact on the overall ionic strength of the pore fluid ( $\text{HCO}_3^-$ : 0.6;  $\text{H}^+$ : 0.6%;  $\text{Na}^+$ : 49.4%;  $\text{Cl}^-$ : 49.6%).

At the opposite, the overall reaction leading to the dissolution of calcite in presence of  $\text{CO}_2$  lead to high concentrations of  $\text{Ca}^{2+}$  and  $\text{HCO}_3^-$  with respect to what is expected from equation (21) in the pore fluid



**Figure 8.** Water relative permeability data during drainage using  $\text{N}_2$  (blue circles) and  $\text{CO}_2$  (red circles) for the limestone L2. In each case, the Corey exponent ( $N_w$ ) is equal to 4.5 and determined using the least squares method.



**Figure 9.** Streaming potential during drainage in the limestone L2. (a, b) Voltage (black line) and pressure difference (orange line) relationships in a brine-N<sub>2</sub> (a) and a brine-CO<sub>2</sub> (b) system, for different saturation steps at NaCl brine concentrations, respectively, equal to 8.55 mMol/L and 17.1 mMol/L. (c, d) Voltage during the drainage associated to the theoretical voltage calculated using equation (24) (red line) under (c) N<sub>2</sub> and (d) CO<sub>2</sub> conditions. The red dashed line represents the theoretical model based on equations (10), (14), and (15). The potential difference  $\delta\Delta\psi$  between the two models is equal to 20.5 mV.

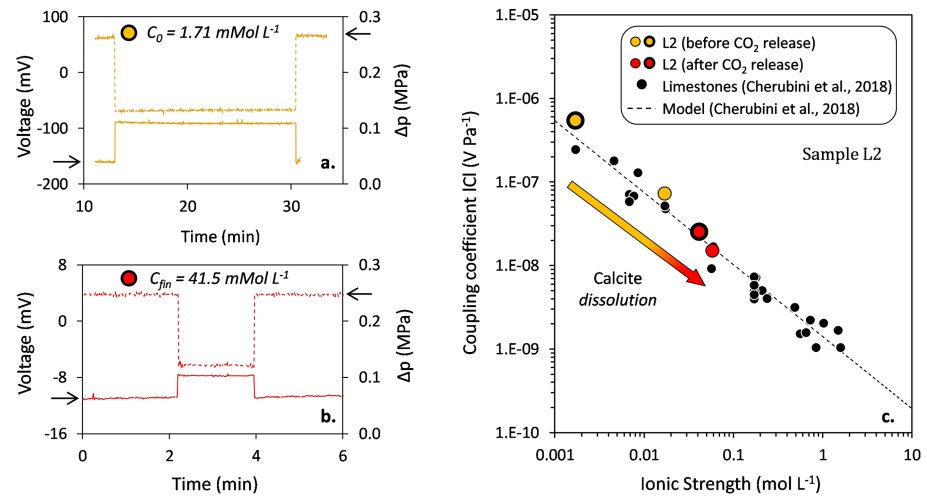
(Figure 6b). The simulation seems applicable to our system. Data (obtained using ionic chromatography) and simulated value of Ca<sup>2+</sup> dissolution are consistent in the limestone L2 (Figure 6b). Ca<sup>2+</sup> and HCO<sub>3</sub><sup>-</sup> have a high impact on the overall ionic strength of the pore fluid (Ca<sup>2+</sup>: 46.2%; HCO<sub>3</sub><sup>-</sup>: 25.0%; H<sup>+</sup>: negligible; Na<sup>+</sup>: 14.4%; Cl<sup>-</sup>: 14.4%). Therefore, the conductivity of pore water is controlled by the calcite dissolution especially by the high contribution of Ca<sup>2+</sup> on the overall ionic strength of the pore water.

Moreover, the electrical rock conductivity is related to the saturation by equation (5) when the Ca<sup>2+</sup> equilibrium is reached. At the saturation  $S_w \approx 0.863$ , we determine the saturation exponent  $n = 2.62$ . This value of the saturation exponent is close than the value of the saturation determined under nitrogen conditions ( $n = 2.30$ ).

#### 4.2.4. Pore Water Conductivity

In section 4.1, we observe that the porosity does not change in a carbonate-brine system in presence of CO<sub>2</sub>, at ambient temperature and with a pore pressure in the range 0.1–0.4 MPa. The formation factor ( $F$ ) and the saturation exponent ( $n$ ) remain also the same. The formula proposed by Le Roux et al. (2013) is applicable in our case when surface conductivity can be neglected (see section 4.2.1). We can use this model to calculate and monitor the pore water conductivity (equation (7)) relative to the experiments presented in Figure 4. In presence of N<sub>2</sub> or CO<sub>2</sub>, the pore water conductivity is quite the same (we just observe a low decrease of conductivity from 0.1 to 0.08 S/m) during drainage in the sandstone S1 (Figure 7a). With an initial NaCl brine concentration ( $C_0$ ) equal to 8.55 mMol/L, the initial pore water conductivity is equal to 0.1 S/m at  $S_w = 1$ , whereas it is also around 0.1 S/m at  $S_w = 0.74$  in each case.

For limestone L1, the pore water conductivity is also around 0.1 S/m in presence of N<sub>2</sub> in the saturation range  $S_w = 0.75$ –1. This is not the case in presence of CO<sub>2</sub>. As shown above (see section 4.2.3) the pore water conductivity is controlled both by the calcite dissolution and the presence of Ca<sup>2+</sup> in pore water. Its value



**Figure 10.** Streaming potential coupling coefficient at full saturation in the limestone L2 (a) before CO<sub>2</sub> release ( $C_{\text{sat}} = -5.46 \times 10^{-7}$  V/Pa) and (b) after CO<sub>2</sub> release ( $C_{\text{sat}} = -2.55 \times 10^{-8}$  V/Pa). The initial NaCl brine concentration is equal to 1.71 mMol/L, whereas the total ionic strength after dissolution equals 41.5 mMol/L. (c) These results are compared to the absolute values of streaming potential coupling coefficient data (black circles) and model (dashed line) obtained in carbonate rocks by Cherubini et al. (2018), using NaCl brines, at full saturation.

increases in the saturation range 1–0.82 to reach a stable value around 0.34 S/m in the saturation range 0.82–0.67. Then, the rock conductivity is related to the saturation by the same power law in the sandstone S1 (Figure 7b) and in the limestone L1 (Figure 7c), in presence of N<sub>2</sub> or CO<sub>2</sub>, adding pore fluid conductivity corrections.

### 4.3. CO<sub>2</sub> Effects on Electrokinetic Properties

We evaluate now the influence of CO<sub>2</sub> on the streaming potential in porous media. The fitting parameters are the saturation Archie's exponent ( $n$ ) and the Corey exponent ( $N_w$ ), which are both determined using equation (5) (assuming that the surface conductivity is negligible) and water relative permeability curves with drainage data (Figure 8), respectively. These parameters will be used to establish a predictive curve for the streaming potential based on the Helmholtz-Smoluchowski equation (section 4.3.1) and the relative streaming potential coupling coefficient model (equation (15)).

#### 4.3.1. Streaming Potential Model

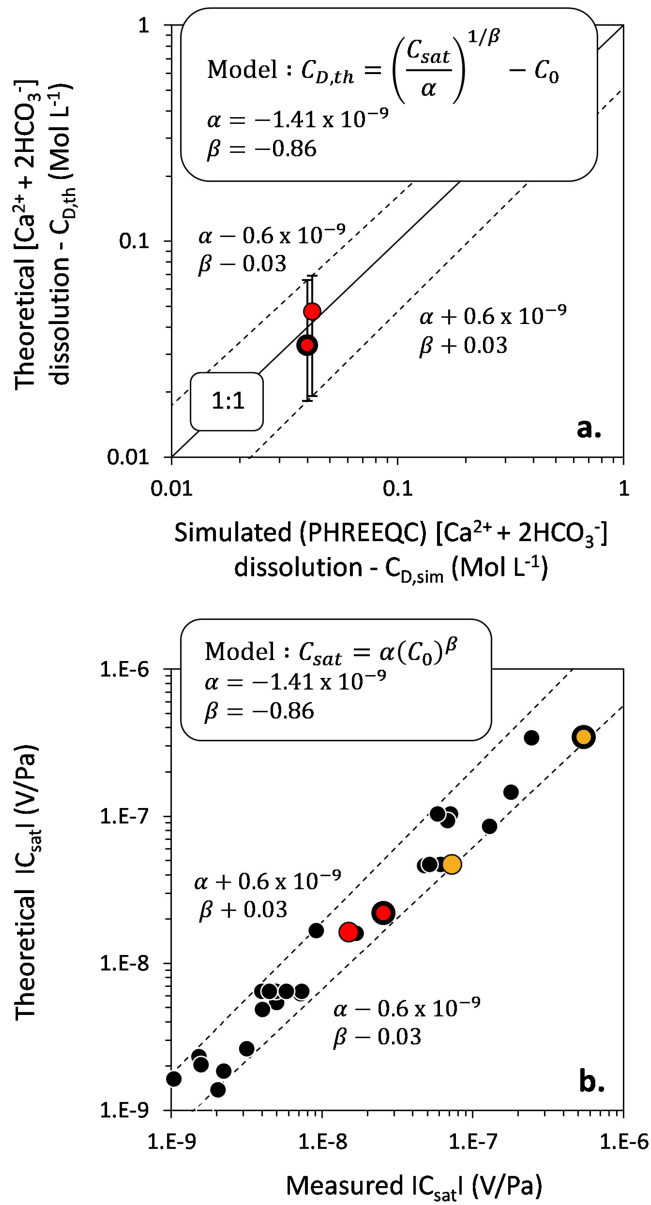
Drainage processes under N<sub>2</sub> and CO<sub>2</sub> conditions have been performed on the core samples L1 and L2 (Figure 8) to determine their relative permeability ( $k_r$ ) curves. In turn, these curves are fitted with the relative permeability model ( $k_r = S_e^{N_w}$ ) of Brooks and Corey (1964) and we obtain the same value of the Corey exponent  $N_w = 4.5$  for the two core samples. At each saturation step, the voltage and the pressure difference between the inlet and the outlet face are recorded (Figures 9a and 9b). The ratio between the pressure difference and the voltage (in other words, the streaming potential coupling coefficient, see equation (10)) is negative. We focus on streaming potential data under unsaturated conditions in a limestone (L1 with N<sub>2</sub> and L2 with CO<sub>2</sub>). We propose to modify equation (10) as

**Table 2**

*Ionic Strength, Streaming Potential Coupling Coefficient ( $C_{\text{sat}}$ ), and Zeta Potential ( $\zeta$ ) of the Saturated Sample L2 for Two Drainage Experiments, Before and After the CO<sub>2</sub> Injection*

| Sample L2  | Ionic strength (mMol/L) |                       | $C_{\text{sat}}$ (nV/Pa) |                       | $\zeta$ (mV)           |                       |
|------------|-------------------------|-----------------------|--------------------------|-----------------------|------------------------|-----------------------|
|            | Before CO <sub>2</sub>  | After CO <sub>2</sub> | Before CO <sub>2</sub>   | After CO <sub>2</sub> | Before CO <sub>2</sub> | After CO <sub>2</sub> |
| Drainage 1 | 1.7                     | 41.5                  | −546.0                   | −25.5                 | −23.9                  | −18.7                 |
| Drainage 2 | 17.1                    | 58.6                  | −72.6                    | −15.0                 | −22.1                  | −15.6                 |

*Note.* The zeta potential is calculated from equation (11).



**Figure 11.** Measured versus theoretical  $[Ca^{2+} + 2HCO_3^-]$  dissolution after  $CO_2$  release in the limestone L2. (a) The theoretical dissolution is calculated using equation (26), with  $\alpha = -1.41 \times 10^{-9}$  and  $\beta = -0.86$  (values from Cherubini et al., 2018). Using two initial NaCl brine concentrations ( $C_0 = 1.71$  mMol/L (orange circle, thick border) and  $C_0 = 17.1$  mMol/L (orange circle, thin border)), the streaming potential coupling coefficient is respectively equals to  $C_{sat} = -25.5$  nV/Pa and  $C_{sat} = -15.0$  nV/Pa after  $CO_2$  release. (b) The error bars in the upper figure are determined using the streaming potential coupling coefficient data of Cherubini et al. (2018) with their uncertainty.

the ionic strength in carbonates (Cherubini et al., 2018). As expected, the use of  $CO_2$  has a real impact on the value of  $C_{sat}$  because of calcite dissolution. Although we lead drainage in carbonates using  $CO_2$  only twice, we saw at a first glance that the streaming potential coupling coefficient could be an interesting tool, alternatively to costly methods to determine the dissolution rate after  $CO_2$  injection in carbonates. By subtracting the initial brine concentration to the final brine concentration after  $CO_2$  release calculated from streaming potential measurements, we obtain,

$$C(S_w) = \left(\frac{\Delta\Psi}{\Delta p}\right)_{J=0} = \frac{\Delta\Psi_{t-1} - \Delta\Psi_t}{\Delta p_{t-1} - \Delta p_t}, \quad (22)$$

$$\Delta\Psi_t = \Delta\Psi_{t-1} - [C(S_w)(\Delta p_{t-1} - \Delta p_t)]. \quad (23)$$

Combining equations (15) and (23) leads to

$$\Delta\Psi_t = \Delta\Psi_{t-1} - \left[\frac{C_{sat} S_w^{N_w}}{S_w^{n+1}} (\Delta p_{t-1} - \Delta p_t)\right]. \quad (24)$$

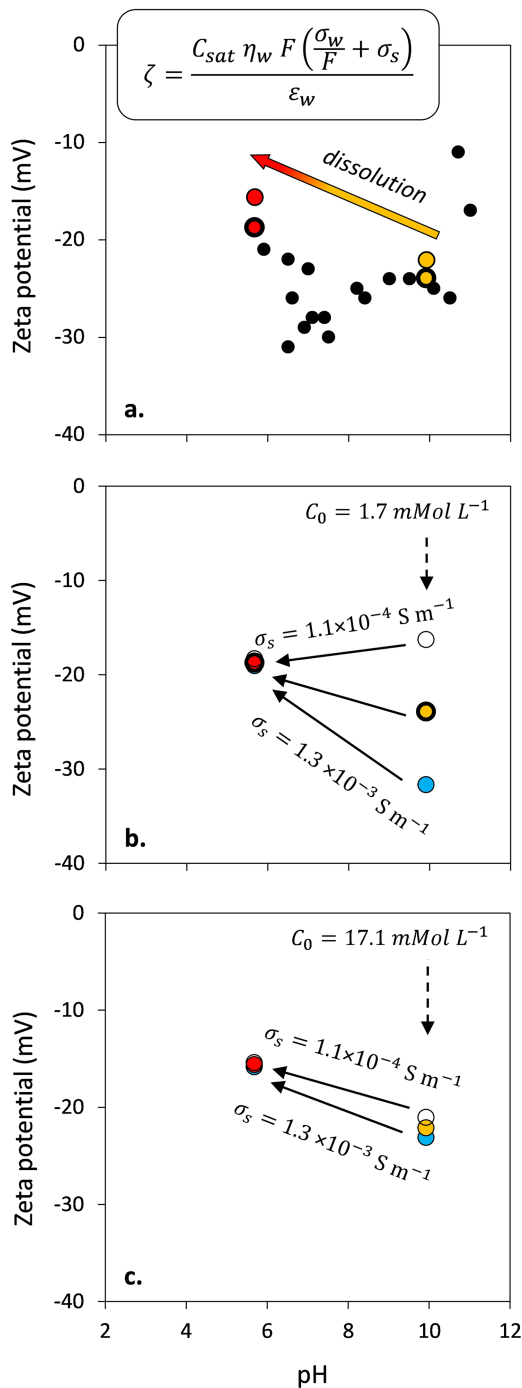
The streaming potential at a given time  $\Delta\Psi_t$  (V) is now connected to the water saturation  $S_w$  and the streaming potential coupling coefficient at full saturation  $C_{sat}$ . This model is fitted with experimental data in Figures 9c and 9d, under nitrogen and carbon dioxide conditions. On one hand, the model fits quite well with data when the nonpolar phase is nitrogen (red line, Figure 9c). The initial streaming potential ( $\Delta\Psi_0$ , at  $S_w = 1$ ) is equal to  $-40$  mV for a brine salinity equal to 8.55 mMol/L. On the other hand, the model does not fit with data when carbon dioxide is injected (red line, Figure 9d). The model seems only valid before and few seconds after the  $CO_2$  injection. There is a voltage drop immediately after the injection shown by a significant increase of the observed streaming potential. However, it is possible to translate the model and fit it with data after the  $CO_2$  injection, using a given streaming potential difference  $\delta\Delta\Psi$  equal to  $+20.5$  mV (red dashed line in Figure 9d) as,

$$\Delta\Psi_t(CO_2) = \Delta\Psi_t + \delta\Delta\Psi. \quad (25)$$

We saw in section 4.2.4 that the pore water conductivity increases after the  $CO_2$  release until it reaches a stable value quickly. This effect probably explains the voltage drop after the injection. The model fits quite well when the pore water is at the equilibrium with the solid phase. That means that the relative streaming potential coupling coefficient model of Revil et al. (2007) is also valid using  $CO_2$  and not only with an inert gas (such as  $N_2$ ) as tested in Cherubini et al. (2018).

#### 4.3.2. Influence of Calcite Dissolution on $C_{sat}$

The streaming potential coupling coefficient has been calculated before and after  $CO_2$  release during two drainage experiments on the sample L2 (Figure 10). Its magnitude decreases from  $-546.0$  to  $-25.5$  nV/Pa and from  $-72.6$  to  $-15.0$  nV/Pa when the initial brine concentrations are respectively equal to 1.7 and 17.1 mMol/L (Table 2). The initial brine concentration is known (brine prepared in laboratory), whereas the final concentration has been determined using ionic chromatography and PHREEQC. Initial (yellow circles) and final (red circles) data of the streaming potential coupling coefficient (at  $S_w = 1$ ) are represented in Figure 10c and are closed to the trend connecting  $C_{sat}$  and



**Figure 12.** Variations of the zeta potential as a function of pH. (a) Our data are presented in colored circles showing the influence of dissolution on the magnitude of the zeta potential. Data of Guichet et al. (2006) are also represented (black circles) at constant ionic strength of the aqueous solution ( $2.0 \times 10^{-3} \text{ Mol/L}$ ) using a  $\text{CaCl}_2$  electrolyte. Their measurements were performed with a sand composed of 98% of quartz and 2% of calcite. (b, c) Check of the sensitiveness of zeta potential for various surface conductivities in the pH range 5.5–10 for initial brine concentrations ( $C_0$ ) equal to 1.7 and 17.1 mMol/L<sup>-1</sup> respectively. The values of  $1.1 \times 10^{-4} \text{ S/m}$  is taken from Guichet et al. (2006), whereas the value of  $1.3 \times 10^{-3} \text{ S/m}$  is from Li et al. (2016). Note that the surface conductivity of our sample L2 before drainage (orange circle) is equal to  $7.0 \times 10^{-4} \text{ S/m}$ .

$$C_{D,\text{th}} = \left( \frac{C_{\text{sat}}}{\alpha} \right)^{1/\beta} - C_0, \quad (26)$$

with  $C_0$  (Mol/L) the initial brine concentration and  $C_{D,\text{th}}$  (Mol/L) the theoretical dissolution rate calculated from streaming potential measurements. The fitting parameters  $\alpha = -(1.41 \pm 0.60) \times 10^{-9}$  and  $\beta = -0.86 \pm 0.03$  are defined from the study of Cherubini et al. (2018) (see Figure 11b). The dissolution rate is in the same range in spite of the initial brine concentration ( $C_{D,\text{th}} = 42 \text{ mMol/L}$  and  $C_{D,\text{th}} = 40 \text{ mMol/L}$  for  $C_0 = 17.1 \text{ mMol/L}$  and  $C_0 = 1.7 \text{ mMol/L}$ , respectively). In both cases, the theoretical dissolution rate calculated from streaming potential measurements is in the same range as the values obtained using PHREEQC (Figure 11a).

#### 4.3.3. Zeta Potential on the Calcite-Water Interface

The zeta potential calculated from streaming potential coupling coefficient data (equation (11)) presents a decrease in magnitude after dissolution, when the ionic strength increases (Table 2) in both drainage cases. The magnitude of the zeta potential decreases from  $-23.9$  to  $-18.7 \text{ mV}$  with an initial brine concentration of  $1.7 \text{ mMol/L}$  and from  $-22.1$  to  $-15.6 \text{ mV}$  for an initial concentration equal to  $17.1 \text{ mMol/L}$ . This decrease in magnitude was expected for limestones (Cherubini et al., 2018; Li et al., 2016). Indeed, it is expected to increase with the pH in the pH range 5.5–10 (Figure 12a). However, it is complicated to dissociate the effects of pH and ionic strength on the zeta potential magnitude. Li et al. (2016) showed a slight decrease of the zeta potential magnitude at constant brine concentration, assuming that the influence of the calcite dissolution is negligible under atmospheric  $\text{CO}_2$  partial pressure conditions ( $p_{\text{CO}_2} = 10^{-1.43} \text{ kPa}$ ). We question this assumption using PHREEQC. Indeed, at  $p_{\text{CO}_2} = 10^{-1.43} \text{ kPa}$ , the ionic strength of the dissolution term  $[\text{Ca}^{2+} + 2\text{HCO}_3^-]$  is equal to  $1.5 \text{ mMol/L}$ , which represents 60% of the total ionic strength of the aqueous solution ( $2.5 \text{ mMol/L}$ ) when the initial  $\text{NaCl}$  concentration is equal to  $1.0 \text{ mMol/L}$ , hence the need to store the aqueous solutions in glassware to avoid equilibrating chemical processes with atmospheric components (particularly the  $\text{CO}_2$ ). In comparison, Guichet et al. (2006) showed a quite stable zeta potential (however with a significant data dispersion) in the pH range 5.5–10 (Figure 12a) on a silica sand containing 2% of calcite.

Guichet et al. (2006) measured scattered zeta potential values for pH higher than 10.5 and assumed that this scattering is related to calcite precipitation (Figure 12a). As seen previously in this study and in Li et al. (2016) this scattering does not seem to be due to the calcite-water interface, but from the decrease in permeability, as they suggested. This assumption is consistent with the following equation obtained by combining equations (11) and (13), as

$$\zeta = -\frac{k\hat{Q}_V F}{\varepsilon_w}. \quad (27)$$

However, they performed their experiments on a sand characterized by a very low value of the surface conductivity ( $\sigma_s = 1.1 \times 10^{-4} \text{ S/m}$ ) using a  $\text{CaCl}_2$  electrolyte with a concentration equal to  $2.0 \times 10^{-3} \text{ Mol/L}$ . We represent the values of the zeta potential of the sample L2 assuming surface conductivity variations (Figures 12b and 12c). We take the values of  $\sigma_s = 1.1 \times 10^{-4} \text{ S/m}$  (Guichet et al., 2006; represented by the white

circles) and  $\sigma_s = 1.3 \times 10^{-3}$  S/m (Li et al., 2016; represented by the blue circles) for initial brine concentrations equal to  $1.7 \times 10^{-3}$  Mol/L (Figure 12b) and  $17.1 \times 10^{-3}$  Mol. L (Figure 12c). For low concentrations (Figure 12b), the magnitude of the zeta potential is highly influenced by the surface conductivity effects, while this is not the case for higher salinities (Figure 12c). This phenomenon could explain the scattered zeta potential values measured by Guichet et al. (2006) because of the likely decrease of the electrolyte concentration due to calcite precipitation at high pH values ( $>10.5$ ).

## 5. Conclusion

Electrical and electrokinetic measurements were performed to investigate the impact of CO<sub>2</sub> during drainage in carbonate rocks. Electrical conductivity measurements are also performed during drainage with N<sub>2</sub> (considered as a neutral gas) and CO<sub>2</sub> in a sandstone (considered as a chemically neutral rock with respect to CO<sub>2</sub>) and limestones (reactive rocks).

In a pure limestone, the rock conductivity increases when the saturation decreases until the pore water equilibrium is reached, when the nonwetting phase is composed of CO<sub>2</sub>. The pore water conductivity does not vary during drainage in a pure sandstone whatever the nonwetting phase (N<sub>2</sub> or CO<sub>2</sub>). This assumption is also true in a limestone in presence of N<sub>2</sub> only. The presence of CO<sub>2</sub> has a drastic impact regarding the conductivity of the pore water. Two main phenomena are tested to explain this increase of pore water conductivity in presence of CO<sub>2</sub> in carbonates. The change in Ca<sup>2+</sup> in the pore water during drainage with CO<sub>2</sub> in a limestone explains the change in the pore water conductivity. Indeed, the ionic strength of water is dominated by the presence of Ca<sup>2+</sup> when calcite dissolution occurred.

Electrokinetics properties are also impacted by the presence of CO<sub>2</sub> in carbonates. The dissolution of calcite after a CO<sub>2</sub> release leads to a decrease of the magnitude of the streaming potential coupling coefficient due to the increase of the water ionic strength (and thereby the increase of brine conductivity) in agreement with the prediction of the Helmholtz-Smoluchowski theory. The streaming potential coupling coefficient seems to be a potentially interesting tool to estimate brine concentrations after a CO<sub>2</sub> release in carbonates, and thereby, a tool to estimate dissolution rates. Moreover, we can expect to use this parameter in order to detect CO<sub>2</sub> leakages from storage sites.

## Acknowledgments

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# Zeta potential of artificial and natural calcite in aqueous solution



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## ABSTRACT

Despite the broad range of interest and applications, controls on calcite surface charge in aqueous solution, especially at conditions relevant to natural systems, remain poorly understood. The primary data source to understand calcite surface charge comprises measurements of zeta potential. Here we collate and review previous measurements of zeta potential on natural and artificial calcite and carbonate as a resource for future studies, compare and contrast the results of these studies to determine key controls on zeta potential and where uncertainties remain, and report new measurements of zeta potential relevant to natural subsurface systems.

The results show that the potential determining ions (PDIs) for the carbonate mineral surface are the lattice ions  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{CO}_3^{2-}$ . The zeta potential is controlled by the concentration-dependent adsorption of these ions within the Stern layer, primarily at the Outer Helmholtz Plane (OHP). Given this, the Iso-Electric Point (IEP) at which the zeta potential is zero should be expressed as pCa (or pMg). It should not be reported as pH, similar to most metal oxides.

The pH does not directly control the zeta potential. Varying the pH whilst holding pCa constant yields constant zeta potential. The pH affects the zeta potential only by moderating the equilibrium pCa for a given  $\text{CO}_2$  partial pressure ( $p\text{CO}_2$ ). Experimental studies that appear to yield a systematic relationship between pH and zeta potential are most likely observing the relationship between pCa and zeta potential, with pCa responding to the change in pH. New data presented here show a consistent linear relationship between equilibrium pH and equilibrium pCa or pMg irrespective of sample used or solution ionic strength. The surface charge of calcite is weakly dependent on pH, through protonation and deprotonation reactions that occur within a hydrolysis layer immediately adjacent to the mineral surface. The Point of Zero Charge (PZC) at which the surface charge is zero could be expressed as pH, but surface complexation models suggest the surface is negatively charged over the pH range 5.5–11.

Several studies have suggested that  $\text{SO}_4^{2-}$  is also a PDI for the calcite surface, but new data presented here indicate that the value of  $p\text{SO}_4$  may affect zeta potential only by moderating the equilibrium pCa. Natural carbonate typically yields a more negative zeta potential than synthetic calcite, most likely due to the presence of impurities including clays, organic matter, apatite, anhydrite or quartz, that yield a more negative zeta potential than pure calcite. New data presented here show that apparently identical natural carbonates display differing zeta potential behaviour, most likely due to the presence of small volumes of these impurities. It is important to ensure equilibrium, defined in terms of the concentration of PDIs, has been reached prior to taking measurements. Inconsistent values of zeta potential obtained in some studies may reflect a lack of equilibration.

The data collated and reported here have broad application in engineering processes including the manufacture of paper and cement, the geologic storage of nuclear waste and  $\text{CO}_2$ , and the production of oil and gas.

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## 1. Introduction

The properties of the calcite mineral surface and the interface between calcite and aqueous solution are of broad interest in many areas of science and engineering. Calcite is a common mineral, comprising approximately 4% of the Earth's crust, and surface reactions on calcite play an important role in many geochemical and environmental systems, as well as many areas of industry, including the manufacture of paper and cement (e.g. [5]), the geologic storage of nuclear waste and CO<sub>2</sub> (e.g. [30]), and the production of oil and gas (e.g. [9,28,86,88]). However, despite the broad range of interest and applications, controls on calcite surface charge in aqueous solution, especially at conditions relevant to natural systems, remain poorly understood. Numerous papers have reported inconsistent or contradictory data and models, and there is still active debate over the relationship between calcite surface charge and electrolyte pH, the concentration of ions such as Ca<sup>2+</sup>, Mg<sup>2+</sup>, CO<sub>3</sub><sup>2−</sup> and SO<sub>4</sub><sup>2−</sup> in aqueous solution, the partial pressure of CO<sub>2</sub>, the difference between natural and artificial calcite, and the role of dissolution and/or precipitation.

The focus of this paper is the zeta potential of artificial and natural calcite in aqueous solution. The zeta potential is an important measure of the electrical potential at the mineral surface, and the magnitude and sign of the zeta potential control the electrostatic interactions between the mineral surface and polar species in aqueous solution, between the mineral surface and other charged interfaces such as the water–air and water–oil interfaces, and between mineral particles in suspension including flocculation and dispersion. Measurements of zeta potential in low concentration solutions and at laboratory conditions are relatively straightforward, and most studies of the calcite mineral surface have reported measurements of zeta potential or the closely related property of electrophoretic mobility. Other approaches to determine surface charge, such as potentiometric titration, are challenging to apply in calcite because rapid dissolution kinetics and the buffering effect of carbonate ions in solution can affect the results. Thus the primary data source to understand calcite surface charge comprises

measurements of zeta potential. Where relevant, we use the results of spectroscopic, microscopic, surface diffraction, modelling and theoretical studies to help explain experimentally determined values of zeta potential. However, a comprehensive review of these studies is beyond the scope of the paper; a companion paper of comparable length would be required to do justice to this work.

The aims of the paper are therefore threefold; (1) to collate and review previous measurements of zeta potential on natural and artificial calcite and carbonate as a resource for future studies, (2) to compare and contrast the results of these studies to determine key controls on zeta potential and where uncertainties remain, and (3) to report new measurements of zeta potential relevant to natural subsurface systems. There has been no comprehensive review of zeta potential measurements in calcite and natural carbonate to date, although Wolthers et al. [85] collated published zeta potential data to constrain their new surface complexation model. Moreover, there are a lack of data which can be applied to natural systems owing to the comparatively high ionic strength (typically >0.01 M and often >2 M) and complex compositions (including Na<sup>+</sup>, Ca<sup>2+</sup>, Mg<sup>2+</sup>, Cl<sup>−</sup>, CO<sub>3</sub><sup>2−</sup>, SO<sub>4</sub><sup>2−</sup>) of natural brines, compared to the simple, dilute aqueous solutions typically used in laboratory experiments.

We report new data obtained using (intact) natural samples saturated with electrolytes relevant to natural systems and an experimental methodology specifically designed to allow this parameter space to be explored. Data obtained in this parameter space are still very scarce, despite the broad range of interest. Our results demonstrate that apparently identical natural carbonates can exhibit differing zeta potential behaviour, suggesting that small variations in the type and content of impurities in the mineral lattice can significantly impact the surface charge. However, the effect of the lattice ions Ca<sup>2+</sup> and Mg<sup>2+</sup> on the zeta potential of a given carbonate is identical within experimental error. The presence of apparently indifferent ions such as Na<sup>+</sup> and Cl<sup>−</sup> at the high concentrations typical of natural brines can significantly shift the isoelectric point expressed as pCa and pMg. We demonstrate experimentally that equilibrium pH is strongly correlated to

equilibrium pCa and pMg, and apparent trends in zeta potential with pH reflect trends in zeta potential with pCa and pMg.  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  are the potential determining ions (PDIs) for the calcite surface. We provide experimental evidence suggesting that  $\text{SO}_4^{2-}$ , which has previously been suggested as a PDI, may not directly control the zeta potential of calcite; rather, varying  $\text{pSO}_4$  causes variations in pCa which can be correlated with observed changes on zeta potential. These new results have broad application to subsurface engineering processes such as the geologic storage of nuclear waste and  $\text{CO}_2$ , and the production of oil and gas.

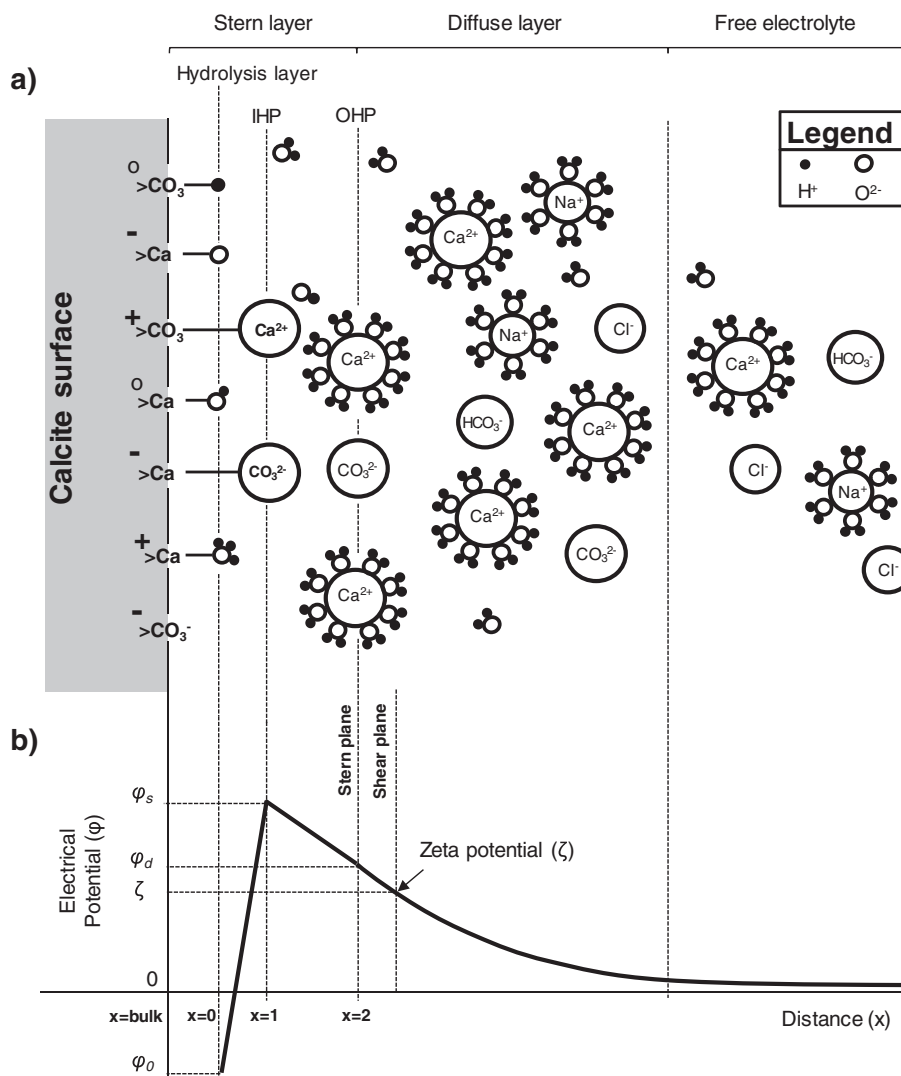
## 2. Zeta potential and the electrical double layer

### 2.1. The electrical double layer

The immersion of a calcite mineral in aqueous solution leads to a separation of electrical charge at the mineral–solution interface. An excess of charge at the mineral surface is balanced by a region of equal but opposite charge in the adjacent solution, in an arrangement often termed the electrical double layer (EDL; e.g. [25,26]; see Fig. 1). The charge at the mineral surface is balanced by a relative decrease in the concentration of co-ions (i.e. ions with the same charge as the surface)

and increase in the concentration of counter-ions (i.e. ions with the opposite charge as the surface) in the adjacent solution. The region immediately adjacent to the mineral surface is typically termed the Stern layer, and contains ions that are attached to the mineral surface; the Stern layer may be further divided into the inner and outer Helmholtz layers, defined by the inner and outer Helmholtz planes (the IHP and OHP shown in Fig. 1a; note the inner and outer Helmholtz layers and associated planes are also sometimes termed the inner and outer Stern layers and planes). The IHP defines the location of ions that closely approach the mineral surface and are attached to surface sites (Fig. 1a). The OHP defines the location of larger, typically hydrated ions that cannot enter the inner Helmholtz plane but which are nevertheless attached to the mineral surface [6].

In most cases, the charge in the Stern layer does not exactly balance the surface charge, giving rise to a ‘diffuse’ or ‘Gouy–Chapman’ layer that contains the remaining excess charge in the solution required to ensure electrical neutrality of the EDL (e.g. [43]). The difference between the Stern and diffuse layers is that the co- and counter-ions in the diffuse layer are not attached to the mineral surface. At low ionic strength ( $\leq 0.1\text{M}$ ), the ion concentration in the diffuse layer decreases exponentially with distance from the OHP (Fig. 1a). The electrical



**Fig. 1.** (a) A schematic representation of the electrical double layer at the calcite–water interface. The surface speciation sites ( $>\text{CO}_3\text{H}^0$ ,  $>\text{CaOH}^+$ ,  $>\text{CO}_3\text{Ca}^+$ ,  $>\text{CaOH}^0$ ,  $>\text{CaCO}_3^-$ ,  $\text{CaOH}_2^+$  and  $>\text{CO}_3^-$ ) are from [57]. The Stern layer is described by three planes. The 0-plane ( $x = 0$ ) corresponds to the hydrolysis layer where H and OH are chemi-bonded to the bulk ions [71]. The 1-plane ( $x = 1$ ) denotes inner-sphere complexes and corresponds to the inner Helmholtz plane (IHP), while the 2-plane ( $x = 2$ ) denotes outer-sphere complexes and corresponds to the outer Helmholtz plane (OHP; [85]). (b) A schematic representation of the variation in the electrical potential with distance from the mineral surface. Here the mineral surface is negatively charged, consistent with the surface complexation model of [29], but the zeta potential is positive because of adsorption of the lattice PDIs  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$ .

potential corresponding to the charge distribution within the EDL decreases linearly with distance from the mineral surface through the Stern layer, although there may be a difference in gradient between the inner and outer Helmholtz layers, and exponentially with distance from the OHP through the diffuse layer, falling to zero in the uncharged solution (Fig. 1b).

## 2.2. Electrokinetic phenomena

Electrokinetic phenomena arise when there is a relative motion between the excess charge in the diffuse layer of the EDL and the charged surface [34]. It is typically assumed that the excess charge in the diffuse layer is mobile only beyond a plane termed the 'shear' or 'slip' plane that lies a small distance away from the OHP (Fig. 1). The zeta potential is the electrical potential at the shear plane. In some models of the EDL, it is assumed that the shear plane and the OHP are identical, in which case the zeta potential also represents the electrical potential at the OHP. Methods to determine the zeta potential make use of electrokinetic phenomena, most typically (1) electrophoresis, which is the mobilisation of charged particles, relative to a stationary solution, under the influence of an applied electrical field, and (2) streaming potential, which is the potential difference that arises when a solution is moved relative to a stationary solid under the influence of an applied pressure gradient [15].

The electrophoretic method (termed here EPM) used to determine zeta potential is conducted on powdered samples suspended in the solution of interest. An electrical field  $E$  is applied across the suspension and the resulting velocity  $v_e$  of the charged particles is measured with respect to the solution. The velocity is then normalized by the electrical field to yield the electrophoretic mobility  $u_e$ , which is typically treated as an isotropic quantity that can be related to the zeta potential by the Helmholtz–Smoluchowski equation for electrophoresis [15]

$$u_e = \frac{\varepsilon \zeta}{\mu}, \quad (1)$$

where  $\varepsilon$  is the permittivity ( $\text{F m}^{-1}$ ) and  $\mu$  the viscosity ( $\text{Pa s}$ ) of the solution. The zeta potential obtained using the EPM is an effective value across all the suspended particles; the zeta potential of individual particles (and, indeed, between the different faces of a given particle) may differ from the single value interpreted from EPM measurements.

The streaming potential method (termed here SPM) used to determine zeta potential is conducted using intact samples of porous materials, packed beds of particles, or surfaces through, or across, which the solution of interest is caused to flow. A pressure difference  $\Delta P$  is applied across the sample and the resulting electrical potential difference  $\Delta V$  is measured. The potential difference is then normalized by the pressure difference to yield the streaming potential coupling coefficient, which is typically treated as an isotropic quantity that can be related to the zeta potential by the Helmholtz–Smoluchowski equation for streaming potential (e.g. [17,34,52])

$$C = \frac{\varepsilon \zeta}{\sigma_s \mu}, \quad (2)$$

where  $\sigma_s$  is the electrical conductivity of the solution of interest ( $\text{S m}^{-1}$ ). When streaming potential measurements are obtained in porous media, it may be necessary to use a version of Eq. (2) modified to account for the effect of enhanced electrical conductivity through the EDL (often termed surface electrical conductivity; [49])

$$C = \frac{\varepsilon \zeta}{\sigma_{ss} \mu F}, \quad (3)$$

where  $\sigma_{ss}$  is the electrical conductivity of the system saturated with the solution of interest and  $F$  is the so-called formation factor (dimensionless) which is defined as the ratio of the solution conductivity  $\sigma_s$  to

the system conductivity  $\sigma_{ss}$  when surface electrical conductivity is negligible. The formation factor is typically measured on the system saturated with a concentrated solution with high electrical conductivity (e.g. [35,40,83]).

## 3. Development of surface charge on calcite in aqueous solution

### 3.1. Effect of pH

The development of surface charge on calcite in aqueous solution is still not fully understood [71,79,85]. In most metal oxides, charge development occurs when mineral surfaces sites are hydrated to form amphoteric groups  $>\text{MeOH}$  (where Me denotes a metal ion and  $>$  denotes the crystal lattice) and these react either with  $\text{H}^+$  or  $\text{OH}^-$  ions according to (e.g. [61,77])



The relative abundance of negatively and positively charged surface sites depends on the relative concentration of  $\text{H}^+$  and  $\text{OH}^-$  in solution; these ions behave as 'potential determining ions' (PDIs) for the surface. Consequently, the net surface charge depends on the solution pH. It is therefore common to see surface charge and/or zeta potential plotted as a function of pH, with the surface charge and/or zeta potential becoming more negative as the pH increases and the deprotonation reaction (4) is favoured, and more positive as the pH decreases and the protonation reaction (5) is favoured (e.g. [13,32]). When the number of positively and negatively charged surface sites exactly balances, a surface charge of zero is observed and the corresponding pH is termed the 'point of zero charge' (PZC). A zeta potential of zero may also be observed and the corresponding pH is termed the 'isoelectric-point' (IEP) (e.g. [34]). The IEP and PZC may not coincide, for reasons discussed in more detail in the next section.

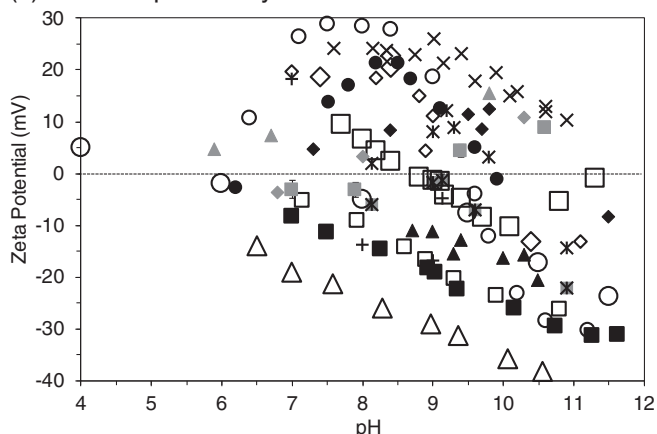
The surface of calcite differs from the metal oxides in several important respects. First, the mineral is soluble in aqueous solution and the lattice ions  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  can be released into solution or deposited on the surface depending upon the solution pH [66,75]. Second, atmospheric  $\text{CO}_2$  in open system experiments can dissolve into solution, affecting the pH and the equilibrium concentrations of  $\text{Ca}^{2+}$ ,  $\text{CO}_3^{2-}$  and  $\text{HCO}_3^-$  (e.g. [18]). Third, the hydrated calcite surface contains protonated anion surface sites  $>\text{CO}_3\text{H}^0$  as well as hydroxylated cation sites  $>\text{CaOH}^0$  (e.g. [57,79]). Evidence to support the presence of these sites has been provided by X-ray photoelectron spectroscopic (XPS) and infrared (IR) spectroscopic measurements [56,72,73]. The protonation and deprotonation reactions at these surface sites, and therefore the relative abundance of positively and negatively charged sites on the mineral surface, is pH dependent according to the following reactions (e.g. [29, 57,79])



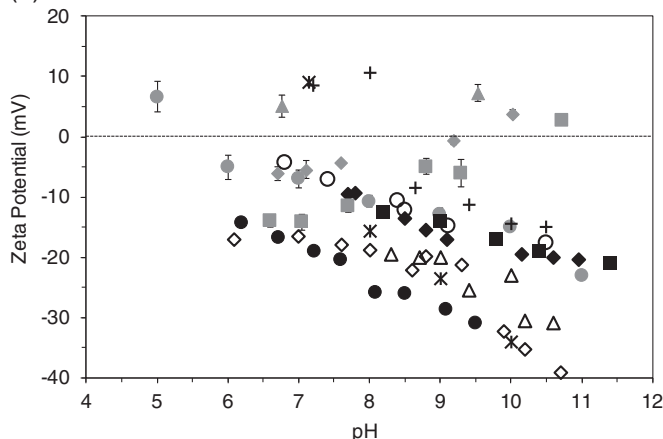
Thus it would be expected that the surface charge, and hence the zeta potential, of calcite in aqueous solution are both determined by the pH, with  $\text{H}^+$  and  $\text{OH}^-$  behaving as the PDIs. Numerous studies have determined the variation in zeta potential with pH for a variety of calcite types and solution compositions (e.g. Fig. 2).

However, this analysis ignores the contribution of the lattice ions  $\text{Ca}^{2+}$ ,  $\text{CO}_3^{2-}$  in solution. Several experimental studies have shown that

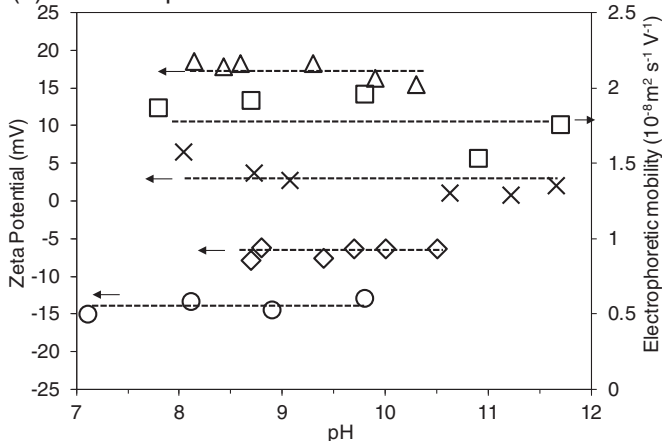
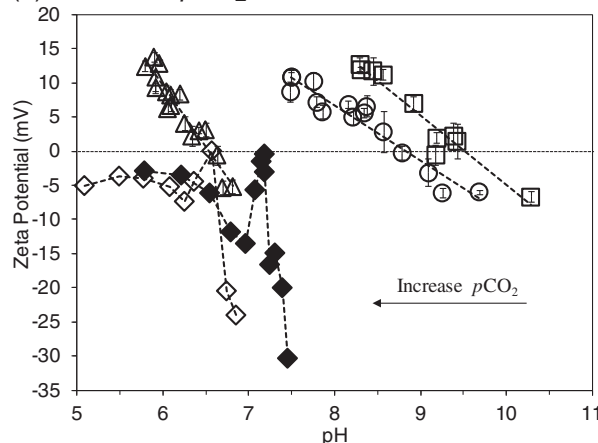
(a) Iceland Spar and synthetic calcites



(b) Natural calcites and carbonates



(a) Constant pCa

(b) Controlled pCO<sub>2</sub>

**Fig. 2.** Zeta potential of (a) Iceland spar and synthetic calcite and (b) natural calcite and carbonate rocks in indifferent electrolytes as a function of pH. Measurements were obtained using the electrophoretic mobility method (EPM), with the exception of Somasundaran and Agar [67] (diamonds) and Thompson and Pownall [78] (squares). Error bars are shown when reported by the source. In (a), data from Somasundaran and Agar [67] in deionized water after different mixing times (♦: no mixing; ◇: after 1 week; ○: after two months of mixing); Sampat Kumar et al. [60] in deionized water (+); Siffert and Fimbel [63] in  $10^{-2}$  M NaCl electrolyte (\*: mass dispersed = 50 mg; grey \*: 30 mg); Thompson and Pownall [78] in various electrolytes (■: NaCl ( $5 \times 10^{-3}$  M); □: NaCl ( $5 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $1 \times 10^{-3}$  M); □: NaCl ( $5 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $1 \times 10^{-3}$  M)/H<sub>2</sub>CO<sub>3</sub>/Ca(OH)<sub>2</sub>; Cicerone et al. [12] in  $10^{-3}$  M KCl electrolyte (▲); Vdović and Biščan [81] in  $10^{-3}$  M NaCl electrolyte (○); Vdović [80] in  $10^{-3}$  M NaCl electrolyte (●); Eriksson et al. [18] in  $5.56 \times 10^{-3}$  M NaCl electrolyte (×); Sondi et al. [70] in  $10^{-3}$  M NaCl electrolyte (Δ) and Kasha et al. [42] in deionized water (○). Grey symbols show data from Mahani et al. [47] in various electrolytes (▲: formation brine; ♦: seawater; ■: 25 times diluted seawater). In (b), data from Mishra [48] in  $2 \times 10^{-2}$  M NaClO<sub>4</sub> electrolyte (+); Rao et al. [58] in  $2 \times 10^{-3}$  M NaClO<sub>4</sub> electrolyte (♦); Cicerone et al. [12] in  $10^{-3}$  M KCl electrolyte (Δ); Vdović and Biščan [81] in  $10^{-3}$  M NaCl electrolyte (◇); [53] in  $2 \times 10^{-3}$  M NaNO<sub>3</sub> electrolyte (\*); Vdović [80] in  $10^{-3}$  M NaCl electrolyte (○); limestone (●); lake sediment (●) and Somasundaran et al. [69] in  $10^{-3}$  M KNO<sub>3</sub> electrolyte (■). Grey symbols show data on natural carbonate rocks in various electrolytes (●: deionized water containing 0.09 M CaCl<sub>2</sub> from [11]; ▲: formation brine; ♦: seawater; ■: 25 times diluted seawater, all from [47]).

**Fig. 3.** Zeta potential and electrophoretic mobility of synthetic calcite and Iceland spar as a function of pH in experiments with (a) constant pCa and (b) controlled pCO<sub>2</sub>. Error bars are shown when reported by the source. In (a), all data are from synthetic calcite. Data from Foxall et al. [21] in  $10^{-2}$  M NaCl electrolyte with pCa = 2.1 (□); Thompson and Pownall [78] in  $5 \times 10^{-4}$  M CaCl<sub>2</sub> electrolyte with pCa = 3.3 (×); Cicerone et al. [12] in  $10^{-3}$  M KCl electrolyte with pCa = 2 (Δ) and pCa = 3 (◇); and Sondi et al. [70] in  $10^{-3}$  M NaCl electrolyte with pCa = 2 (○). In (b), all data are from powdered Iceland spar in deionized water with various pCO<sub>2</sub>. Data from Moulin and Roques [51] (♦: pCO<sub>2</sub> = 0.023; ◇: pCO<sub>2</sub> = 0.234) and Heberling et al. [29] (Δ: pCO<sub>2</sub> = 1; ○ pCO<sub>2</sub> =  $10^{-3.44}$ ; □: pCO<sub>2</sub> =  $10^{-5.2}$ ).

oxides (e.g. [71]). Instead, the lattice ions Ca<sup>2+</sup> and CO<sub>3</sub><sup>2-</sup> adsorb onto the mineral surfaces through surface complexation reactions such as (e.g. [29,71,79,85])



and, in the presence of CO<sub>2</sub>, reactions such as

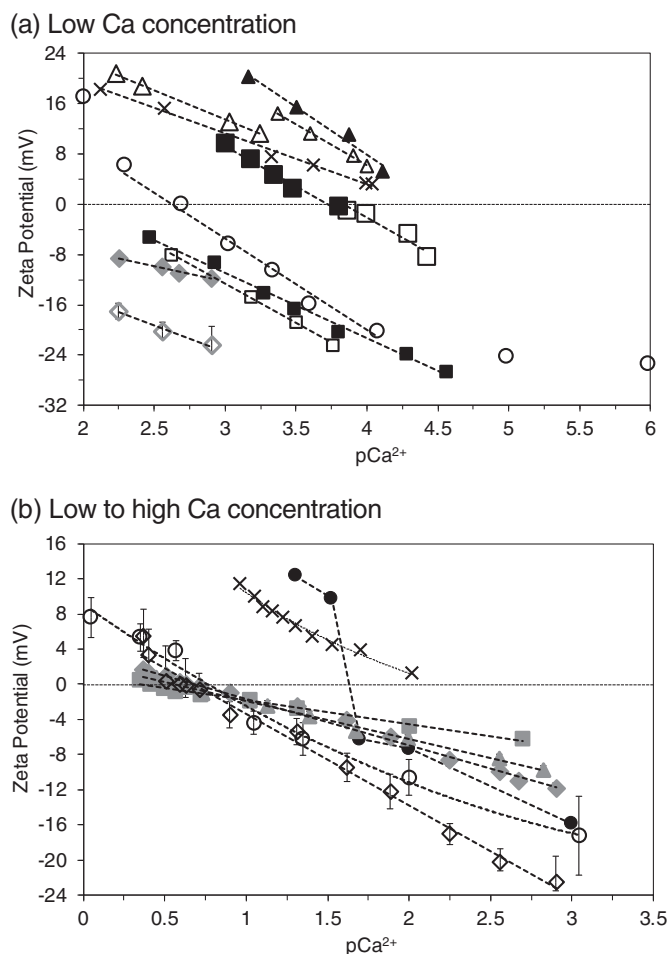


As discussed below, the concentration dependent adsorption of these lattice ions (and other potential determining ions) is the primary control on the zeta potential of calcite.

### 3.2. Potential determining ions

Surface diffraction studies have revealed well-ordered water layers a few Å (=  $1 \times 10^{-10}$  m) above the calcium and carbonate ions on the mineral surface [19,20,24,29,45]. This is the hydrolysis

the zeta potential of calcite is independent of pH if the concentration of calcium (expressed in this paper as pCa, where p represents the negative logarithm) is kept constant (e.g. [12,21,70]; see Fig. 3a). Moreover, other studies have demonstrated a relationship between zeta potential and pCa (e.g. [4,12,54,78]; see Fig. 4), and a strong dependence between the zeta potential of calcite and the excess of Ca<sup>2+</sup> ions at the mineral surface has been observed [33]. These data show that the development of charge on the calcite surface is more complex than the simple protonation and deprotonation reactions that occur at the surface sites of metal



**Fig. 4.** Zeta potential of synthetic calcite and natural carbonate in indifferent electrolytes as a function of pCa, at (a) low Ca concentration only and (b) low to high Ca concentration. Error bars are shown when reported by the source. In (a), all data are from powdered synthetic calcite with the exception of Alroudhani et al. [4] (grey symbols). Data from Thompson and Pownall [78] in different aqueous compositions (■: NaCl ( $2 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $10^{-2}$  M); □: NaCl ( $2 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $10^{-2}$  M)/Ca(OH)<sub>2</sub> ( $10^{-2}$  M); ■: NaCl ( $2 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $10^{-2}$  M)/H<sub>2</sub>CO<sub>3</sub> ( $10^{-2}$  M)); Foxall et al. [21] in three different NaCl concentrations (▲:  $1 \times 10^{-2}$  M; △:  $5 \times 10^{-2}$  M; big △:  $1.5 \times 10^{-1}$  M); Huang et al. [33] in 0.01 CaCl<sub>2</sub> electrolyte (×) and Cicerone et al. [12] in  $10^{-3}$  M KCl electrolyte (○). Grey diamonds show data from Alroudhani et al. [4] on natural carbonate in 0.05 M NaCl electrolyte at high pCa (♦: SPM data; ◇: EPM data); the full span of these data is shown in Fig. 2b. In (b), all data are from natural calcite and carbonate rocks, with the exception of Kasha et al. [42] (filled symbols). Data from Zhang and Austad [88] on powdered Stevns Klint chalk in 0.571 M NaCl electrolyte (×); Chen et al. [11] in deionized water (○); Kasha et al. [42] in synthesized brines equivalent to seawater composition (●) and Alroudhani et al. [4] in 0.05 M NaCl (◇). Grey symbols show data from Alroudhani et al. [4] in intact natural carbonate rocks in various NaCl concentrations (♦: 0.05 M; ▲: 0.5 M; ■: 2 M; all from; error bars are the same size or smaller than the symbols).

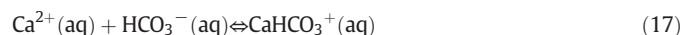
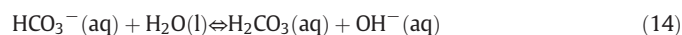
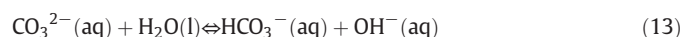
plane of Stipp [71] and the 0-plane invoked in various surface complexation models (SCMs) of the calcite–water interface ([29, 85]; see also Fig. 1a). Protonation and deprotonation reactions occur in this plane, but it does not contain the lattice ions Ca<sup>2+</sup> and CO<sub>3</sub><sup>2-</sup> or other adsorbed ion complexes. These are confined to the 1- and 2-planes (or the a- and b-planes), which here we associate with the inner and outer Helmholtz planes respectively (Fig. 1a). Most SCMs include ordered water layers at the 0-plane, and the pH-dependent protonation and deprotonation reactions (6)–(8) that occur at the sites defining the 0-plane [29,71,85]. However, the SCM of Heberling et al. [29] predicted that the charge at the 0-plane is only weakly pH dependent. The dominant surface species are >CaOH<sup>0</sup> and >CaCO<sub>3</sub><sup>-</sup>, causing the 0-plane to be negatively

charged across the pH range 5.5–11. Thus, unlike the >MeOH groups on oxide mineral surfaces, the calcite surface species >CaOH<sup>-</sup> and >CaCO<sub>3</sub><sup>-</sup> do not determine the pH dependence of the zeta potential. Rather, the potential at the OHP and, hence, the zeta potential, is instead controlled by adsorption of the potential-determining lattice ions Ca<sup>2+</sup> or CO<sub>3</sub><sup>2-</sup> at the 1- or 2-planes. The available evidence suggests that most of the adsorbed lattice PDIs are located at the 2-plane [29,85].

Dissolution of calcite in aqueous solution is described by the following reaction:



The equilibrium conditions are defined by the pH, the concentration of the ionic species and, in open system conditions, the partial pressure of CO<sub>2</sub> (pCO<sub>2</sub>) in solution [66,68]. Moreover, the concentration of lattice ion species available to adsorb onto the calcite surface is controlled by the reactions [18]:



It can be seen from (12)–(17) that pH (and pCO<sub>2</sub>) determines the Ca<sup>2+</sup> and CO<sub>3</sub><sup>2-</sup> concentrations in the solution at equilibrium. Apparent trends between zeta potential and pH therefore reflect the fact that pH and pCa are directly related at fixed pCO<sub>2</sub> (e.g. [64]). We confirm this experimentally in a later section. Thus, H<sup>+</sup> is not the key PDI for the calcite surface; rather, it is the lattice ion Ca<sup>2+</sup> [4,12,21,33,51,54]. Moreover, the PZC and IEP are not closely related. In principle, the PZC could be defined in terms of pH, as pH controls the protonation and deprotonation reactions at the 0-plane. However, in practice, the PZC is not observed over the pH range 5.5–11 [29]. The IEP must be defined in terms of pCa, or in terms of the concentration of other PDIs for the calcite surface such as Mg<sup>2+</sup> as discussed below.

#### 4. Experimental measurements of zeta potential

Published measurements of zeta potential in calcite and natural carbonates in aqueous solution range from +29 mV to −39 mV (Fig. 2). As discussed below, the zeta potential depends on the electrolyte concentration and composition, the nature of the calcite, sample preparation, experimental conditions and measurement method. However, limited reporting of the nature of the samples and experimental method, especially sample history, preparation and cleaning, and experimental conditions (especially pH, pCa and pCO<sub>2</sub>), makes it challenging to systematically compare experimental data (see Table 1 for a summary). In this section, we review and discuss the data published to date, and summarize the key controls on zeta potential.

##### 4.1. Electrolyte concentration and composition

###### 4.1.1. Effect of pH

Many studies have reported more negative values of zeta potential with increasing pH (e.g. [12,18,54,78,80]). For example, Thompson and Pownall [78] reported a close-to-linear relationship between zeta potential and pH (square data in Fig. 2a) using packed synthetic calcite and NaCl solution of ionic strength  $5 \times 10^{-3}$  M

**Table 1**

Summary of published zeta potential of calcite and carbonate. Table 1a reports studies focusing on the effect of varying pH. Table 1b reports studies focusing on the effect of varying pCa.

| 1a                           |        |                                                                                                                                |                               |                         |                                    |                                        |                       |                   |                   |           |
|------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------|------------------------------------|----------------------------------------|-----------------------|-------------------|-------------------|-----------|
| Material                     | Method | Background electrolyte                                                                                                         | Equilibrium condition         | Equilibration time (hr) | Equilibrium pH (pH <sub>EQ</sub> ) | Open vs. closed system                 | Zeta potential at EQ. | pH <sub>IEP</sub> | IEP determination | Reference |
| Iceland spar                 | SPM    | Deionized water (HNO <sub>3</sub> /NaOH)                                                                                       | Calcite/water/air             | 1440                    | 8.2                                | Open                                   | Positive              | 8.2               | Extrapolated      | [67]      |
| Iceland spar                 | SPM    | SiO <sub>2</sub> /Na <sub>2</sub> O                                                                                            | Calcite/water                 | N/A                     | 9.8                                | Closed                                 | Positive              | 10.8              | Direct            | [22]      |
| Iceland spar                 | SPM    | Deionized water                                                                                                                | Calcite/water/CO <sub>2</sub> | N/A                     | 6.55                               | pCO <sub>2</sub> = 0.23                | Negative              | 6.55              | Direct            | [51]      |
|                              |        |                                                                                                                                |                               |                         | 7.05                               | pCO <sub>2</sub> = 0.023               | Negative              | 7.05              |                   |           |
| Iceland spar                 | SPM    | Deionized water                                                                                                                | Calcite/water/CO <sub>2</sub> | 24–720                  | 5.8                                | pCO <sub>2</sub> = 1                   | Negative              | 6.5               | Direct            | [29]      |
|                              |        |                                                                                                                                |                               |                         | 7.5                                | pCO <sub>2</sub> = 10 <sup>-3.44</sup> | Negative              | 8.8               |                   |           |
|                              |        |                                                                                                                                |                               |                         | 8.3                                | pCO <sub>2</sub> = 10 <sup>-5.2</sup>  | Negative              | 9.4               |                   |           |
| Iceland spar                 | EPM    | Deionized water                                                                                                                | Calcite/water                 | 24                      | N/A                                | Closed                                 | Negative              | 5.4               | Extrapolated      | [42]      |
| Iceland spar                 | EPM    | Deionized water                                                                                                                | Calcite/water/CO <sub>2</sub> | 1–2                     | N/A                                | Open                                   | Negative              | 7.2–7.8           | Extrapolated      | [60]      |
| Iceland spar                 | EPM    | Formation water                                                                                                                | Calcite/water                 | 24                      | N/A                                | Closed                                 | Positive              | N/A               | Direct            | [47]      |
|                              |        | Seawater (SW)                                                                                                                  |                               |                         | N/A                                |                                        | Negative              | 7.5               |                   |           |
|                              |        | 25 times diluted SW                                                                                                            |                               |                         | 9.2                                |                                        | Negative              | 8.5               |                   |           |
| Synthetic calcite            | EPM    | 1 × 10 <sup>-2</sup> ,<br>5 × 10 <sup>-2</sup> ,<br>1.5 × 10 <sup>-1</sup> M NaCl                                              | Calcite/water/air             | N/A                     | N/A                                | Open                                   | Positive              | N/A               | N/A               | [21]      |
| Synthetic calcite            | EPM    | 10 <sup>-2</sup> M NaCl                                                                                                        | Calcite/water                 | 10                      | 9.1                                | Closed                                 | Pos – Neg             | 10–11             | N/A               | [63]      |
| Synthetic calcite            | SPM    | 5 × 10 <sup>-3</sup> M NaCl                                                                                                    | Calcite/water                 | 48                      | 9.11–9.87                          | Closed                                 | Negative              | N/A               | N/A               | [78]      |
| Synthetic calcite            | SPM    | 5 × 10 <sup>-3</sup> M NaCl/<br>10 <sup>-3</sup> M NaHCO <sub>3</sub>                                                          | Calcite/water                 | 48                      | 9.11–9.87                          | Closed                                 | Negative              | N/A               | N/A               | [78]      |
| Synthetic calcite            | SPM    | 5 × 10 <sup>-3</sup> M NaCl/<br>10 <sup>-3</sup> M NaHCO <sub>3</sub> /<br>H <sub>2</sub> CO <sub>3</sub> /Ca(OH) <sub>2</sub> | Calcite/water                 | 48                      | 9.11–9.87                          | Closed                                 | Negative              | 8.9               | Extrapolated      | [78]      |
| Synthetic calcite            | SPM    | 5 × 10 <sup>-4</sup> M CaCl <sub>2</sub>                                                                                       | Calcite/water                 | 48                      | 9.11–9.87                          | Closed                                 | Positive              | N/A               | N/A               | [78]      |
| Synthetic calcite            | EPM    | 10 <sup>-2</sup> M NaCl                                                                                                        | Calcite/water                 | 24                      | 10                                 | Closed                                 | Positive              | 9.6               | Direct            | [54]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M KCl                                                                                                         | Calcite/water                 | 168                     | 9.8–10                             | Closed                                 | Negative              | N/A               | N/A               | [12]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M KCl/10 <sup>-3</sup> M<br>CaCl <sub>2</sub>                                                                 | Calcite/water                 | 168                     | 9.8–10                             | Closed                                 | Negative              | N/A               | N/A               | [12]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M KCl/10 <sup>-2</sup> M<br>CaCl <sub>2</sub>                                                                 | Calcite/water                 | 168                     | 9.8–10                             | Closed                                 | Positive              | N/A               | N/A               | [12]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.6                                | Closed                                 | Positive              | 9.8               | Direct            | [81]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.4                                | Closed                                 | Positive              | 9.8               | Direct            | [80]      |
| Synthetic calcite            | EPM    | 5.56 × 10 <sup>-3</sup> M NaCl                                                                                                 | Calcite/water/CO <sub>2</sub> | 1                       | 8–8.5                              | Open                                   | Positive              | N/A               | N/A               | [18]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.3                                | Closed                                 | Negative              | N/A               | N/A               | [70]      |
| Natural calcite              | EPM    | 2 × 10 <sup>-3</sup> M NaClO <sub>4</sub>                                                                                      | Calcite/water                 | 0.5                     | 9–10                               | Closed                                 | Negative              | N/A               | N/A               | [58]      |
| Natural calcite              | EPM    | 2 × 10 <sup>-2</sup> M NaClO <sub>4</sub>                                                                                      | Calcite/water/air             | 0.5                     | 8.2                                | Open                                   | Negative              | 8.2               | Extrapolated      | [48]      |
| Natural calcite              | EPM    | 10 <sup>-2</sup> M NaCl                                                                                                        | Calcite/water                 | 24                      | 10                                 | Closed                                 | Negative              | 9.4               | Direct            | [54]      |
| Natural (biogenetic) calcite | EPM    | 10 <sup>-3</sup> M KCl                                                                                                         | Calcite/water                 | 168                     | 9.8–10                             | Closed                                 | Negative              | N/A               | N/A               | [12]      |
| Natural calcite              | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.6                                | Closed                                 | Negative              | N/A               | N/A               | [81]      |
| Natural calcite              | EPM    | 2 × 10 <sup>-3</sup> M NaNO <sub>3</sub>                                                                                       | Calcite/water/air             | 2.2                     | 8.3                                | Open                                   | Negative              | 7.8               | Extrapolated      | [53]      |
| Natural calcite              | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.4                                | Closed                                 | Negative              | N/A               | N/A               | [80]      |
| Natural calcite              | EPM    | 10 <sup>-3</sup> M KNO <sub>3</sub>                                                                                            | Calcite/water/air             | 1                       | N/A                                | Open                                   | Negative              | N/A               | N/A               | [69]      |
| Natural carbonate            | EPM    | Deionized water                                                                                                                | Calcite/water                 | 1                       | N/A                                | Closed                                 | Negative              | 5.2–5.8           | Extrapolated      | [11]      |
| Natural carbonate            | EPM    | Formation water                                                                                                                | Calcite/water                 | 24                      | N/A                                | Closed                                 | Positive              | N/A               | Direct            | [47]      |
|                              |        | Seawater (SW)                                                                                                                  |                               |                         | N/A                                |                                        | Negative              | 9.25              |                   |           |
|                              |        | 25 times diluted SW                                                                                                            |                               |                         | 9.2                                |                                        | Negative              | 10.25             |                   |           |

| 1b                |                         |                                                                                                                          |                       |                         |                                    |                        |                        |                    |                   |           |
|-------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------|------------------------------------|------------------------|------------------------|--------------------|-------------------|-----------|
| Material type     | Method                  | Background electrolyte                                                                                                   | Equilibrium condition | Equilibration time (hr) | Equilibrium pH (pH <sub>EQ</sub> ) | Open vs. closed system | dζ/dpCa                | pCa <sub>IEP</sub> | IEP determination | Reference |
| Iceland spar      | Electro osmosis         | NaCl/Na <sub>2</sub> CO <sub>3</sub> /CaCl <sub>2</sub>                                                                  | Calcite/water         | 2                       | 9                                  | Closed                 | N/A                    | 3.33               | Extrapolated      | [16]      |
| Iceland spar      | SPM on packed particles | Deionized water (HNO <sub>3</sub> /NaOH)                                                                                 | Calcite/water/air     | 1440                    | 8.2                                | Open                   | N/A                    | 4.25               | Extrapolated      | [67]      |
| Iceland spar      | EPM                     | 0.571 M NaCl                                                                                                             | Calcite/water         | 24                      | N/A                                | Closed                 | N/A                    | 1.65–1.75          | Extrapolated      | [42]      |
| Synthetic calcite | EPM                     | 1 × 10 <sup>-2</sup> ,<br>5 × 10 <sup>-2</sup> ,<br>1.5 × 10 <sup>-1</sup> M NaCl                                        | Calcite/water/air     | N/A                     | N/A                                | Open                   | –15.2<br>–12.8<br>–9.3 | 4.5                | Extrapolated      | [21]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> M NaCl                                                                                              | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –12.7                  | 2.02               | Extrapolated      | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> M<br>(NaCl/NaHCO <sub>3</sub> )                                                   | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –10.2                  | 1.92               | Extrapolated      | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> M<br>(NaCl/CaCl <sub>2</sub> )                                                    | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –10.1                  | 3.4                | Extrapolated      | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> M<br>(NaCl/H <sub>2</sub> CO <sub>3</sub> )                                       | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –10.8                  | 4                  | Extrapolated      | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> /10 <sup>-2</sup> M<br>(NaCl/NaHCO <sub>3</sub> /H <sub>2</sub> CO <sub>3</sub> ) | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –12.9                  | 3.8                | Direct            | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> /10 <sup>-2</sup> M                                                               | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –12.2                  | 3.8                | Direct            | [78]      |

Table 1 (continued)

| Material type       | Method                | Background electrolyte                                                     | Equilibrium condition | Equilibration time (hr) | Equilibrium pH (pH <sub>EQ</sub> ) | Open vs. closed system | dζ/dpCa                           | pCa <sub>IEP</sub>   | IEP determination | Reference |
|---------------------|-----------------------|----------------------------------------------------------------------------|-----------------------|-------------------------|------------------------------------|------------------------|-----------------------------------|----------------------|-------------------|-----------|
| 1b                  |                       |                                                                            |                       |                         |                                    |                        |                                   |                      |                   |           |
| Synthetic calcite   | EPM                   | (NaCl/NaHCO <sub>3</sub> /Ca(OH) <sub>2</sub> )<br>10 <sup>−2</sup> M NaCl | Calcite/water         | 24                      | 10                                 | Closed                 | N/A                               | 3.3                  | Direct            | [54]      |
| Synthetic calcite   | SPM                   | 10 <sup>−5</sup> –0.01 M CaCl <sub>2</sub>                                 | Calcite/water         | 120                     | 9.3–9.9                            | Closed                 | −8.34                             | 4.38                 | Extrapolated      | [33]      |
| Synthetic calcite   | EPM                   | 10 <sup>−3</sup> M KCl                                                     | Calcite/water         | 168                     | 9.8–10                             | Closed                 | −14.5<br>(in the vicinity of IEP) | 2.7                  | Direct            | [12]      |
| Natural calcite     | EPM                   | 10 <sup>−2</sup> M NaCl                                                    | Calcite/water         | 24                      | 10                                 | Closed                 | N/A                               | 4                    | Direct            | [54]      |
| Stevens Klint chalk | EPM                   | CaCl <sub>2</sub> and NaSO <sub>4</sub> in<br>0.571 M NaCl                 | Chalk/water/air       | 48                      | Controlled at 8.4                  | Closed                 | N/A                               | N/A                  | N/A               | [88]      |
| Natural carbonate   | EPM                   | Deionized water (DI)                                                       | Calcite/water         | 1                       | Controlled at 8.4                  | Closed                 | N/A                               | 0.2–0.75             | Extrapolated      | [11]      |
| Natural carbonate   | SPM in intact samples | 0.05 M NaCl<br>0.5 M NaCl<br>2 M NaCl                                      | Calcite/water/air     | 168–720                 | 8.2                                | Closed                 | −5.29<br>−4.35<br>−2.75           | 0.61<br>0.49<br>0.41 | Direct            | [4]       |

(filled squares) and NaCl/NaHCO<sub>3</sub> of ionic strength  $5 \times 10^{-4}$  M (open squares). Similarly, Vdovic [80] obtained a linear relationship between zeta potential and pH using natural carbonate powder samples suspended in  $10^{-3}$  M NaCl electrolyte (filled and open circles in Fig. 2b). However, there are numerous exceptions (e.g. [11,42,46–48,63,67,81]). For example, Siffert and Fimbel [63] found that the zeta potential of synthetic calcite at equilibrium pH (9.1; Table 1a) could be either positive or negative, dependent on the dispersed mass. Away from the equilibrium pH, the zeta potential decreased with either increasing or decreasing pH (open and grey stars in Fig. 2a). Vdović and Bišćan [81] (open circles in Fig. 2a) and Vdovic [80] (solid circles in Fig. 2a) observed similar behaviour. The latter reported zeta potential values that changed polarity with both increasing and decreasing pH, thus apparently defining two IEPs. Mahani et al. [46] [47] reported an increasingly negative zeta potential with increasing pH (grey triangles, diamonds and squares in Fig. 2a–b), contrary to the trends reported above. However, as discussed in the previous section, SCMs for calcite suggest the electrical potential at the mineral surface is only weakly pH dependent, and experimental data have shown that the zeta potential is independent of pH if pCa is held constant (Fig. 3). These data confirm that zeta potential is not directly controlled by pH, but rather by the concentration of lattice ions such as Ca<sup>2+</sup> and other PDI's discussed below.

#### 4.1.2. Effect of pCa and pCO<sub>2</sub>

Many authors have concluded that the principal PDI's for calcite are the lattice ions Ca<sup>2+</sup> and CO<sub>3</sub><sup>2−</sup> (e.g. [7,12,21,33,48,54,62,63,67,78,80,81]). Consequently, the effect of Ca<sup>2+</sup> on the zeta potential has been the topic of several experimental studies (e.g. [4,12,33,54,78]; see Fig. 4). The consistent observation in these studies is that decreasing pCa yields more positive zeta potential, consistent with increased adsorption of Ca<sup>2+</sup> onto the calcite surface (Fig. 1). However, significant differences in the magnitude and polarity of the zeta potential, in the gradient of zeta potential vs. pCa (dζ/dpCa), and in the IEP (expressed as pCa), have been reported (Fig. 4; Table 1b). The zeta potential varies between +20 mV and −26 mV, depending on calcite type and experimental conditions. Most studies report a linear or close-to-linear relationship between zeta potential and pCa, especially for pCa values close to the IEP, with the gradient dζ/dpCa in the range −8 to −15 mV/decade (Fig. 4). A linear relationship is consistent with Nernstian behaviour of the calcite surface and suggests the electrical double layer can be

reasonably described by the simple Gouy–Chapman–Grahame model close to the IEP (e.g. [34]). This model is valid only when ions other than the lattice ions show no specific adsorption at the surface. The relationship between zeta potential and pCa can then be expressed as [21]:

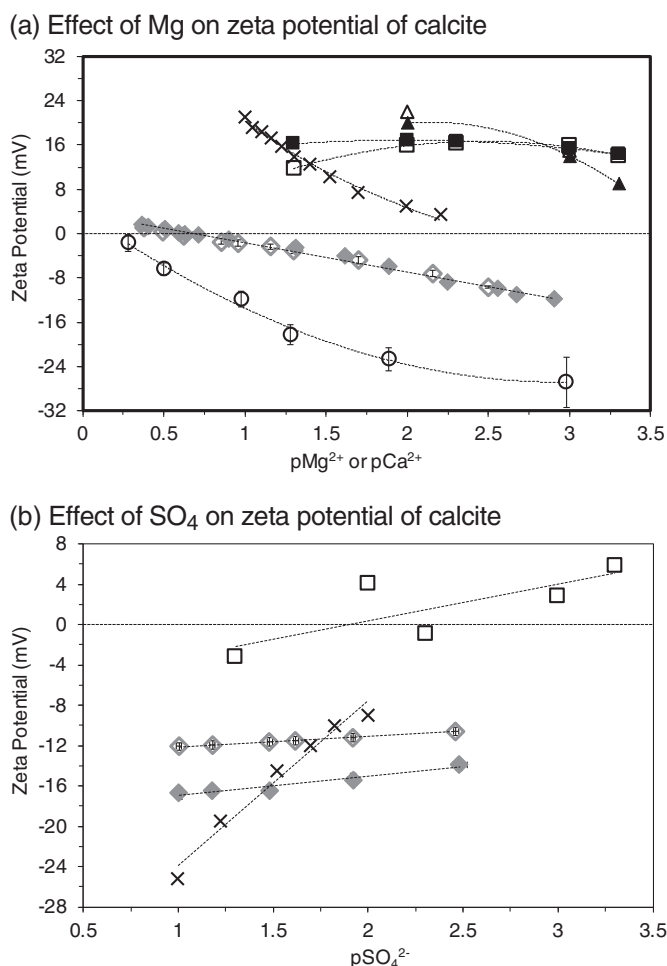
$$\left. \frac{d\zeta}{dpCa} \right|_{\zeta \rightarrow 0} = \frac{-2.303 \frac{kT}{ze}}{\left(1 + \frac{C_d}{C_s}\right) \exp(\kappa\Delta)}, \quad (18)$$

where  $k$  is Boltzmann's constant,  $T$  is the temperature,  $z$  is the valence of the PDI,  $e$  is the charge on an electron,  $\kappa$  is the Debye–Hückel reciprocal length,  $C_d$  and  $C_s$  are the capacitance per unit area of the diffuse and Stern layers respectively,  $\kappa$  is the Debye–Hückel reciprocal length, and  $\Delta$  is the distance of the shear plane from the Stern plane (see Fig. 1).

Zeta potential data obtained at low ionic strength using the EPM method show a broadly consistent value of dζ/dpCa of approximately −13 mV/decade (Fig. 4a). Eq. (18) can be used to match these data using one value of  $C_s \sim 0.4 \text{ Fm}^{-2}$  and  $\Delta = \text{zero}$ . For comparison, Heberling et al. [29] fitted their zeta potential data using a more sophisticated SCM with  $C_s = 0.45 \text{ Fm}^{-2}$  and  $\Delta = 0.33 \text{ nm}$ . Previous SCMs for calcite used unrealistically high  $C_s$  values of 10–100  $\text{Fm}^{-2}$  [57,79]. However, a number of recent studies have reported non-linear relationships between zeta potential and pCa (e.g. [11,42,88]; see Fig. 4b). Moreover, a wide range of IEP (expressed as pCa) values have been reported (pCa of 1.92–4.5; Table 1b). As discussed in the following sections, the variability of these results might be attributed to differences in experimental conditions, including pCO<sub>2</sub>, calcite type, measurement technique and the establishment of equilibrium. The effect of the carbonate ion on the zeta potential has received less attention than the calcium ion. The very limited data suggest that addition of CO<sub>3</sub><sup>2−</sup> makes the zeta potential more negative, consistent with increased adsorption of CO<sub>3</sub><sup>2−</sup> onto the calcite surface (e.g. [16,51,64]).

#### 4.1.3. Effect of pMg and pSO<sub>4</sub>

The magnesium ion is also compatible with the calcite crystal structure [8]. It has a direct influence on geochemical processes involving natural carbonates of mixed mineralogy such as dolomite [50] and is also abundant in natural brines, yet has received much less attention than Ca<sup>2+</sup>. De Groot and Duyvis [14] suggested that Ca<sup>2+</sup> and Mg<sup>2+</sup> influence the zeta potential of calcite to a similar extent, but their dataset



**Fig. 5.** Zeta potential of synthetic calcite and natural carbonate as a function of (a) pMg and (b) pSO<sub>4</sub>. Error bars are shown when reported by the source. In (a), open symbols show zeta potential as a function of pMg. Data from De Groot and Duyvis [14] in deionized water (Δ); Smallwood [64] in deionized water (□); Zhang et al. [87] in 0.571 M NaCl electrolyte (×); Chen et al. [11] in deionized water (○) and Alroudhani et al. [4] in intact natural Portland carbonate in 0.05 M NaCl electrolyte (◇; error bars for these data are the same size or smaller than the symbols). For comparison, filled symbols show measurements as a function of pCa when available from the same studies. In (b), data from Smallwood [64] on synthetic calcite in deionized water (□); Zhang and Austad [88] on Stevns Klint chalk in 0.571 M NaCl electrolyte (×) and Alroudhani et al. [4] in intact natural Portland carbonate in various NaCl electrolytes (◇: 0.05 M; ◇: 0.5 M; error bars are the same size or smaller than the symbols; data from).

was very sparse, comprising just five measurements (two for Mg<sup>2+</sup> and three for Ca<sup>2+</sup>) at high pMg (i.e. at very low concentration; compare filled and open triangles in Fig. 5a). Smallwood [64] obtained a similar response (compare filled and open squares in Fig. 5a) over the same concentration range. More recent studies have found that the zeta potential becomes more positive with decreasing pMg ([4,11,87]; see data represented by crosses, circles and diamonds in Fig. 5a). Alroudhani et al. [4] found that Mg<sup>2+</sup> behaved identically to Ca<sup>2+</sup> within experimental error over a wide range of pCa and pMg (0.4–3; compare filled and open diamonds). Moreover, they were the first to report an IEP in terms of pMg, which was identical to pCa<sub>IEP</sub> within experimental error. However, data showing the effect of pMg on zeta potential are still very scarce, especially at low pMg.

A number of studies have also shown that the zeta potential is affected by the concentration of SO<sub>4</sub><sup>2-</sup>, observing that the zeta potential becomes more negative with decreasing pSO<sub>4</sub> ([4,64,88]; Fig. 5b). Smallwood [64] argued that SO<sub>4</sub><sup>2-</sup> is adsorbed onto the calcite mineral

surface, similar to the lattice ions Ca<sup>2+</sup>, Mg<sup>2+</sup> and CO<sub>3</sub><sup>2-</sup>, as indicated by its ability to reverse the polarity of the zeta potential (Fig. 5b). However, no other papers have reported an IEP in terms of pSO<sub>4</sub>. Zhang and Austad [88] and Alroudhani et al. [4] both observed a linear relationship between zeta potential and pSO<sub>4</sub> for chalk and natural carbonate respectively, but the IEP was not encountered over the concentration range investigated (Fig. 5b). Data showing the effect of pSO<sub>4</sub> on zeta potential are still scarce and do not conclusively show that SO<sub>4</sub><sup>2-</sup> is a PDI for the calcite surface.

#### 4.1.4. Indifferent ions and total ionic strength

Ions other than Ca<sup>2+</sup>, Mg<sup>2+</sup>, CO<sub>3</sub><sup>2-</sup> and SO<sub>4</sub><sup>2-</sup> are generally assumed to be indifferent to the calcite mineral surface; they have little or no tendency to adsorb onto the surface. It is well known that the zeta potential decreases in magnitude with increasing concentration of indifferent ions such as Na<sup>+</sup>, K<sup>+</sup>, Cl<sup>-</sup>, reflecting contraction of the electrical double layer (EDL; e.g. [36,83]). Pierre et al. [54] obtained the same IEP (expressed as pH) on natural calcite irrespective of the concentration of NaCl (0, 10<sup>-3</sup>, 10<sup>-2</sup> M), indicating that Na and Cl are indifferent ions. Alroudhani et al. [4] found that the sensitivity of zeta potential to pCa (i.e. dζ/dpCa) decreased as NaCl concentration increased, consistent with contraction of the EDL reducing the absolute magnitude of the zeta potential (Fig. 4b and Table 1b). However, they also found that the IEP (as pCa) decreased with increasing NaCl concentration, which is not expected for indifferent ions. They suggested that at the high NaCl concentrations investigated (up to 2 M), the ability of Ca<sup>2+</sup> ions to interact with calcite surface was reduced due to contraction of the EDL and increasing occupancy of hydrated Na<sup>+</sup> ions in the diffuse layer. However, data to support this hypothesis are scarce; few studies have obtained zeta potential measurements on calcite at high ionic strength (>0.1 M).

#### 4.2. Partial pressure of CO<sub>2</sub>

Variations in pCO<sub>2</sub> across different experiments may be one factor that explains the variation in reported values of IEP. In closed system experiments with controlled pCO<sub>2</sub>, the very limited data available suggest that the IEP (expressed as pH) decreases with increasing pCO<sub>2</sub> [29,51]. Moreover, in open system experiments, Somasundaran and Agar [67] postulated that pH<sub>IEP</sub> was a function of mixing time of the sample powder in solution, due to an increase in CO<sub>2</sub> concentration in the suspension. They used this mechanism to explain the decrease in pH<sub>IEP</sub> they observed with increased mixing time. The reported pH<sub>IEP</sub> in closed system experiments is generally higher than the value in open system experiments, consistent with reduced access to CO<sub>2</sub> (Table 1a). The absence of, or limited access to, CO<sub>2</sub> in closed system experiments causes a decrease in carbonic acid formation (Eq. (16)) and in turn increases pH<sub>IEP</sub> [66].

The effect of pCO<sub>2</sub> on the IEP expressed as pCa or pMg is less well understood. No studies have varied pCO<sub>2</sub> and pCa or pMg, or measured pCa or pMg whilst varying pCO<sub>2</sub> and pH. Studies investigating the effect of pCa on zeta potential have typically been conducted in simple open or closed system conditions. In open system experiments, measured pCa<sub>IEP</sub> lies in the range of 4.25–4.5 ([21,33,67]; Table 1b). In closed system experiments, reported pCa<sub>IEP</sub> values have a wider range of 0.4–4 and are always lower than the pCa<sub>IEP</sub> values reported in open system experiments (Table 1b). The broad range of pCa<sub>IEP</sub> observed in closed system experiments might be attributed to different amounts of CO<sub>2</sub> dissolved into solution during preparation (e.g. [78]) or the type of calcite used (synthetic vs. natural; [54]; see also the next section). Recent studies of zeta potential on natural carbonates in closed system experiments consistently show the lowest values of pCa<sub>IEP</sub> in the range 0.4–0.75 [4, 11,42]. In closed system experiments, pCO<sub>2</sub> is fixed and, as we show later, lower pCa corresponds to lower pH. Lower pH is equivalent to higher pCO<sub>2</sub> (e.g. [29]; see also reactions 14–18).

### 4.3. Natural versus synthetic calcite

The zeta potential has been measured on a wide variety of calcite samples with differing nature and origin which can be broadly subdivided into three types: (1) synthetic calcite (pure precipitated crystalline calcium carbonate), (2) Iceland spar (pure crystalline calcite of natural origin) and (3) natural carbonate (composed mostly of calcite with inorganic material or organic matter). Data for Iceland spar and synthetic calcite show both positive and negative values for a given value of pH (Fig. 2; Table 1a) and pCa (Fig. 4a, Table 1b). The broad range of zeta potential values observed for a given pH may reflect differences in the concentrations of PDIs such as  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  that were not measured in the experiments; they may also reflect differences in  $p\text{CO}_2$  or, as discussed below, the equilibration method. The range of values observed for a given pCa is more difficult to explain, but most likely reflects differences in  $p\text{CO}_2$  or the equilibration method; it may also reflect differences in the concentrations of other PDIs such as  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$ .

Data for natural calcite and carbonate rocks show less variability, with most studies returning negative values of zeta potential irrespective of pH; the few positive values in Fig. 2 were mostly obtained using natural brines rich in  $\text{Ca}^{2+}$  where, as we show later, positive zeta potential is expected. Berlin and Khabakov [7] reported negative zeta potentials on 115 natural carbonate samples irrespective of pH. Likewise, with relatively few exceptions, the zeta potential on natural samples has been found to be negative at high pCa and positive at low pCa, although there is some variability in  $p\text{Ca}_{\text{IEP}}$  as discussed above (Fig. 4b). It has been suggested that the more negative zeta potentials typically observed in natural carbonates are due to the presence of organic matter incorporated in their structure during formation [80,81]. For example, Vdovic [80] obtained negative zeta potential on two natural carbonate samples over the whole pH range (filled and open circles in Fig. 2b); however, they obtained a positive zeta potential on synthetic calcite (filled circles in Fig. 2a). Likewise, Cicerone et al. [12] found that natural (biogenetic) calcite (open triangles in Fig. 2b) bore a more negative zeta potential than pure calcite (filled triangles in Fig. 2a). However, the role of organic material in controlling the zeta potential of natural carbonates remains poorly understood.

### 4.4. Measurement technique

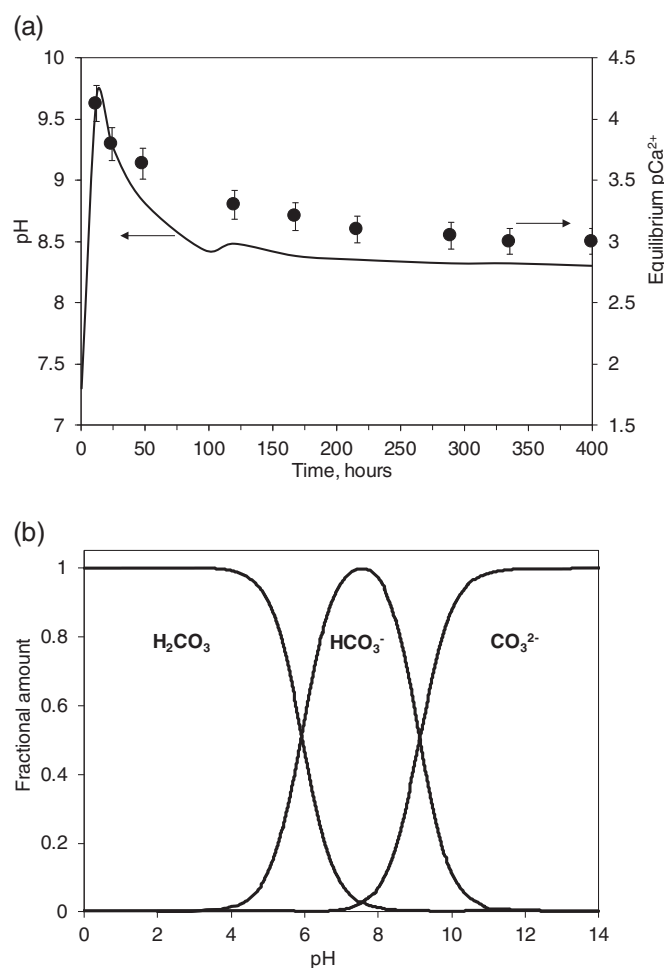
Measurements of zeta potential generally use either the electrophoretic mobility (EPM) method or the streaming potential method (SPM) introduced earlier. There appears to be no systematic difference in the reported IEP and both techniques have yielded a linear Nernstian relationship between zeta potential and pCa (Table 1; Fig. 4). However, only Alroudhan et al. [4] have compared the two techniques on the same (natural) calcite samples and electrolytes. Using the EPM method and natural (powdered) carbonate suspended in 0.05 M NaCl electrolyte, they obtained a linear regression between zeta potential and pCa with a gradient of  $-10.45$  mV/decade, which they fitted using Eq. (18) and values of  $C_s = 1.13 \text{ Fm}^{-2}$  and  $\Delta = 0$ . Note that this gradient is lower than obtained in previous studies on similar samples (as discussed above; see also Fig. 4a) and the value of  $C_s$  used to obtain a match is higher. However, Alroudhan et al. [4] investigated electrolytes of higher total ionic strength than previous studies and their results suggest that  $C_s$  increases with increasing total ionic strength.

Using the SPM method and the same natural (intact) carbonates and electrolytes, Alroudhan et al. [4] obtained the same IEP (expressed as pCa) within experimental error, but the gradient of a linear regression through the data was approximately two times smaller at  $-5.10$  mV/decade. They argued that the shear plane is further away from the mineral surface in intact natural samples as opposed to powdered samples owing to the complex topology of the pore-space (see also Vernhet et al. [82] for similar comparisons on other materials), and fitted the SPM data using Eq. (18) with the same value of  $C_s$  but a higher value of

$\Delta = 0.245$  nm. Thus differences in the observed sensitivity of zeta potential to pCa (or the concentration of other PDIs) may reflect differences in the measurement technique and/or the nature of the sample (intact versus powdered).

### 4.5. Establishment of equilibrium between sample and electrolyte

Most studies equilibrate the calcite–electrolyte systems of interest for 48 h or less prior to measuring the zeta potential (Table 1). Only a few studies have investigated longer equilibration times (e.g. [4,12,29,33,67]). Some studies have reported equilibration times of an hour or less (e.g. [11,18,48,58,60]). Heberling et al. [29] investigated the difference between equilibrium and non-equilibrium measurements. In a study using natural intact limestone, Alroudhan et al. [4] found that the equilibrium pH in open system conditions was reached after 100 h and sometimes more, and that it could take a factor of 2–3 times longer to reach the equilibrium pCa (e.g. Fig. 6). Thus there is the risk that short equilibration times may not yield a stable equilibrium in pH and/or pCa, which will affect the measured zeta potential. Moreover, the equilibrium pH may be achieved before the equilibrium pCa, so assessing equilibration based on pH measurements alone may not be sufficient. As discussed above, values of the zeta potential on calcite are primarily controlled by pCa and the concentration of other PDIs such as  $\text{Mg}^{2+}$ , rather than pH. Therefore, it is important to ensure that equilibrium in



**Fig. 6.** Calcite–water– $\text{CO}_2$  equilibrium. Plot (a) shows calcium concentration (expressed as pCa) and pH measured as a function of time during equilibration of natural Portland rock samples with deionized water. Plot (b) shows carbonate speciation into  $\text{H}_2\text{CO}_3$ ,  $\text{HCO}_3^-$ , and  $\text{CO}_3^{2-}$  as a function of pH. (Modified from Alroudhan et al. [4]).

the concentration of these PDIs has been reached prior to taking measurements. Inconsistent values of zeta potential obtained in some studies may reflect a lack of equilibration.

#### 4.6. Summary of published zeta potential data on calcite

Published measurements to date have typically observed linear, Nernstian relationships between zeta potential and pH, and between zeta potential and pCa, especially for values of pH and pCa close to the IEP. However, despite the clear relationship between zeta potential and pH obtained in such studies, the proton is not a PDI for calcite, as varying pH whilst holding pCa constant yields constant zeta potential within experimental error. Moreover, surface complexation models suggest that the calcite surface remains negatively charged over the pH range 5.5–11, and the surface charge is only weakly dependent on pH. The zeta potential is controlled instead by the concentration-dependent adsorption of lattice ions  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  in the Stern layer, and also by the adsorption of  $\text{Mg}^{2+}$  ions, although experimental data testing the effect of pMg are scarce. The few data available suggests that  $\text{Mg}^{2+}$  behaves identically to  $\text{Ca}^{2+}$  within experimental error. The  $\text{SO}_4^{2-}$  ion has also been suggested as a PDI for the calcite surface, but the evidence is not conclusive.

A wide range of values of IEP (expressed as pCa) have been observed, depending upon whether the sample is natural or artificial, whether the experiments are open or closed to atmospheric  $\text{CO}_2$ , and the approach taken to equilibrate the sample and electrolyte prior to obtaining data. In open system experiments, measured  $\text{pCa}_{\text{IEP}}$  lies in the range of 4.25–4.5, but in closed system experiments,  $\text{pCa}_{\text{IEP}}$  has a wider range of 0.4–4 and is always lower than values reported in open system experiments. Natural samples generally show more negative values of zeta potential than artificial samples for a given value of pCa, and therefore yield lower values of  $\text{pCa}_{\text{IEP}}$ . Recent studies of zeta potential on natural carbonates in closed system experiments consistently show the lowest values of  $\text{pCa}_{\text{IEP}}$  in the range 0.4–0.75.

### 5. Measurements of zeta potential on natural limestone rock samples: Impact of rock type, PDI concentration and ionic strength

As outlined in the previous section, few studies have reported measurements of zeta potential in carbonates at conditions relevant to natural subsurface systems. Most have explored synthetic calcite and dilute electrolytes with much lower total ionic strength and PDI concentration than subsurface brines. Moreover, most did not employ an experimental method that established equilibrium conditions of pH,  $\text{pCO}_2$  and PDI concentration relevant to subsurface carbonates. The aim of this section is to report new measurements of zeta potential relevant to natural systems, with a particular emphasis on determining (i) the relationship between pH and pPDI ( $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ ) in equilibrated calcite–electrolyte systems, given that no studies have measured pPDI whilst varying pH, or reported pH whilst varying pPDI, (ii) the effect of high concentrations of indifferent ions such as NaCl on the sensitivity of the zeta potential to

pPDI, (iii) whether  $\text{SO}_4^{2-}$  is a PDI for the calcite mineral, and (iv) how the zeta potential is affected by pPDI over the concentration range found in natural brines. Only Alroudhani et al. [4] have probed this parameter space, and they used only a single rock type. Here we investigate two further natural carbonates to determine whether the results are rock-type specific.

We used the SPM described by Jackson and co-workers [4,35,38,39,83] to measure the zeta potential. The SPM is applicable to intact natural samples, can be used to measure zeta potential at high ionic strength ( $>2\text{ M}$ ), and can also be used during multiphase flow and at elevated temperature [1,2,4,35,37,38,83,84]. The SPM measurements were complemented by chemical analysis of the effluent electrolyte, to monitor adsorption/desorption of ionic species during the experiments. The results have broad application to earth engineering processes such as hydrocarbon production and geological  $\text{CO}_2$  storage, and we use them here to investigate the mechanisms that underpin the use of controlled injection brine compositions during waterflooding of hydrocarbon reservoirs [41,74,86,87].

#### 5.1. Materials and experimental method

We used two intact natural carbonate core samples that in x-ray diffraction (XRD) analysis appear to be identical pure limestone that differ only in age and permeability (Table 2). Also shown in Table 2 are the properties of the Portland Limestone samples investigated by Alroudhani et al. [4], which we compare to the new results presented here when possible. Samples were cleaned using the enhanced method reported in Alroudhani et al. [4]. The brines used were synthetic solutions of reagent-grade NaCl,  $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ ,  $\text{Na}_2\text{SO}_4$  and  $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$  salts (Sigma Aldrich, UK) in deionized water (DIW) from a filtered system with electrical conductivity below  $1\text{ }\mu\text{S/cm}$ .

The equilibrium condition of carbonate/electrolyte/ $\text{CO}_2$  discussed previously (see Fig. 6) was replicated following the approach of Alroudhani et al. [4]. NaCl electrolytes of varying concentrations (0.05–5 M) were equilibrated by leaving offcuts of the tested carbonate samples in a closed beaker containing an air gap to allow  $\text{CO}_2$  to dissolve into solution, replicating the open-system conditions of carbonate deposition [76]. The pH of the NaCl electrolyte (measured using a Five-Go Mettler-Toledo pH meter with their 3-in-1 pH electrode LE438, implementing where necessary the manufacturer's recommended calibration and correction procedures at high ionic strength) and  $\text{Ca}^{2+}$  concentration (measured using Inductively Coupled Plasma Atomic Emission Spectroscopy, ICP-AES) were monitored until both reached constant value within experimental error. The equilibrium pH was found to be  $8.3 \pm 0.1$  for both Ketton and Estailades, in agreement with Alroudhani et al. [4], and the equilibrium pCa was found to be  $2.8 \pm 0.1$  and  $3.1 \pm 0.1$ , respectively. The equilibrated NaCl solutions were termed NaCl-EQ and were then used directly in zeta potential measurements, or were modified by addition of  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and/or  $\text{SO}_4^{2-}$ .

The apparatus used to measure zeta potential in our SPM is closed to the atmosphere, and the second stage of equilibration prior to measuring

**Table 2**  
Properties and mineralogy of rock samples used in this study compared to Portland used by Alroudhani et al. [4].

| Property/rock                  | Ketton                                                           | Estailades                                                       | Portland                                                         |
|--------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|
| Description                    | Middle Jurassic Oolitic limestone from UK                        | Upper Cretaceous limestone from France                           | Upper Jurassic Oolitic limestone from UK                         |
| Porosity                       | $23\% \pm 0.5$                                                   | $28\% \pm 0.5$                                                   | $20\% \pm 0.5$                                                   |
| Permeability                   | $1.4\text{ Darcy} \pm 0.4$                                       | $0.13\text{ Darcy} \pm 0.2$                                      | $0.005\text{ Darcy} \pm 0.001$                                   |
| Intrinsic formation factor (F) | $13.87 \pm 0.5$                                                  | $12.92 \pm 0.5$                                                  | $22.04 \pm 0.5$                                                  |
| Compositions                   | 97% calcite ( $\text{CaCO}_3$ )<br>3% magnesium*                 | 97% calcite ( $\text{CaCO}_3$ )<br>3% magnesium*                 | 96.6% calcite ( $\text{CaCO}_3$ )<br>3.4% quartz                 |
| Dimensions                     | Length (L) = $0.076\text{ m}$<br>Diameter (D) = $0.038\text{ m}$ | Length (L) = $0.076\text{ m}$<br>Diameter (D) = $0.038\text{ m}$ | Length (L) = $0.076\text{ m}$<br>Diameter (D) = $0.038\text{ m}$ |

\* Magnesium ( $\text{Mg}^{2+}$ ) is likely incorporated into the calcite structure as  $\text{MgCO}_3$  or  $\text{CaMg}(\text{CO}_3)_2$ .

the zeta potential was to ensure equilibrium between the electrolyte of interest (NaCl–EQ with or without the addition of  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  or  $\text{SO}_4^{2-}$ ) and the rock sample at the closed-system conditions pertaining to a rock–brine system at depth. The rock sample was pre-saturated with the selected electrolyte at open-system conditions and then confined in the core holder at closed-system conditions, and the electrolyte was pumped through the sample from the (closed) inlet reservoir to the (closed) outlet reservoir and back again. The repeated flow of the electrolyte through the sample at closed system conditions mimics migration of the electrolyte into the carbonate rock at depth. At regular intervals, the electrical conductivity and pH of the electrolyte in the reservoirs was measured, and equilibrium was assumed to have been reached when the conductivity and pH of the electrolyte in each reservoir differed by <5%.

Electrolyte samples were analysed before and after the SPM experiments for key ions ( $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$ ). We used ICP-AES for cations and Ion Chromatography (IC) for anions with appropriate dilution where necessary. The ICP-EAS analysis was carried out in the Analytical Chemistry Laboratory at the Natural History Museum London; the IC analysis was carried out in the TOTAL Laboratory for Reservoir Physics at Imperial College London. Instrument error was determined by using certified solutions containing  $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  and the error calculated from the standard deviation of five repeat measurements of each solution.

## 5.2. Design of experiments

We conducted two sets of experiments. In the first, the sensitivity of zeta potential to variations in NaCl concentration was tested using the equilibrated NaCl–EQ solutions described above. In these experiments, the concentration of  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  was dictated by the equilibration process; no additional salts containing these ionic species were added to the NaCl–EQ solutions. In the second, the sensitivity of

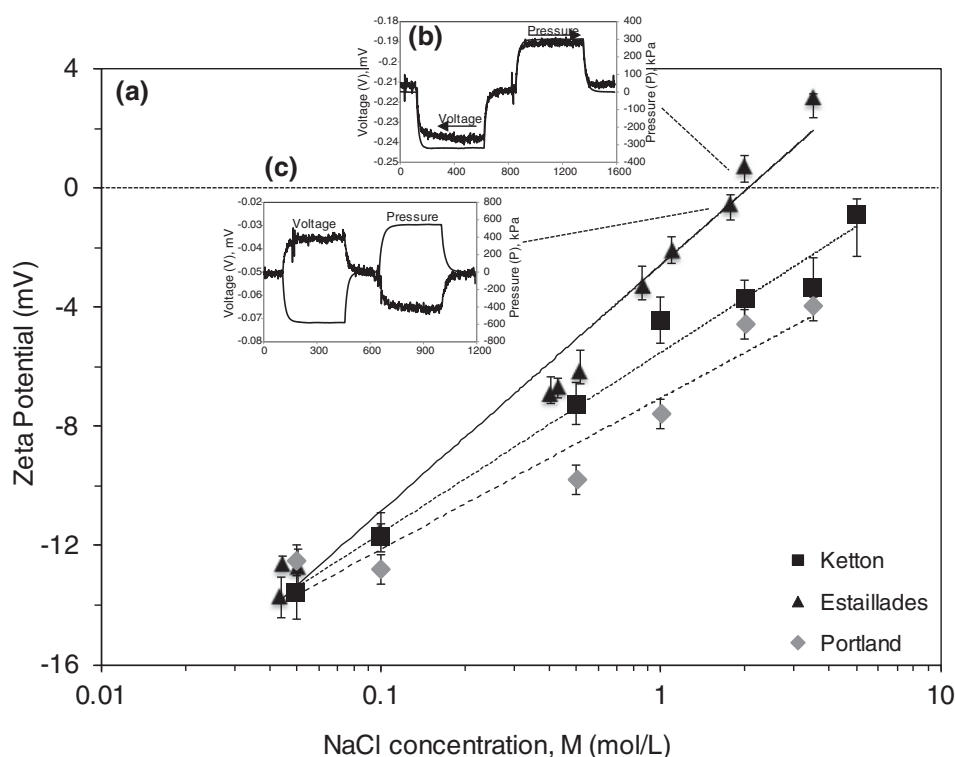
the zeta potential to variations in  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  concentration was tested by adding salts containing these ionic species over the range found in natural brines (0.007–0.42 M for  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ ; 0.002–0.096 M for  $\text{SO}_4^{2-}$ ) to NaCl–EQ solutions of two different ionic strengths (0.5 M and 2 M). The 0.5 M NaCl concentration represents seawater and is similar to the ‘ZP brine’ of Zhang and Austad [88] and Zhang et al. [87], allowing direct comparison of results. The 2 M NaCl concentration represents the saline brines found in many saline aquifers and hydrocarbon reservoirs (e.g. [59]). In the second set of experiments, we investigated for the first time the relationship between pCa (or pMg) and pH. Note that in the experiments reported here, precipitation of salts such as  $\text{CaSO}_4$  and  $\text{MgCO}_3$  in the pore-space was prevented because each ion of interest ( $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  or  $\text{SO}_4^{2-}$ ) was added to NaCl–EQ electrolyte containing only trace or zero concentration of cations or anions other than  $\text{Na}^+$  and  $\text{Cl}^-$ .

## 5.3. Results

Streaming potentials were measured using the paired-stabilization method of Vinogradov et al. [83] and typical raw results are shown in Fig. 7b,c. The stabilized voltage was plotted against the stabilized pressure difference across the sample for four different flow rates and a linear regression through the data yielded the streaming potential coupling coefficient and, via Eq. (3), the zeta potential. The uncertainty in the streaming potential coupling coefficient arising from the range of linear regressions that can be forced through the stabilized voltage and pressure data was used to determine the associated experimental error in zeta potential reported in the following sections.

### 5.3.1. Impact of $\text{Na}^+$ concentration on zeta potential

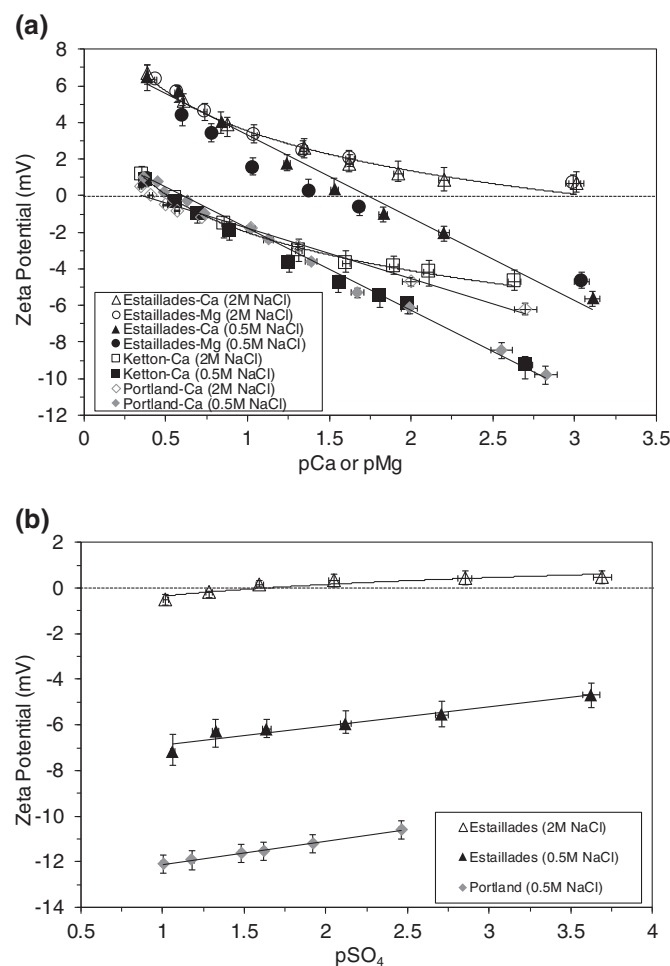
We begin by reporting the effect of NaCl concentration on zeta potential. When possible, we compare results from the Ketton and



**Fig. 7.** (a) Effect of NaCl concentration on zeta potential on the Ketton (squares) and Estailades (triangles) samples. Data obtained on the Portland carbonate sample by Alroudhani et al. [4] are shown for comparison (grey diamonds). The inset plots (b) and (c) show typical example measurements of voltage and pressure against time from the SPM used to determine the zeta potential. When pressure and voltage respond in the same sense the zeta potential is positive and vice-versa. The polarity of the zeta potential is accurately determined even when its value is close to zero.

Estailades samples investigated here against the Portland sample studied by Alroudhan et al. [4]. Fig. 7a shows the zeta potential as a function of NaCl concentration (in M) on a log-log scale for all three natural carbonate samples. A linear regression can be fitted to the data from each rock type (with  $R^2 > 0.97$ ) but the gradient of the regression differs between samples. The Estailades sample (filled triangles) shows the highest gradient ( $-8.08 \pm 0.52$  mV/M) and the Portland sample (filled grey diamonds) the lowest gradient ( $-5.08 \pm 0.47$  mV/M) with the Ketton sample (filled squares) in between ( $-6.11 \pm 0.49$  mV/M).

At low NaCl concentration ( $<0.1$  M) the zeta potential is negative and identical (within experimental error) for all three samples. However, as NaCl concentration increases, the zeta potentials diverge with  $|\zeta_{\text{Portland}}| > |\zeta_{\text{Ketton}}| > |\zeta_{\text{Estailades}}|$ . Moreover, the zeta potential of the Estailades samples becomes positive at NaCl concentration  $> 1.9$  M, whereas the other two samples yield negative zeta potential over the entire concentration range investigated. The inset plots 7b and 7c show there is no ambiguity in the polarity of the zeta potential: for a NaCl concentration of 1.8 M the streaming potential, and hence zeta potential, of the Estailades sample is clearly negative ( $-0.56$  mV  $\pm$  0.33) and becomes positive ( $0.72$  mV  $\pm$  0.33) for a NaCl concentration of 2 M. As discussed above,  $\text{Na}^+$  and  $\text{Cl}^-$  are believed to be indifferent ions to the calcite mineral surface and their presence is expected to affect the magnitude, but not the polarity, of the zeta potential (e.g. [23,34,54]).



**Fig. 8.** Zeta potential versus (a) pCa and pMg, and (b) pSO<sub>4</sub> for the Ketton (squares) and Estailades (triangles and circles) samples at different NaCl concentrations. Data obtained on the Portland carbonate sample by Alroudhan et al. [4] are shown for comparison (diamonds).

### 5.3.2. Impact of $\text{pCa}^{2+}$ , $\text{pMg}^{2+}$ and $\text{pSO}_4^{2-}$ on zeta potential

We next investigate the impact of the concentration of the confirmed and suggested PDIs  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  on the zeta potential. We plot concentration as pPDI and, in all cases, the highest pPDI values reported correspond to the equilibrium concentrations in the NaCl-EQ electrolytes. Fig. 8a shows the impact of pCa and pMg on the zeta potential of Estailades, and the impact of pCa on the zeta potential of Ketton, for two different NaCl concentrations (0.5 M and 2 M) typical of subsurface brines; also shown for comparison are the data from Portland obtained by Alroudhan et al. [4]. We find that the zeta potential consistently becomes more positive with decreasing pCa or pMg. Moreover, the response to pCa and pMg is identical within experimental error for the Estailades carbonate, irrespective of NaCl concentration. Alroudhan et al. [4] also found that pCa and pMg behaved identically for the Portland carbonate, although they investigated a lower NaCl concentration of 0.05 M and their data are not shown here.

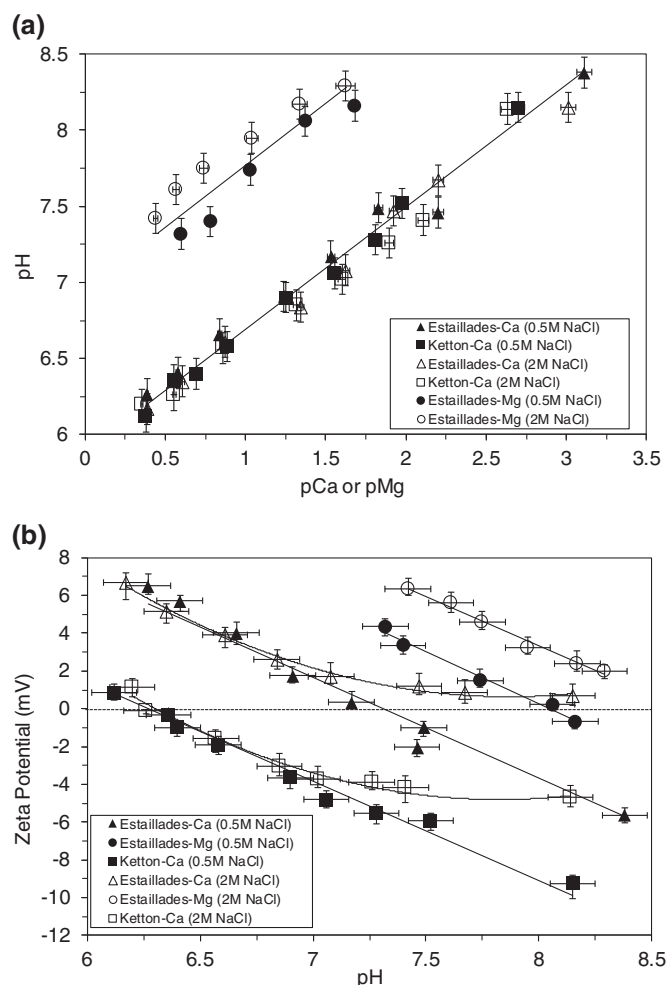
The response of the zeta potential to pCa or pMg is generally sample specific and depends on the NaCl concentration. At the lower (0.5 M) NaCl concentration investigated here, Ketton and Portland exhibit identical behaviour over the entire pCa range investigated: a linear regression provides an excellent fit to the data ( $R^2 > 0.98$ ) with a gradient of  $-4.5 \pm 0.25$  mV/decade and an  $\text{pCa}_{\text{IEP}}$  of  $0.5 \pm 0.03$ . A linear regression with the same gradient also provides an excellent fit to the Estailades data at the lower NaCl concentration investigated; however,  $\text{pCa}_{\text{IEP}}$  is much higher at  $1.7 \pm 0.03$ . Nernstian (linear) behaviour is therefore always observed at the lower NaCl concentration over the entire pCa and pMg range investigated, but the IEP is different for the Estailades sample. Fitting the Gouy–Chapman–Grahame model (Eq. 18) to these data yields  $C_s = 1 \text{ Fm}^{-2}$  and  $\Delta = 0.245 \text{ nm}$ , similar to the values obtained for Portland carbonate by Alroudhan et al. [4] and within the range used by Heberling et al. [29].

At the higher (2 M) NaCl concentration investigated here, the Ketton carbonate continues to show linear, Nernstian behaviour close to the IEP with a gradient and IEP identical to that obtained using 0.5 M NaCl within experimental error, but the behaviour becomes non-linear (non-Nernstian) away from the IEP at high pCa. The Portland sample shows linear, Nernstian behaviour over the entire pCa range investigated with an identical IEP within experimental error but a lower gradient of  $-2.86 \pm 0.25$  mV/decade. The Estailades carbonate exhibits notably different behaviour, with the zeta potential remaining positive over the entire pCa/pMg range investigated, and non-Nernstian behaviour at high pCa. Cicerone et al. [12] also observed non-Nernstian behaviour far from IEP and related that to variations in the Stern layer capacitance or to breakdown of the Gouy–Chapman–Grahame model.

Increasing pSO<sub>4</sub> consistently yields more positive zeta potential, although the sensitivity is much less than that observed when varying pCa or pMg (Fig. 8b). A linear regression yields a good fit to the data at the lower NaCl concentration investigated here ( $R^2 > 0.92$ ) with a consistent gradient of  $0.84 \pm 0.2$  mV/decade for both the Estailades and Portland carbonates. A key feature is that an apparent IEP of  $\text{pSO}_4 = 1.59 \pm 0.2$  is observed for the Estailades sample in 2 M NaCl electrolyte. It is also notable that the difference in absolute zeta potential between the different carbonate samples observed in Fig. 7 persists in Fig. 8, with  $|\zeta_{\text{Portland}}| > |\zeta_{\text{Ketton}}| > |\zeta_{\text{Estailades}}|$ .

### 5.3.3. Relationship between pH and pCa or pMg

Fig. 9a shows the relationship between the equilibrium pH and pCa or pMg obtained across all of the experiments reported here. A linear regression provides an excellent fit to the data ( $R^2 > 0.98$ ) with a gradient of 0.81, irrespective of NaCl concentration, rock sample or PDI varied, although the value of pH is consistently higher in the experiments varying pMg as compared to those varying pCa. These experimental data confirm that equilibrium pH and pPDI are related at fixed  $\text{pCO}_2$  via Eqs. (14)–(17). It is possible to plot the data obtained here against pH and obtain an apparently strong correlation (Fig. 9b) even though pCa and pMg were the controlled parameters. The



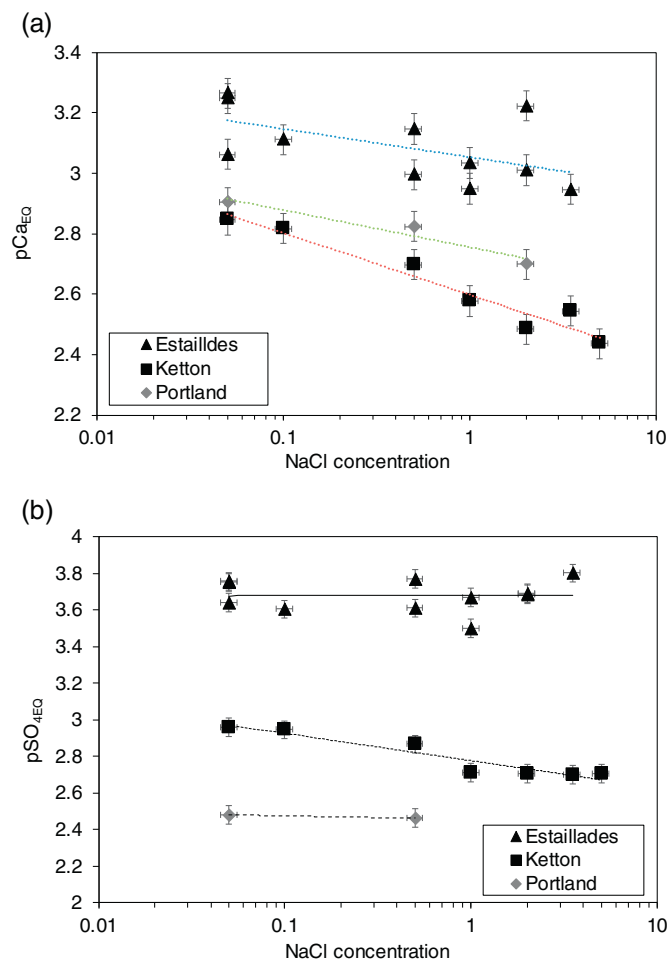
**Fig. 9.** (a) Relationship between pH and pCa or pMg obtained from all the experiments shown in Fig. 8, along with linear regressions to the data. (b) Data shown in Fig. 8 replotted against pH using the linear regression shown in (a).

relation between pH and pCa or pMg obtained here is consistent with model predictions (e.g. PHREEQC; [55]). However, it has not previously been demonstrated in properly equilibrated experiments where zeta potential was also measured, or over such a wide range of pCa.

#### 5.4. Discussion

##### 5.4.1. Impact of indifferent ions

The results presented here show a clear relationship between the concentration of indifferent ions  $\text{Na}^+$  and  $\text{Cl}^-$  and the zeta potential on natural carbonate (Fig. 7). Such a relationship is expected owing to collapse of the double layer with increasing ionic strength, but if double layer collapse is the only control on zeta potential, then the zeta potential should be proportional to the logarithm of NaCl concentration (as a proxy for ionic strength; e.g. [27]). Consequently, the dependence of zeta potential on NaCl concentration observed in Fig. 7 cannot result only from double layer collapse. Fig. 10 shows the equilibrium pCa and  $\text{pSO}_4$  as a function of NaCl concentration, corresponding to the zeta potential data shown in Fig. 7. Note that pMg remained large and approximately constant at  $4.19 \pm 0.03$  for all samples irrespective of NaCl concentration (i.e. the  $\text{Mg}^{2+}$  concentration is small). pCa generally decreases with increasing NaCl concentration (Fig. 10a); hence, given that zeta potential becomes more positive with decreasing pCa, these data are consistent with the more positive zeta potential observed



**Fig. 10.** Equilibrium pCa (a) and  $\text{pSO}_4$  (b) versus NaCl concentration for the Ketton (squares) and Estailades (triangles) samples. Data obtained on the Portland carbonate sample by Alroudhay et al. [4] are shown for comparison (diamonds).

with increasing NaCl concentration in Fig. 7. However, the sensitivity decreases at high NaCl concentration, moreover, these data do not explain the polarity change of the zeta potential observed for the Estailades carbonate. This remains a curious feature of the new data obtained here and may suggest that  $\text{Na}^+$  is not in fact an indifferent ion to the calcite surface in some circumstances, as suggested by [29,44].

##### 5.4.2. Impact of carbonate type

The results obtained here also show that natural carbonate samples that appear identical in XRD display different zeta potential behaviour. The Estailades carbonate, in particular, behaves differently to the Ketton and Portland carbonates, showing inversion of the zeta potential with increasing NaCl concentration (Fig. 7), higher values of  $\text{pCa}_{\text{IEP}}/\text{pMg}_{\text{IEP}}$  compared to Ketton and Portland (Fig. 8a), and also an apparent IEP expressed as  $\text{pSO}_4$  (Fig. 8b). The Estailades carbonate released the least  $\text{SO}_4^{2-}$  during equilibration with the NaCl solutions, while the Portland sample released the most  $\text{SO}_4^{2-}$  (Fig. 10b), consistent with the observed differences in zeta potential ( $\zeta_{\text{Portland}} > \zeta_{\text{Ketton}} > \zeta_{\text{Estailades}}$ ) in Figs. 7 and 8. We hypothesize that Ketton and Portland contain undetected sources of  $\text{SO}_4^{2-}$  which affects the pristine zeta potential, yielding more negative values and lower  $\text{pCa}_{\text{IEP}}$  than Estailades. Natural carbonates can incorporate a number of different minerals, including clays, organic matter, apatite, anhydrite or quartz, that yield a more negative zeta potential than pure calcite [12,65,66]. Our data therefore suggest that the behaviour of the natural Ketton and Portland carbonate samples investigated here is affected by the presence of impurities such as

anhydrite. In contrast, the behaviour of the Estailades sample is consistent with the ionic strength effect as calculated with a Pitzer model for a calcite-atmospheric  $\text{CO}_2$  equilibrium solution [55].

The contrasting behaviour of Estailades carbonate may also reflect differences in the affinity of  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  for the mineral surface. Pierre et al. [54] suggested that the IEP is governed by the relative magnitude of the equilibrium constants  $K_{\text{Ca}}$  and  $K_{\text{CO}_3}$  governing the adsorption of  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  ions on the calcite mineral surface. The IEP shifts to lower pCa if  $K_{\text{CO}_3} > K_{\text{Ca}}$ ; that is, if the calcite surfaces show greater affinity for  $\text{CO}_3^{2-}$  than  $\text{Ca}^{2+}$ . Pierre et al. [54] found the IEP differed for synthetic and natural calcite and argued that this reflected the differing affinity for Ca and  $\text{CO}_3$ . The Pierre et al. model suggests that the Estailades carbonate investigated here has a much lower affinity to  $\text{CO}_3^{2-}$  (and greater affinity to  $\text{Ca}^{2+}$ ) than the Ketton and Portland carbonates at equilibrium conditions.

#### 5.4.3. $\text{SO}_4$ as a PDI for calcite

Several studies have suggested that the  $\text{SO}_4^{2-}$  ion is a PDI for calcite (e.g. [64,88]). However, these studies did not measure changes in pCa or pMg in response to changing p $\text{SO}_4$ . Here, we find that  $\text{SO}_4^{2-}$  has only a minor influence on the zeta potential of natural carbonate (Fig. 8b). At low ionic strength (0.5 M NaCl), the change in p $\text{SO}_4$  correlates to a change in pCa (Fig. 11a) which can explain the change in zeta potential (Fig. 11b). We observe a linear relationship between zeta

potential and pCa with a similar gradient (5.1 mV/decade;  $R^2 > 0.85$ ) to that obtained in the experiments where pCa was varied (4.5 mV/decade). However, at high ionic strength (2 M NaCl), pCa is relatively insensitive to p $\text{SO}_4$  (Fig. 11a), yet the zeta potential still varies in response to changing p $\text{SO}_4$  and changes polarity from positive to negative. Moreover, there is not a clear correlation between pCa and zeta potential (Fig. 11b). It may be that small changes in pCa, pMg and even pNa within experimental error all contribute to the observed change in zeta potential, as it is only c. 1 mV in magnitude and close to the IEP. Geochemical modelling beyond the scope of the paper is required to test this hypothesis.

#### 5.4.4. Implications for wettability alteration and hydrocarbon recovery in carbonate reservoirs

In carbonate rock formations with positively charged mineral surfaces, an attractive electrostatic force will act between the mineral surfaces and the negatively charged oil-brine interface, promoting wettability alteration to oil-wet conditions (e.g. [10,31]). Several studies have suggested that carbonate rocks saturated with natural brines rich in  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  ions have positively charged mineral surfaces so are likely to be oil-wet (e.g. [4,11,38,46,47]). The wettability of a reservoir plays a key role in the efficiency with which the oil can be produced. However, we find here considerable variability in the IEP expressed as pCa, suggesting that some carbonates will be positively charged in contact with natural brines, and others will not. The SPM used here is a suitable technique to determine the zeta potential of intact reservoir rock samples saturated with natural brine (e.g. [38]).

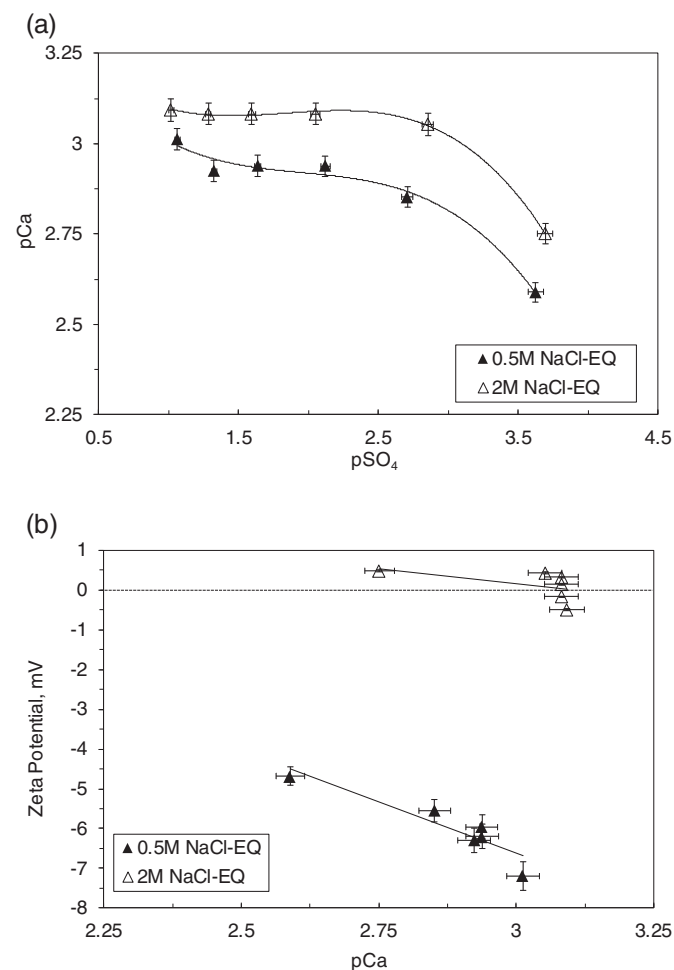
In many reservoirs, water is injected to maintain the pressure and displace oil towards production wells. If the concentration of  $\text{Ca}^{2+}$  or  $\text{Mg}^{2+}$  in the injected water can be modified to fall below the IEP, either selectively or by bulk dilution in a process termed ‘controlled salinity waterflooding’ (CSW), the zeta potential will change polarity from positive to negative. The polarity change leads to electrostatic repulsion between the mineral surfaces and the negatively charged oil-brine interface. This may promote wettability alteration to water-wet conditions, releasing previously adsorbed oil from the calcite mineral surfaces and therefore improving oil recovery (e.g. [47]).

Most previous reported values of the IEP expressed as pCa suggest that considerable reduction in Ca concentration is required to change the polarity of calcite (Table 1). However, our results suggest that far less dilution may be required in some natural carbonates (e.g. Ketton and Portland); in others, the IEP may never be encountered and the mineral surfaces may remain positively charged (e.g. Estailades). The carbonate-rock-specific relationship between zeta potential and PDI concentration observed here may explain why CSW in some cases yields increased recovery and in other cases does not (e.g. [3,59]). Future work relevant to CSW should focus on testing the link between brine composition, zeta potential and increased oil recovery using integrated experiments with consistent materials and experimental conditions (e.g. [37]).

## 6. Conclusions

Experimental studies of the zeta potential of calcite and carbonate have reported a broad range of values with numerous inconsistent results. Yet the primary data source to understand calcite surface charge comprises measurements of zeta potential, so it is important to understand the key controls on zeta potential and why studies have reported apparently contradictory values. A comprehensive review of the literature, supplemented by experimental data new to this study, suggests that:

1. The potential determining ions (PDIs) for the carbonate mineral surface are the lattice ions  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{CO}_3^{2-}$ . The zeta potential is controlled by the concentration-dependent adsorption of these ions within the Stern layer, primarily at the Outer Helmholtz Plane (OHP).



**Fig. 11.** Impact of p $\text{SO}_4$  on pCa and zeta potential measured on the Estailades sample. (a) pCa measured during the experiments shown in Fig. 8b in which  $\text{SO}_4^{2-}$  was added to pre-equilibrated NaCl brine plotted against measured p $\text{SO}_4$ . (b) Zeta potential from the experiments shown in Fig. 8b plotted against the measured pCa recorded in (a).

- Given (1), the Iso-Electric Point (IEP) at which the zeta potential is zero should be expressed as pCa (or pMg). It should not be reported as pH, similar to most metal oxides.
- The pH does not directly control the zeta potential. Varying the pH whilst holding pCa constant yields constant zeta potential. The pH affects the zeta potential only by moderating the equilibrium pCa for a given CO<sub>2</sub> partial pressure (pCO<sub>2</sub>).
- Experimental studies that appear to yield a systematic relationship between pH and zeta potential are most likely observing the relationship between pCa and zeta potential, with pCa responding to the change in pH. New experimental data presented here show a consistent linear relationship between equilibrium pH and equilibrium pCa or pMg irrespective of sample or solution ionic strength.
- The surface charge is weakly dependent on pH through protonation and deprotonation reactions that occur within a hydrolysis layer immediately adjacent to the mineral surface. The Point of Zero Charge (PZC) at which the surface charge is zero could be expressed as pH, but surface complexation models suggest the surface is negatively charged over the pH range 5.5–11.
- Several studies have suggested that SO<sub>4</sub><sup>2−</sup> is also a PDI for the calcite surface, but new experimental data presented here indicate that the value of pSO<sub>4</sub> may affect zeta potential only by moderating the equilibrium pCa. Modelling will be required to provide a more thorough interpretation of the new data.
- Natural carbonate typically yields a more negative zeta potential than synthetic calcite, most likely due to the presence of impurities including clays, organic matter, apatite, anhydrite or quartz, that yield a more negative zeta potential than pure calcite. New data presented here show that apparently identical natural carbonates display differing zeta potential behaviour, most likely due to the presence of small volumes of these impurities.
- It is important to ensure that equilibrium, defined in terms of the concentration of PDIs, has been reached prior to taking measurements. Inconsistent values of zeta potential obtained in some studies may reflect a lack of equilibration.
- The data collated and reported here have broad application in engineering processes including the manufacture of paper and cement, the geologic storage of nuclear waste and CO<sub>2</sub>, and the production of oil and gas.

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# **Zeta Potential of Intact Natural Limestone: Impact of Potential-Determining Ions Ca, Mg and SO<sub>4</sub>**

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## **Abstract**

We report measurements of the zeta potential on intact limestone samples obtained using the streaming potential method (SPM), supplemented by the more ubiquitous electrophoretic mobility method (EPM). The effect of the potential-determining ions (PDI) Ca, Mg and SO<sub>4</sub>, and the total ionic strength controlled by NaCl concentration, is investigated over the range typical of natural brines. We find that the zeta potential varies identically and linearly with calcium and magnesium concentration expressed as pCa or pMg. The zeta potential also varies linearly with pSO<sub>4</sub>. The sensitivity of the zeta potential to PDI concentration, and the IEP expressed as pCa or pMg, both decrease with increasing NaCl concentration. We report considerably lower values of IEP than most previous studies, and the first observed IEP expressed as pMg. The sensitivity of the zeta potential to PDI concentration is lower when measured using the SPM compared to the EPM, owing to the differing location of the shear

plane at which the zeta potential is defined. SPM measurements are more appropriate in natural porous samples because they reflect the mineral surfaces that predominantly interact with the adjacent fluids. We demonstrate that special cleaning procedures are required to return samples to a pristine zeta potential after exposure to PDIs. We apply our results to an engineering process: the use of modified injection brine composition to increase oil recovery from carbonate reservoirs. We find a correlation between an increasingly negative zeta potential and increased oil recovery.

## Introduction

The zeta potential of natural carbonates plays a role in many subsurface processes, governing the electrostatic interactions between mineral surfaces and polar species in both aqueous and non-aqueous phase liquids (NAPLs). For example, the self-freshening often observed when brackish water invades a freshwater aquifer depends on preferential adsorption of aqueous salt species such as Ca and Mg (e.g. Appelo, 1994), while contaminated carbonate aquifers may be remediated through sequestration of the contaminant by co-precipitation with the mineral phase (Meece and Benninger, 1993). Uptake of contaminants such as heavy metals is related to their reactivity as a function of the ionic strength and pH of the aqueous electrolyte (Reeder et al., 2001). The wetting state of carbonate oil reservoirs is believed to be influenced by the zeta potential (Buckley et al., 1998; Gomari et al., 2006), as is the success of enhanced oil recovery by modification of injection brine composition and/or ionic strength (Zhang and Austad, 2006; Yousef et al., 2010). Moreover, solubility of CO<sub>2</sub> in brine as a trapping mechanism in saline aquifers is an important component of carbon capture and storage (Riley, 2010). Compared to sandstones, aqueous CO<sub>2</sub> solubility is greatly enhanced in the presence of carbonate minerals such as calcite (Rosenbauer et al., 2005). The increase in CO<sub>2</sub> concentration has a profound

effect on pH (Pokrovsky et al., 2005), which in turn alters the zeta potential of calcite and leads to its dissolution (Eriksson et al., 2007). The zeta potential is also an important control on the use of self-potential measurements to monitor subsurface fluid flow (e.g. Saunders et al. 2008; Gulamali et al., 2011; Jackson et al., 2012a, b).

The calcite-water interface is electrically charged with the calcite crystal lattice constituents  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  being the main potential determining ions (PDIs, being those ions whose concentration in aqueous solution controls the polarity and density of electrical charge on the mineral surface; Somasundaran and Agar, 1967). However, it is well known that divalent ions such as  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  are also PDIs (Pierre et al., 1990). Figure 1 shows that there are numerous papers reporting measurements of the zeta potential on calcite. The zeta potential is modified by the concentration of both indifferent and potential-determining ions, with the concentration of indifferent ions controlling the thickness of the electrical double layer, and the concentration of PDIs controlling both the double layer thickness and the mineral surface charge. Previous studies of the zeta potential on calcite have highlighted the difference between natural and artificial calcite samples (e.g. Cicerone et al., 1992, Vdovic, 2001), the importance of controlling  $\text{CO}_2$  partial pressure ( $\text{pCO}_2$ ) in open or closed-system experiments (Thompson and Pownall, 1989; Heberling et al., 2011), the impact of wetting state in the presence of NAPLs (e.g. Jackson and Vinogradov, 2012; Kasha et al., 2015), and the effect of PDI concentration (Pierre et al., 1990; Zhang and Austad, 2006; Strand et al., 2006; Alotaibi et al., 2011; Chen et al., 2014; Alotaibi and Yousef, 2015; Mahani et al., 2015a,b). However, few report measurements of zeta potential in carbonates at conditions relevant to natural subsurface systems. Most explore only dilute electrolytes, with much lower total ionic strength and PDI concentration than subsurface brines, and crushed rather than intact rock samples. Moreover, most do not employ an experimental method that establishes equilibrium conditions of pH,  $\text{pCO}_2$  and PDI concentration relevant to subsurface carbonates. Many use artificial calcite,

open system measurements with uncontrolled  $p\text{CO}_2$ , or vary pH and/or  $p\text{CO}_2$  over a broad range not relevant to subsurface brines.

Most previous studies utilised measurements of electrophoretic mobility (EPM) to determine the zeta potential (Madsen, 2002). In this approach, the sample is crushed to a fine powder and suspended in a solution of the electrolyte of interest. An electrical potential field is applied across the suspension (the field typically oscillates at a controlled frequency, inducing an alternating current through the suspension) and the resulting movement of the solid particles is used to interpret the zeta potential via the Helmholtz-Smoluchowski equation (see Delgado et al., 2007). EPM measurements may not reflect the natural conditions of interest for several reasons. First, the samples are crushed, which creates ‘fresh’ mineral surfaces that may have different properties to ‘aged’ surfaces that have been previously exposed to fluids in the pore-space. Second, the ratio of electrolyte volume to mineral surface area is significantly changed compared to the natural porous medium, which may be important in systems such as carbonates where dissolution and precipitation and/or adsorption and desorption may simultaneously modify surface charge and electrolyte composition (Thompson and Pownall, 1989; Pierre et al., 1990). Third, the EPM method is limited to representing only one solid or fluid phase in addition to the supporting electrolyte. Hence, it cannot be used to obtain multiphase measurements when both NAPLs and water are present within the rock pore-space, as is often the case in subsurface carbonates.

The aim of this study is to determine the zeta potential in *intact* natural carbonate samples saturated with aqueous electrolytes containing PDIs at similar concentration to natural brines, and with total ionic strength similar to natural brines. We are particularly interested in determining how the zeta potential is affected by the concentration of PDIs such as Ca, Mg and  $\text{SO}_4$  over the range found in natural brines. Several previous studies have investigated the relationship between Ca concentration and zeta potential, but these typically probed

concentration ranges much lower than natural brines (e.g. Foxall et al., 1978; Thompson and Pownall, 1989; Pierre et al., 1990; Cicerone et al., 1992). Much less attention has been paid to the role of Mg and  $\text{SO}_4$  as PDIs yet these ions are also abundant in natural brines such as seawater (e.g. Zhang and Austad, 2006;). We also wish to determine how the zeta potential is affected by the concentration of these PDIs in the presence of Na and Cl ions over the range found in natural brines. Na and Cl are by far the most common ionic species found in such brines and are believed to be indifferent to the calcite mineral surface; nonetheless, it has not yet been determined whether the effect of the known PDIs (Ca, Mg and  $\text{SO}_4$ ) on carbonate surface charge is modified by the presence of Na and Cl at high concentration.

Rather than the EPM used in most previous studies, we used the streaming potential method (SPM) described by Jackson and Vinogradov and co-workers (Jaafar et al., 2009; Vinogradov et al., 2010; Vinogradov and Jackson, 2011; Jackson and Vinogradov, 2012). The advantage of the SPM is that it can be used with intact samples, including the cylindrical core plugs that are ubiquitously obtained from subsurface reservoirs (Jaafar et al., 2009), is suitable for high ionic strength ( $>2\text{M}$ , where M represents moles/litre) electrolytes (Vinogradov et al., 2010), can be used to measure zeta potential during multiphase flow and for varying wettability (Vinogradov and Jackson, 2011; Jackson and Vinogradov, 2012), and can be extended to the elevated temperatures (up to  $150^\circ\text{C}$ ) often found in deep saline aquifers, hydrocarbon reservoirs, and geothermal systems (Vinogradov and Jackson, 2015). The SPM measurements were complemented by pH and electrical conductivity measurements, and chemical analysis of the effluent electrolyte, to ensure accuracy of the reported compositions and monitor any adsorption/desorption and/or dissolution/precipitation that occurred during the experiments. In addition, we report zeta potential measurements using EPM for the same materials (rock and electrolytes) and preparation procedure, which allows direct comparison between these two electrokinetic methods.

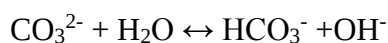
Our approach contrasts with many previous studies because the experimental method is specifically designed to ensure the equilibrium achieved between sample and electrolyte is consistent with natural processes. The results are directly applicable to a wide variety of natural subsurface carbonates. Here we apply our results to a key engineering process relevant to subsurface carbonates: the use of modified injection brine composition to increase oil recovery in a process termed controlled salinity waterflooding (CSW).

## Methodology

### Materials and sample preparation

The rock samples used in the experiments are Portland limestone from the Portland quarry on the south coast of the UK (Table 1). We used two different types of electrolyte. The first comprised reagent-grade NaCl,  $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ ,  $\text{Na}_2\text{SO}_4$  (Sigma-Aldrich),  $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$  (Fluka Analytical) solutions in deionized water (DIW) from a Thermo Scientific filtered system with electrical conductivity below  $1 \mu\text{S}/\text{cm}$ . In these electrolytes, the maximum concentration probed was 2M for NaCl, 0.42M for  $\text{CaCl}_2$  and  $\text{MgCl}_2$ , and 0.13M for  $\text{Na}_2\text{SO}_4$ . The second comprises natural seawater (SW) from the Arabian Gulf, collected from Dammam, Saudi Arabia. The natural seawater sample was treated with UV light and then filtered through  $5 \mu\text{m}$  filter paper. Table 2 lists the compositions of the electrolytes used, including the natural seawater and synthetic formation brine (FMB) typical of oil reservoirs and deep saline aquifers (e.g. Romanuka et al., 2012).

The majority of limestone formations are directly deposited in seawater (Morse, 1986; Morse and Mackenzie, 1990) and the seawater is in equilibrium with both calcite and CO<sub>2</sub> (Stumm and Morgan, 1996). Any factor, such as organic activity, CO<sub>2</sub> partial pressure, or temperature that modifies the equilibrium will result in precipitation or dissolution of calcite. Carbonate minerals are soluble in water and dissolution yields carbonate ions (CO<sub>3</sub><sup>2-</sup>) that can react to form bicarbonate (HCO<sub>3</sub><sup>-</sup>) and hydroxide (OH<sup>-</sup>) ions according to the equilibrium reaction



If the system is open, atmospheric CO<sub>2</sub> dissolves into the water, reacting directly with hydroxide to form bicarbonate and hence reducing the pH according to the equilibrium reaction



Equilibrium between calcite and water in the presence of CO<sub>2</sub> is reached when most of the carbonate ions are turned into bicarbonate (Krauskopf, 1989); this corresponds to a minimum aqueous concentration of carbonate and carbonic acid, and a maximum for bicarbonate (Figure 2b). The equilibrium is affected by the total ionic strength, including the concentration of Na and Cl ions that are often found in natural brines. In simple solutions of CaCO<sub>3</sub> exposed to CO<sub>2</sub> at 25°C the equilibrium pH is 8.3-8.4 (Garrels and Christ, 1965, Stumm and Morgan, 1996; Figure 2a).

These conditions of carbonate/water/CO<sub>2</sub> equilibrium were replicated here using the following two-stage equilibration procedure. For DIW-based electrolytes, we began by preparing a NaCl solution of the desired concentration in DIW. In the first stage of the equilibration procedure, this solution was placed in a beaker with offcuts of the Portland limestone, maintaining an air layer in the beaker to provide a source of atmospheric CO<sub>2</sub> but sealing the beaker to prevent

evaporation. Monitoring of the pH (using a Five-Go Mettler-Toledo pH meter with their 3-in-1 pH electrode LE438, implementing where necessary the manufacturer's recommended calibration and correction procedures at high ionic strength) and Ca concentration (described below) confirmed the dissolution of calcite and associated pH changes discussed above (e.g. Figure 2b). The initial increase in pH reflects the formation of hydroxide ions according to the equilibrium reaction (1). The subsequent decrease in pH reflects the formation of bicarbonate according to the equilibrium reaction (2). In this case the final pH of the equilibrated solution was c. 8.2, consistent with the predicted value for a simple open system  $\text{CaCO}_3$  solution (Figure 2a). Dissolution of calcite is demonstrated by the increase in Ca concentration from zero to approximately  $0.001 \pm 0.0001$  M (Figure 2a). The resulting equilibrated NaCl solution was termed NaCl-EQ, and this first stage of equilibration mimics the open-system conditions when the carbonate sediments are first deposited. For the experiments reported below, equilibrated solutions of three different NaCl concentrations (0.05M, 0.5M, and 2M) were prepared. Equilibrium was assumed to have been reached when the measured change in both pH and pCa (where p represents the negative logarithm) was zero within experimental error over a timespan >350 hours (two weeks). The equilibrium pH was found to be consistently  $8.2 \pm 0.2$  for 0.05M and 0.5M NaCl electrolytes, and  $8.3 \pm 0.3$  for 2M NaCl electrolyte. The NaCl-EQ solution was then used directly in zeta potential measurements, or was modified by addition of PDIs. This first stage of the equilibration procedure is essential to ensure equilibrium between calcite, water and atmospheric  $\text{CO}_2$  and prevent calcite dissolution and associated changes in surface charge during measurements of zeta potential.

The core flooding apparatus used to measure the zeta potential in the SPM (described below) is closed to the atmosphere, and the second stage of equilibration prior to measuring the zeta potential was to ensure equilibrium between the electrolyte of interest (NaCl-EQ after the addition of any PDIs to be studied) and the rock sample at the closed-system conditions

pertaining to a rock-brine system at depth. The rock sample was pre-saturated with the selected electrolyte at open-system conditions and then confined in the core holder at closed-system conditions, and the electrolyte was pumped through the sample from the (closed) inlet reservoir to the (closed) outlet reservoir and back again. The repeated flow of the electrolyte through the sample at closed system conditions mimics migration of the electrolyte into the carbonate rock at depth. At regular intervals, the electrical conductivity and pH of the electrolyte in the reservoirs was measured, and equilibrium was assumed to have been reached when the conductivity and pH of the electrolyte in each reservoir differed by <5%. The pH varied over a small range in all experiments and was consistent with reported values for natural brines in carbonate rocks (pH in the range 7-8; Yousef et al., 2012). Addition of Ca or Mg reduced the pH to the range 7.2-8, while addition of SO<sub>4</sub> caused a smaller change, yielding pH in the range 7.9-8.1. The uncertainty in the measured values of pH was always less than  $\pm 0.3$ .

Prior to a given experiment, the rock sample was cleaned in a Soxhlet apparatus with methanol for 48 hours. It was then dried for at least 12 hours in a vacuum oven at 80°C. Then, it was allowed to cool at room temperature for a minimum of 6 hours. This is a standard core sample cleaning procedure used in many previous studies and was used with fresh samples here (e.g., Jaafar et al., 2009). However, for reasons discussed later in the paper, after a series of experiments using electrolytes with elevated PDI concentration, the rock samples were flooded with at least 2 pore-volumes (PV) of deionized water (DIW) prior to the methanol cleaning step, and were then flooded with a further 4 PV of 0.05M NaCl-EQ electrolyte. The conductivity of the effluent electrolyte was measured in order to confirm it was the same as that obtained on the fresh samples using the same electrolyte within a 5% tolerance.

For comparison, the zeta potential of one selected sample was also measured using the EPM method (described below). Off-cuts of fresh Portland Limestone were cleaned for 48 hours in methanol and then crushed using a jaw crusher. A Tema Mill with an agate vessel was then

used to obtain a fine powder of the sample. NaCl-EQ was used to prepare solutions with different Ca content. Suspensions of 0.05g of Portland powder in 50mL (1 wt. %) of the desired electrolyte were prepared and left for a minimum of 1 hour, to allow the fraction of larger suspended particles to settle out of solution. For each sample, the suspension was injected via a syringe into a capillary cell in order to obtain the zeta potential measurement. Care was taken to ensure no air bubbles were left in the cells.

## **Measurement of Zeta Potential**

### ***Streaming Potential Measurement (SPM)***

The zeta potential was measured using the SPM described by Vinogradov et al. (2010). Only a brief summary of the method is provided here. The carbonate core samples were tightly confined within an embedded rubber sleeve in a stainless steel core holder with non-metallic end caps. A syringe pump was used to induce a fluid pressure difference across the sample, causing the electrolyte to flow through the sample from reservoirs connected to each side of the core holder (Figure 3). Synthetic oil was used to translate the induced pressure from the pump to the brine in the inlet reservoir, which maintains closed-system conditions by preventing exposure of the electrolyte to atmosphere. The pump maintains constant rate to high accuracy and flow can be directed in either direction through the sample.

The pressure difference across the sample was measured using a pair of pressure transducers (calibrated Druck PDCR 810 with accuracy 0.1% of measured value, resolution 70 Pa) and the voltage across the sample was measured using non-polarizing Ag/AgCl electrodes and an NI9219 voltmeter (internal impedance  $>1\text{G}\Omega$ , accuracy 0.18%, resolution 50 nV). The noise level of the measurements is dictated by the stability of the electrodes, rather than the performance of the voltmeter. The electrodes were positioned out of the flow path, in an

electrolyte reservoir of a NaCl solution of the same ionic strength as that used in the experiments.

We used the ‘paired-stabilization’ or PS method of Vinogradov et al. (2010) to measure the streaming potential across the sample, in which flow is induced through the sample at the same rate but in opposing directions. The method eliminates the effect of temporal variations in the static voltage and demonstrates that electrode polarization effects are negligible through confirmation that the change in potential induced by flow in one direction is equal and opposite to the change in potential induced by flow in the opposite direction. To ensure that exclusion-diffusion potentials were eliminated during measurements of the streaming potential, uniform and constant electrolyte conductivity and pH in each reservoir, and uniform and constant temperature (23°C), were maintained within a 5% tolerance. Redox potentials, which may affect the measured voltage if metals such as steel are in contact with the saline electrolyte, were eliminated by electrically isolating all metallic components from the electrolyte except for the Ag/AgCl electrodes.

Interpretation of the results from the PS experiments follows from the observation that at steady-state, the streaming current induced by the flow is balanced by a conduction current to maintain overall electrical neutrality. It is reasonable to assume that the currents follow approximately the same 1-D path along the samples, in which case the streaming potential coupling coefficient can be determined using

$$C_{SPM} = \frac{\Delta V}{\Delta P} \quad (3)$$

where  $\Delta V$  and  $\Delta P$  are the stabilized voltage and pressure measured across the plug, respectively. The coupling coefficient is given by the slope of a linear regression through a plot of voltage against pressure difference obtained for a number of different flow rates (e.g. Figure 4c, d). An effective value for the zeta potential for the sample was obtained using a

modified version of the Helmholtz-Smoluchowski equation that accounts for surface electrical conductivity (e.g. Jackson, 2015)

$$\zeta = \frac{C_{SPM}\mu\sigma_{rw}F}{\varepsilon} \quad (4)$$

where  $F$  is the formation factor, which is the ratio of the conductivity of the electrolyte to the conductivity of the saturated rock sample when surface conductivity is negligible (e.g. Jouniaux and Pozzi, 1995),  $\varepsilon$  is the permittivity of the electrolyte,  $\mu$  is the electrolyte viscosity and  $\sigma_{rw}$  is the electrical conductivity of the saturated rock sample. The formation factor and electrical conductivity were measured following the methodology of Vinogradov et al. (2010) (Table 1). Note that the zeta potential obtained is an effective value because it reflects the average streaming charge density transported by the flow of the electrolyte; at the pore-level, the zeta potential may vary. The viscosity and permittivity of the electrolyte as a function of ionic strength were also determined using the approach of Vinogradov et al. (2010). Uncertainty in the reported value of zeta potential reflects the range of possible regressions that can be fitted to the measured streaming potential data within experimental error (Figure 4).

### *Electrophoretic Mobility Measurement (EPM)*

The zeta potential for one powdered sample in suspension was also obtained for comparison with the SPM using a Brookhaven ZetaPALS zetameter to measure the electrophoretic mobility  $u_e$  of the suspension; this is related to the zeta (shear plane) potential using the Helmholtz-Smoluchowski equation for electrophoresis (Delgado et al, 2007):

$$\zeta = \frac{u_e\mu}{\varepsilon} \quad (5)$$

As noted above, the zeta potential obtained is an effective value because it reflects the average surface charge on the particles in suspension; at the particle level, the zeta potential may vary. The measurement of each sample consisted of 5 runs; each run consisted of 10 cycles. The

mean of all the runs for each sample is reported as the zeta potential and the error bars represent the standard deviation.

### **Measurement of Electrolyte Composition**

Electrolyte composition was determined using inductively coupled plasma atomic emission spectroscopy (ICP-AES). The analysis was carried out in the Analytical Chemistry Laboratory at the Natural History Museum, London.

Electrolyte samples from the SPM measurements were collected from the core holder via a valve on the outlet flow line at the end of a given suite of zeta potential measurements for the chosen electrolyte; each effluent sample had therefore interacted with the rock sample for a minimum volume of 10 PV spread over a minimum of two days. These samples are referred to as the final effluent electrolytes. Appropriate dilutions were prepared for each sample prior to analysis depending on the total ionic strength and relative abundance of the PDIs of interest. All samples were acidified with 2%  $\text{HNO}_3$  to prevent formation of complexes that might affect the interpreted concentrations.

Reference standard solutions at concentrations ranging from 0.5 - 200 ppm containing all the ions of interest (Na, Ca, Mg, and S) were prepared to represent the ion matrix of the effluent samples. The accuracy of the method was determined using certified check solutions and the repeatability by conducting 5 repeat measurements on all the samples whose standard deviation is represented by the error bars.

### **Design of Experiments**

In this work, we investigated the effect of three key PDIs (Ca, Mg and  $\text{SO}_4$ ) on the zeta potential of natural limestone in two ways. The first approach was to systematically vary the concentration of each PDI over the range found in natural brines to establish its effect on the zeta potential. For each range of PDI concentrations, we tested three different NaCl (0.05M,

0.5M and 2M) concentrations to determine whether this changes the relationship between the PDI concentration and surface charge. The 0.5M NaCl concentration represents seawater and is similar to the ‘ZP brine’ of Zhang and Austad (2006) and Zhang et al. (2007) which contained 0.573M NaCl, allowing direct comparison of results. The 0.05M NaCl concentration represents a tenfold dilution of seawater and approximates the injection brine used in controlled salinity waterflooding (CSW) for enhanced oil recovery (Yousef et al, 2010), while the 2M NaCl concentration represents the saline brines found in many deep saline aquifers. The second approach was to combine all three PDIs in the proportions and total concentration typical of (i) natural saline brines, and (ii) natural seawater, and compositions derived from seawater similar to those used in CSW.

## Results

### Measurements of streaming potential and interpretation of zeta potential

Figure 4(a, b) shows typical results for the PS experiments for low and high ionic strength electrolytes respectively. The pressure response to pumping is clear and the pressure difference across the samples reached a stable value (fluctuations <500 Pa around an induced pressure difference of c. 500kPa) in all experiments. The voltage response is also clear and reached a stable value with fluctuations typically below  $\pm 5\mu\text{V}$  at high ionic strength (e.g., FMB) and below  $\pm 50\mu\text{V}$  at low ionic strength (<0.5M NaCl) in all experiments. The interpreted values of stabilized pressure and voltage are denoted by the dashed lines, while the error bars show the interpreted spread. The stabilized voltage was reproducible within  $\pm 25\mu\text{V}$  across three repeat experiments at a given flow rate for high ionic strength and  $\pm 35\mu\text{V}$  for low ionic strength. The voltage fluctuations, and reproducibility of the stabilized voltage measurements, are similar to previous experiments conducted on limestone samples saturated with electrolytes of similar

ionic strength (Jackson and Vinogradov, 2012). An important aspect of the SPM is that the polarity of the surface charge is very clear: if the polarity of the voltage response is in the opposite sense to the pressure response (i.e. a more positive pressure difference yields a more negative voltage difference relative to a common reference pressure and voltage at one end of the sample) then the surfaces are negatively charged, and vice-versa. This allows the isoelectric point (IEP) to be accurately determined even when the zeta potential is close to zero.

Figure 4 (c, d) shows typical plots of the stabilized voltage plotted against the corresponding stabilized pressure difference from each pair of PS experiments for the same electrolytes shown in Figs (a, b) respectively. The error bars represent the reproducibility of (typically) three repeat measurements at each flow rate. The streaming potential coupling coefficient, obtained from a linear regression through the measured data (equation 3), is clearly negative in Fig 4c and positive in Fig. 4d and the linear regression is well constrained by the relatively small error bars associated with each value of stabilized voltage (Fig. 4a,b). We calculate the associated zeta potential using equation (4). The uncertainty in the streaming potential coupling coefficient arising from the range of linear regressions that can be forced through the stabilized voltage and pressure data was used to determine the associated uncertainty in zeta potential reported in the following sections.

### **Impact of Ca, Mg and SO<sub>4</sub> concentration on zeta potential**

We begin by reporting experiments in which the concentration of each PDI was systematically varied in pre-equilibrated 0.05M NaCl electrolytes (NaCl-EQ). Figure 5 shows the zeta potential as a function of calcium, magnesium and sulfate concentration. We plot concentration as pPDI. Note that in all cases the lowest concentration (highest pPDI) investigated corresponds to the equilibrated concentration in the NaCl-EQ electrolyte. We notice first that a linear regression provides an excellent fit to the data for each PDI ( $R^2 > 0.98$ ) and that the gradient of the regression for Ca and Mg is identical within experimental error ( $-5.10 \pm 0.47$  mV/decade).

Moreover, the zeta potential is negative at high pCa or pMg (i.e. low Ca or Mg concentration), becomes less negative with decreasing pCa or pMg, and becomes positive at low pCa or pMg. The IEP (defined as pPDI) is identical for Ca and Mg ( $\text{pPDI} = 0.60 \pm 0.03$ ) within experimental error. However, the behaviour of  $\text{SO}_4$  is very different. The zeta potential remains negative regardless of p $\text{SO}_4$  and becomes increasingly negative with decreasing p $\text{SO}_4$  (i.e. increasing  $\text{SO}_4$  concentration). Moreover, the gradient of the linear regression that best fits the data is much smaller than that observed for Ca and Mg ( $1.9 \pm 0.3$  mV/decade). These results suggest that Ca and Mg behave almost identically as PDIs at room temperature and can have a significant impact on zeta potential, yielding positive zeta potential at  $\text{pPDI} < 0.60$ . However, the zeta potential is much less sensitive to p $\text{SO}_4$ .

### Impact of varying the concentration of NaCl

Figure 6 shows the zeta potential as a function of Ca concentration for each of the three NaCl concentrations investigated (Figure 6a), and as a function of  $\text{SO}_4$  concentration for two of the NaCl concentrations investigated (Figure 6b). Considering first the impact of Ca concentration, we again find that a linear regression provides an excellent fit to the data for each value of NaCl concentration ( $R^2 > 0.98$ ) and that the gradient of the linear regression decreases with increasing NaCl concentration (Figure 6c). Thus, the zeta potential becomes less sensitive to pCa as the NaCl concentration increases. In all cases, the zeta potential is negative at high pCa (i.e. low Ca concentration), becomes less negative with decreasing pCa, and becomes positive at low pCa. The IEP (defined as pCa) decreases with increasing NaCl concentration although the change only exceeds experimental error for the lowest NaCl concentration investigated (Figure 6c). Considering next the impact of  $\text{SO}_4$  concentration, we observe similar behaviour. A linear regression again provides an excellent fit to the data, and the gradient of the regression decreases with increasing NaCl concentration (Figure 6b). However, the zeta potential remains negative over the range of p $\text{SO}_4$  investigated.

### Effect of varying multiple PDIs

In this section, we report measurements of zeta potential using electrolytes containing all three PDIs (Ca, Mg, and  $\text{SO}_4$ ) at the concentrations found in typical formation brine (FMB; Table 2) and seawater (SW; Table 2). The formation brine yields a positive zeta potential, which is the same within experimental error as the zeta potential obtained by adding a comparable amount of Ca to 2M NaCl electrolyte (see the filled square in Fig. 5). The natural seawater yields a negative zeta potential, which is more negative than the zeta potential obtained by adding a comparable amount of Ca to 0.05M NaCl electrolyte (see the open square in Fig. 5). Thus the zeta potential in subsurface saline brine appears to be controlled primarily by the Ca content, with Mg and  $\text{SO}_4$  playing a minor role; in contrast, the presence of  $\text{SO}_4$  in seawater leads to a more negative zeta potential.

We also investigate the effect of diluting seawater and adding  $\text{SO}_4$  to seawater. Both of these approaches to modifying the brine injected into carbonate oil reservoirs have been suggested to yield enhanced oil recovery (Zhang and Austad, 2006; Yousef et al. 2011). In the experiments conducted here, seawater (SW) was diluted twice (1/2SW), ten times (1/10SW) and twenty times (1/20SW), and  $\text{SO}_4$  was added to yield twice (2SW), three times (3SW) and four times (4SW) the natural seawater concentration. In all cases, the measured zeta potential is negative (Figure 7a); however, the least negative (or smallest in magnitude) zeta potential is observed for seawater, and the zeta potential becomes increasingly negative (and larger in magnitude) as the seawater is diluted or  $\text{SO}_4$  is added. The zeta potential increases in magnitude with both increasing and decreasing total ionic strength (Figure 7b); the ionic strength increases as  $\text{SO}_4$  is added, but decreases as the seawater is diluted.

### Effect of sample preparation

Many experimental studies use a limited number of samples that are cleaned before each experiment. However, none have confirmed that the typical laboratory cleaning protocol

(described here in the methodology) restores the zeta potential of natural carbonates to a consistent and repeatable value for a given electrolyte. To confirm the repeatability of zeta potential measurements obtained using the SPM, and determine the effect of sample cleaning, the zeta potential for three selected fresh samples was initially measured using 0.05M NaCl-EQ electrolyte (circles in Fig. 8). The samples were then used in experiments in which the Ca or Mg concentration was increased (triangles in Fig. 8; these data are also shown in Fig. 5). The samples were then cleaned using a standard laboratory cleaning protocol and the zeta potential was measured again (diamonds in Fig. 8). Finally the samples were cleaned using the enhanced cleaning protocol reported here (squares in Fig. 8). It is clear that the standard cleaning procedure fails to return pMe (representing the Ca + Mg concentration) or zeta potential to their original fresh values after the samples are exposed to elevated PDI concentrations. It is important to use the enhanced cleaning procedure reported here to flush PDIs from the mineral surfaces and return the zeta potential to its pristine value.

## Discussion

### Comparison with previous studies of the effect of PDI concentration on zeta potential in natural and synthetic calcite/carbonates

We have demonstrated here that Ca and Mg change the zeta potential of intact natural limestone samples, causing a linear decrease in the magnitude of the negative zeta potential with increasing concentration (expressed as pPDI), and causing polarity inversion to positive zeta potential at high concentration; moreover, the two PDIs behave identically within experimental error. Similarly,  $\text{SO}_4$  changes the zeta potential of natural limestone, causing a linear increase in the magnitude of the negative zeta potential with increasing concentration (expressed as pPDI), but the gradient of the linear regression that best fits the data is lower than that of the cations. We have also demonstrated that the gradient of the zeta potential with respect to pCa

and  $p\text{SO}_4$  decreases with increasing NaCl concentration. The relationship between zeta potential and  $p\text{PDI}$  is linear across the entire range of  $p\text{PDI}$  investigated.

No previous studies have determined the relationship between zeta potential and  $p\text{Mg}$ , but several have reported a linear relationship between zeta potential (or its proxy, electrophoretic mobility) and  $p\text{Ca}$  as observed here (e.g. Foxall et al., 1979; Thompson and Pownall, 1989; Pierre et al. 1990). However, these studies were conducted using electrolytes of much lower ionic strength than those considered here (e.g. Fig. 9a). Other studies have observed a non-linear relationship between zeta potential and  $p\text{Ca}$  (e.g. Cicerone et al. 1992; Chen et al., 2014). Linear behaviour is expected if the calcite surface behaviour is Nernstian and the lattice ions  $\text{Ca}$  and  $\text{CO}_3$  are the PDIs, and the electrical double layer is described by the Gouy-Chapman-Grahame model (e.g. Hunter, 1981). Under these circumstances, the gradient of the zeta potential with respect to  $p\text{PDI}$  can be expressed as (e.g. Foxall et al., 1979)

$$\left. \frac{d\zeta}{dp\text{PDI}} \right|_{\zeta \rightarrow 0} = \frac{-2.303 \frac{kT}{ze}}{\left(1 + \frac{C_d}{C_s}\right) \exp(\kappa\Delta)} \quad (5)$$

where  $k$  is Boltzmann's constant,  $T$  is the temperature,  $z$  is the valence of the PDI,  $e$  is the charge on an electron,  $C_d$  and  $C_s$  are the capacitance per unit area of the diffuse and Stern layers respectively,  $\kappa$  is the inverse Debye length, and  $\Delta$  is the distance of the shear plane from the Stern plane. For low zeta potential,  $C_d$  is given approximately by  $\kappa\epsilon$  where  $\epsilon$  is the permittivity. Cicerone et al. (1992) argued that the relationship between zeta potential and  $p\text{PDI}$  is linear only close to the IEP; away from the IEP, zeta potential values level off, because the Stern layer capacitance  $C_s$  varies, or because the Gouy-Chapman-Grahame model breaks down. We

do not observe this levelling off, despite the broad range of pCa values investigated. Equation 5 can be used to fit our experimental data for pCa (and pMg). However, the decrease in gradient with increasing NaCl concentration can only be matched by adjusting the Stern capacitance (see Table 3; these values are discussed in more detail in the next section). Large values of Stern capacitance are required in the range  $1.13\text{--}2.75 \text{ Fm}^{-2}$ , which are at least twice those determined previously, but these values were obtained at considerably lower ionic strength (Foxall et al., 1979; Thompson and Pownall, 1989; Cicerone et al., 1992). For the 0.05M NaCl electrolyte (the lowest concentration investigated), the predicted diffuse layer thickness at the ionic strength corresponding to the IEP (0.8M) is very small (the Debye length is 0.342nm). Given that the calcium ion has a hydrated diameter of 0.59nm (Diebler et al, 1969), it is not clear whether such a diffuse layer thickness is physically meaningful as it cannot accommodate even a single calcium ion. Vinogradov et al. (2010) suggested that the diffuse layer thickness decreases until it reaches the radius of the hydrated counter-ion, and then remains constant regardless of increasing ionic strength. However, their model does not account for changes in the Stern layer capacitance with changing ionic strength, and cannot explain the data reported here.

Figure 9b shows the effect of varying  $\text{SO}_4$  concentration, comparing our data obtained for the 0.5M NaCl electrolyte against that of Zhang and Austad (2006). These are the only comparable data for  $\text{SO}_4$  reported previously. Both datasets yield a linear relationship between zeta potential and p $\text{SO}_4$ , although the gradient of the linear regression is smaller for the Zhang and Austad data than that obtained here. As discussed in the next section, we suggest this is a consequence of the differing measurement methods: Zhang and Austad used the EPM, in contrast to the SPM used here. Moreover, extrapolating the linear regression in each case to obtain the IEP suggests very different values in terms of p $\text{SO}_4$ .

In the single PDI experiments reported here (Figs. 5, 6), precipitation of salts such as  $\text{CaSO}_4$  and  $\text{MgCO}_3$  in the pore-space was prevented because each PDI (Ca, Mg or  $\text{SO}_4$ ) was added to NaCl-EQ electrolyte containing only trace or zero concentration of cations or anions other than Na and Cl. Moreover, in the experiments utilising seawater (SW) and formation brine (FMB), we saw no evidence that these salts were deposited. There was no decrease in the concentration of Ca, Mg and  $\text{SO}_4$  ions in the equilibrated electrolyte within experimental error, inconsistent with the deposition of salts containing these ions. Indeed, when adding  $\text{SO}_4$  we observed a small increase in Ca concentration during equilibration. We also observed no decrease in permeability, or increase in sample mass, within experimental error, and no solids were observed in the effluent electrolyte. Moreover, we note that the zeta potential obtained using FMB was the same within experimental error as the zeta potential obtained by adding a comparable amount of Ca to 2M NaCl electrolyte, suggesting the presence of Ca, Mg and  $\text{SO}_4$  did not cause any compositional change at the mineral surfaces.

### Effect of electrokinetic measuring technique

A common difference between our data and that reported in previous studies is that we use the SPM to obtain the zeta potential, whereas previous studies have primarily used the EPM. Several studies have suggested that the two methods may yield different results (e.g. Vernhet et al, 1994; Delgado et al., 2007). To test this, we compare zeta potential measurements obtained using both methods on the Portland Limestone, varying pCa in 0.05M NaCl electrolyte (Figure 10). We find that the IEP is identical within experimental error, although uncertainty in the IEP derived from the EPM data is significantly greater than for the SPM data, because positive and negative values of zeta potential were observed across a range of pCa (0.71-0.50). There was no such ambiguity in the SPM data.

Both methods also yield a linear relationship between zeta potential and pCa, although the gradient of the linear regression obtained from the EPM data is twice that obtained from the SPM data ( $-10.45 \pm 0.55$  mV/decade for the EPM versus  $-5.10 \pm 0.47$  mV/decade for the SPM). We fit the EPM data using equation 5 and the values reported in Table 3, assuming  $\Delta = 0$  (i.e. assuming the shear plane corresponds with the Stern plane) in common with previous studies using the EPM on calcite (Foxall et al., 1979; Thompson and Pownall, 1989; Pierre et al. 1990; Cicerone et al., 1992). We then fit our SPM data using the same parameters, but adjusting  $\Delta$  to obtain a match, yielding a value of 0.245nm. This is a very small offset for the shear plane, and reflects the very small thickness of the diffuse layer at the IEP as discussed in the previous section. Nonetheless, the difference in gradient is consistent with that expected when there are differences in the relative position of the shear plane in natural porous media and powder suspensions. The complex geometry of natural pore-spaces, including the presence of sharp-angled corners and crevices, means that the effective location of the shear plane lies further from the mineral surface than in powder suspensions. SPM measurements are more relevant when quantifying the zeta potential of natural samples, because the measurements reflect the mineral surfaces that predominantly interact with the adjacent fluids.

### Effect of NaCl concentration on the IEP

No previous studies have determined the IEP for natural and artificial calcite expressed as pMg, but several have reported values of the IEP expressed as pCa (Table 4). The values observed are typically much higher (i.e. the IEP was observed at lower calcium concentration) than those determined here. Only Chen et al. (2014) have observed the IEP at a comparably low value of pCa; they investigated natural limestone consistent with our study, but employed the EPM method and DIW electrolyte, rather than the SPM and NaCl electrolytes used here. It is not

clear why the IEP for natural Portland limestone occurs at such low values of pCa compared to the majority of previous studies. Pierre et al. (1990) suggested that the IEP is governed by the relative magnitude of the equilibrium constants  $K_{Ca}$  and  $K_{CO_3}$  governing the adsorption of Ca and  $CO_3$  ions on the calcite mineral surface. The IEP shifts to lower pCa if  $K_{CO_3} > K_{Ca}$ ; that is, if the calcite surfaces show greater affinity for  $CO_3$  than Ca. Pierre et al. (1990) found the IEP differed for synthetic and natural calcite and argued that this reflected the differing affinity for Ca and  $CO_3$ .

The Pierre et al. model suggests that the natural Portland limestone investigated here has a much greater affinity for  $CO_3$  than Ca. Thus, the difference may be related to sample type: most previous studies used synthetic calcite or natural chalk, rather than the natural limestone used here. It may also be related to the pH and/or the establishment of the initial equilibrium conditions. Thompson and Pownall (1989) and Cicerone et al. (1992) conducted experiments over the pH range 7-11 and 8.5-10.5 respectively; the higher pH values do not represent equilibrium conditions. Zhang et al. (2006) and Chen et al. (2014) kept the pH fixed at 8.4 and 8 respectively, but do not report the pre-equilibration steps used here. The pH was fixed in our experiments by the procedure used to ensure the sample was in equilibrium with the electrolyte prior to starting the experimental measurements.

We have also found that the IEP for Portland limestone decreases with increasing NaCl concentration over the range 0.05M – 0.5M. Previous studies have argued that the IEP is independent of NaCl concentration, as Na and Cl are indifferent ions to the calcite surface (e.g. Pierre et al., 1990). We suggest that the difference in IEP between the 0.05M and 0.5M/2M NaCl electrolytes observed here is due to the reduced ability of the calcium ions to interact with the calcite surface, owing to (i) the collapse of the double layer and (ii) increasing occupancy of the diffuse part of the double layer by hydrated sodium ions, which have a smaller radius than the calcium ions at 0.47nm (Vinogradov et al., 2010). However, we note this

hypothesis fails to explain the data of Chen et al. (2014), as they observed a comparable IEP to ours at much lower NaCl concentration.

### **Implications for controlled salinity waterflooding (CSW)**

We have shown that the zeta potential of intact natural limestone samples is positive at elevated Ca and Mg concentration below the IEP ( $pCa = pMg = 0.63 - 0.41$  as discussed above) and becomes negative as the Ca and or Mg concentration is decreased; it also becomes increasingly negative as the  $SO_4$  concentration is increased. We have also shown that the zeta potential of natural limestone saturated with formation brine, rich in Ca ions, is positive, consistent with previous studies (Jackson and Vinogradov, 2012; Chen et al., 2014; Mahani et al., 2015; see Figure 9a). In such formations, an attractive electrostatic force will act between the positively charged mineral surfaces and the negatively charged oil-brine interface, promoting wettability alteration to oil-wet conditions (e.g. Buckley et al., 1989). However, if the concentration of Ca or Mg in the injection brine during controlled salinity waterflooding (CSW) is decreased below the IEP, the zeta potential changes polarity to negative leading to electrostatic repulsion, which may lead to wettability alteration to more water-wet conditions, releasing previously adsorbed crude oil from the calcite mineral surfaces and therefore improving oil recovery. It has been shown by Jackson and Vinogradov (2012) that more water-wet conditions in natural carbonate samples correlate with a more positive zeta potential.

Previous reported values of the IEP expressed as  $pCa$  suggest that considerable reduction in Ca concentration is required to change the polarity of calcite (Table 4; see also Fig. 9a); however, our results suggest that reducing the concentration of Ca in the injection brine (selectively or by bulk dilution) by a factor of only 2 relative to the formation brine can lead to inversion of the surface charge. Injection of seawater will also cause inversion of the calcite surface charge, because of the lower Ca concentration and higher  $SO_4$  concentration. This can explain why improved recovery in carbonates during CSW has been observed in response to relatively

minor levels of injection brine dilution, compared to sandstones in which improved recovery is only observed for very low salinity injection brines ( $<0.05\text{M}$ ; see Jackson et al., 2015 for a review).

Previous studies have also suggested that improved oil recovery in corefloods or spontaneous imbibition (SI) experiments can be observed by either diluting seawater as the injection fluid (Yousef et al., 2011), or adding  $\text{SO}_4$  to seawater as the imbibing fluid (Zhang and Austad, 2006). In one case, the total ionic strength is simply decreased; in the other, the ionic strength is increased but the relative concentration of ions is changed. Here we show the change in zeta potential is almost identical; diluting seawater and adding  $\text{SO}_4$  causes the negative zeta potential to increase in magnitude i.e. become more negative (Figure 7). As discussed above, this can cause wettability alteration to more water-wet conditions and release previously trapped oil in coreflooding experiments, or cause increased imbibition in SI experiments. Simple dilution causes expansion of the double layer and hence a more negative zeta potential (Ligthelm et al, 2009; Nasralla and Nasr-El-Din, 2014); addition of  $\text{SO}_4$  yields a more negative zeta potential by increasing the negative charge on the calcite mineral surface (e.g. Fig. 5). Figure 11 shows the incremental recovery observed by diluting seawater, or adding  $\text{SO}_4$  to seawater, in the experiments reported by Yousef et al. (2011) and Zhang and Austad (2006), plotted against the change in zeta potential we observed here by modifying the composition of seawater in the same way. There is a clear correlation between increasingly negative zeta potential change and improved recovery, irrespective of how the seawater composition is changed. We suggest that a similar change in zeta potential occurred during the experiments of Yousef et al. (2011) and Zhang and Austad (2006), but was unrecorded because the zeta potential was not measured. However, we note that the samples and experimental methods and conditions are inconsistent across the three sets of experimental results correlated in Figure 11. Future work relevant to CSW should focus on testing the link between brine composition, zeta

potential and increased oil recovery using integrated experiments with consistent materials and experimental conditions.

One final point relevant to CSW relates to the repeatability of laboratory coreflooding experiments. In many studies, a small number of samples are used repeatedly and are cleaned in between experiments. The cleaning protocol typically focuses on ensuring that crude oil is removed from the pore-space. However, we show here that standard cleaning protocols does not restore the zeta potential to its pristine state. This may impact on how the surfaces interact with PDIs in the aqueous phase and polar species in the oil phase, during aging and subsequent waterfloods. If the zeta potential is not returned to its pristine state then the experiments may not be repeatable. We recommend the zeta potential is measured on intact samples before, during and after controlled salinity waterflooding experiments to constrain the behaviour of this key surface property.

## Conclusions

We report here measurements of the zeta potential on intact Portland limestone obtained primarily using the streaming potential method (SPM), supplemented by a smaller number of measurements of the more widely applied electrophoretic mobility method (EPM). The experiments were designed to determine how the zeta potential is affected by the concentration of Ca, Mg and  $\text{SO}_4$  over the range found in natural brines, and also how the zeta potential is affected by the concentration of these potential-determining ions in the presence of Na and Cl over the range found in natural brines. Our approach contrasts with many previous studies because the experimental method is specifically designed to ensure the equilibrium achieved between rock and electrolyte is consistent with natural processes. The results are directly applicable to a wide variety of natural systems including carbonate oil reservoirs and deep saline aquifers. The key findings can be summarized as follows:

- Ca and Mg change the zeta potential of intact natural limestone samples, causing a decrease in magnitude of the negative zeta potential with increasing concentration and causing polarity inversion to positive zeta potential at high concentration. We show for the first time that the two PDIs behave identically within experimental error, and the zeta potential varies linearly with both pCa and pMg over the broad range found in natural brines.
- $\text{SO}_4$  changes the zeta potential of natural limestone, causing an increase in the magnitude of the negative zeta potential with increasing concentration, and the zeta potential varies linearly with  $\text{pSO}_4$  over the broad range found in natural brines. However, the gradient of the linear regression is lower than for Ca and Mg.
- We show for the first time that the IEP (expressed as pCa or pMg) decreases with increasing NaCl concentration. We report considerably lower values of IEP than most previous studies of calcite and chalk, and suggest that this may result from differences in the mineral surfaces (synthetic and natural calcite, natural chalk) compared to the natural limestone investigated here, and the careful method used to establish the initial equilibrium conditions between sample and electrolyte. We recommend this method in all studies of natural carbonates.
- We show for the first time that the IEP (expressed as pCa) obtained using SPM and EPM measurements on the same Portland Limestone are identical within experimental error, but the error is much larger for the EPM method. Both methods show a linear relationship between zeta potential and pCa, but the gradient is a factor of two larger for the EPM method, consistent with a change in the location of the shear plane. SPM measurements are more relevant when quantifying the zeta potential of natural porous samples, because the measurements reflect the mineral surfaces that predominantly interact with the adjacent fluids.

- Standard laboratory cleaning protocols do not return carbonate mineral surfaces to a repeatable ‘pristine’ state, which may affect the repeatability of subsequent experiments on the same sample, including the coreflooding/spontaneous imbibition experiments used to investigate controlled salinity waterflooding.
- Changes in wettability and oil recovery during controlled salinity waterflooding are consistent with the changes in zeta potential observed here. Carbonates saturated with formation brine rich in Ca are likely to have positively charged mineral surfaces (electrostatic attraction), encouraging wettability alteration to oil-wet conditions. Injecting seawater or diluted formation brine can reduce the Ca and/or Mg concentration below the IEP; note that the lower IEP observed here suggests that much less dilution is required than predicted previously. This yields negatively-charged mineral surfaces (electrostatic repulsion), increasing recovery by releasing previously trapped oil. Diluting seawater, or adding  $\text{SO}_4$ , both yield increasingly negative zeta potential, consistent with experimental studies that report improved recovery in both cases.

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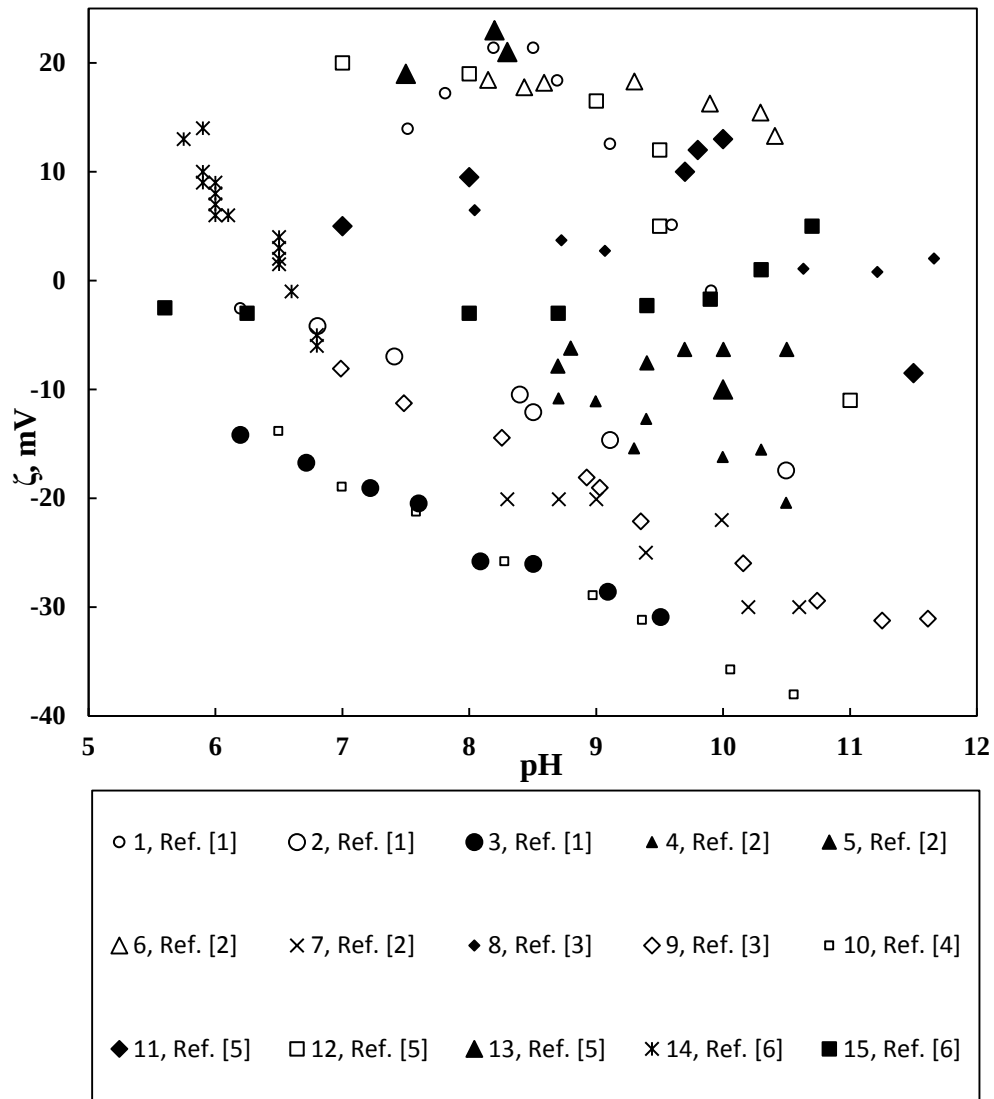
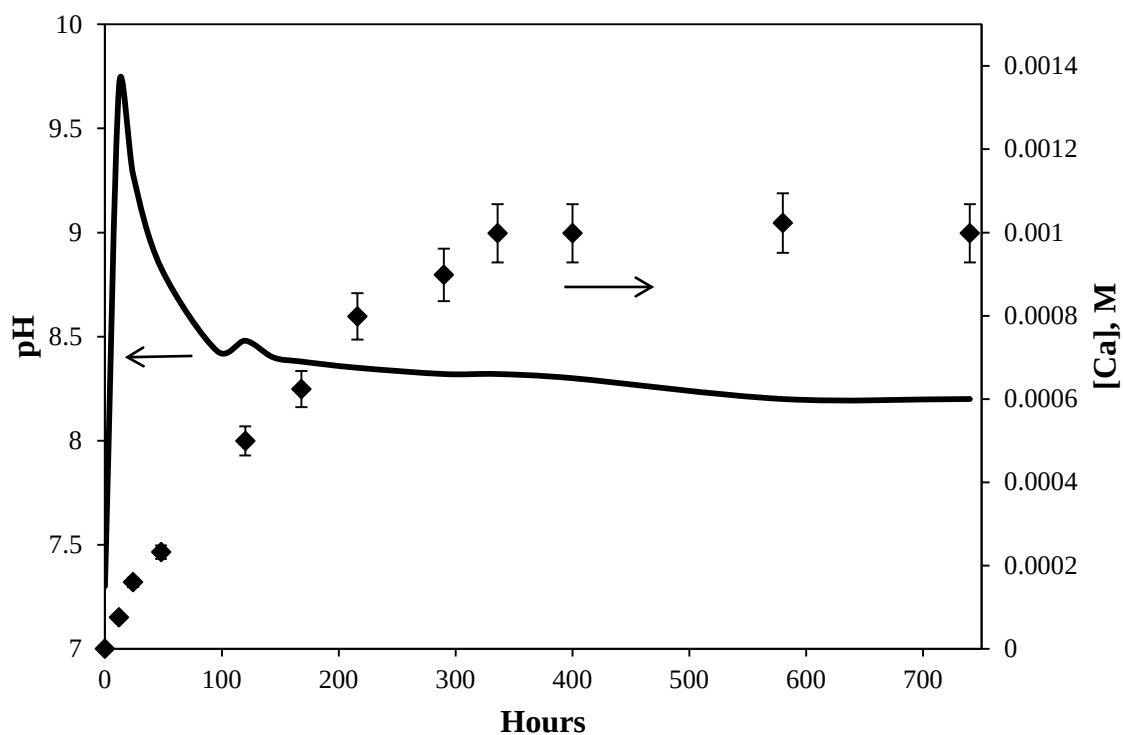


Figure 1. Zeta potential as a function of pH reported on various artificial and natural calcite and limestone samples for various electrolyte compositions and ionic strength. Unless otherwise stated, measurements were obtained using the electrophoretic mobility method (EPM). Vdovic (2001) (Ref. 1) used synthetic calcite (labelled 1), natural limestone (2), and lake sediments (3) in  $10^{-3}$ M NaCl electrolyte. Cicerone et al. (1992) (Ref. 2) used synthetic calcite in 0.03M KCl (4), 0.001M  $\text{CaCl}_2$  (5) and 0.01M  $\text{CaCl}_2$  (6) electrolytes, and natural calcite in 0.03M KCl electrolyte (7). Thompson and Pownall (1989) (Ref. 3) used the streaming potential method (SPM) on synthetic calcite in  $5 \times 10^{-4}$ M  $\text{CaCl}_2$  (8) and 0.005M NaCl (9) electrolytes. They did not correct for surface electrical conductivity. Sondi et al. (2009)

(Ref. 4) used natural calcite in 0.001M NaCl electrolyte (10). Somasundaran and Agar (1967) (Ref. 5) used the SPM on calcite in deionized water after no mixing (11), mixing for one week (12), and mixing for two months (13). They did not correct for surface electrical conductivity. Heberling et al. (2011) (Ref. 6) used the SPM on calcite in non-equilibrated 0.01M NaCl and 0.005M  $\text{CaCl}_2$  electrolytes (14), and the EPM on calcite in 0.1M NaCl in equilibrium with  $p(\text{CO}_2)=1$  bar (15). They did not correct for surface electrical conductivity.

(a)



(b)

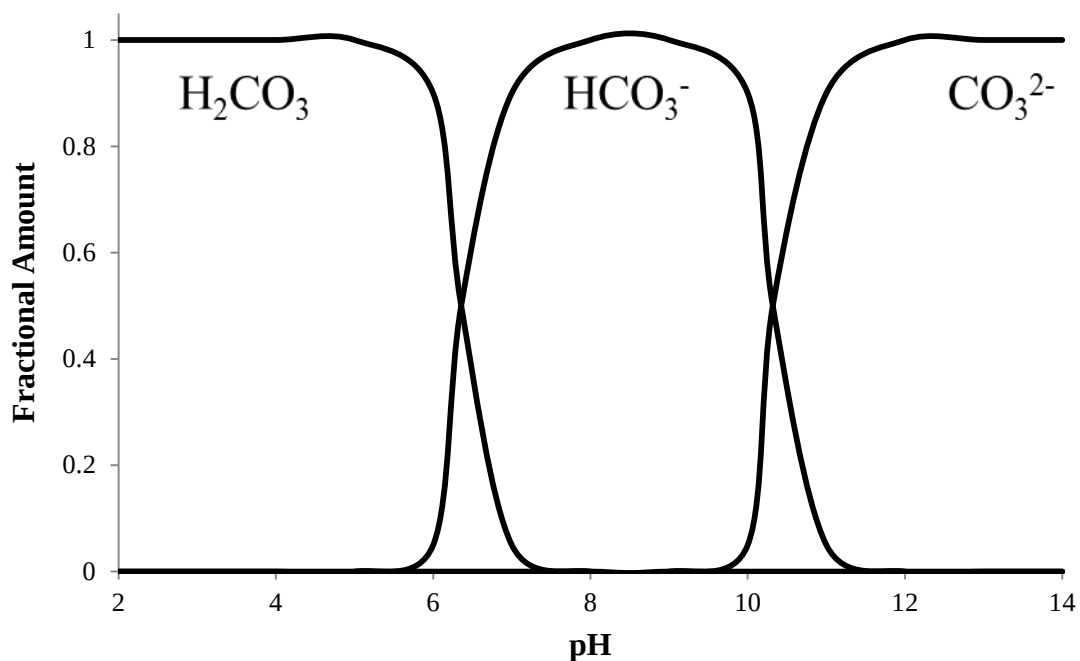


Figure 2. Calcite-water-CO<sub>2</sub> equilibrium for  $I = 0.05$  M NaCl. (a) Calcium concentration and pH measured here as a function of time during equilibration of the natural Portland rock samples with DIW. (b) Carbon speciation into H<sub>2</sub>CO<sub>3</sub>, HCO<sub>3</sub><sup>-</sup>, and CO<sub>3</sub><sup>2-</sup> as a function of pH (modified after Stumm and Morgan, 1996).

Table 1. Properties of Portland rock samples used in this study.

| Sample | Porosity (%) | Permeability (mD) | Formation Factor (F) |
|--------|--------------|-------------------|----------------------|
| P1     | 20           | 3                 | 21.3                 |
| P2     | 19.5         | 2.2               | 22.4                 |
| P3     | 21           | 3.5               | 20.6                 |

Table 2. Composition of the synthetic Formation Brine (FMB) and natural seawater (SW) and derived compositions used in this study. The seawater was twice ( $1/2$ SW), ten times ( $1/10$ SW), and twenty times ( $1/20$ SW) diluted, and also had  $\text{SO}_4$  added to yield twice (2SW), three times (3SW), and four times (4SW) the natural concentration.

| <b>Concentration<br/>(M)</b> | <b>FMB</b> | <b>SW</b> | <b>1/2SW</b> | <b>1/10SW</b> | <b>1/20SW</b> | <b>2SW</b> | <b>3SW</b> | <b>4SW</b> |
|------------------------------|------------|-----------|--------------|---------------|---------------|------------|------------|------------|
| <b>Na</b>                    | 2          | 0.5       | 0.25         | 0.05          | 0.025         | 0.5        | 0.5        | 0.5        |
| <b>Ca</b>                    | 0.42       | 0.012     | 0.006        | 0.0012        | 0.0006        | 0.012      | 0.012      | 0.012      |
| <b>Mg</b>                    | 0.07       | 0.05      | 0.025        | 0.007         | 0.00025       | 0.05       | 0.05       | 0.05       |
| <b>SO<sub>4</sub></b>        | 0.0033     | 0.033     | 0.016        | 0.0033        | 0.0016        | 0.066      | 0.099      | 0.13       |
| <b>pH</b>                    | 7.15       | 8         | 8            | 8             | 8             | 8          | 8          | 8          |
| <b>Total</b>                 | 2.49       | 0.615     | 0.107        | 0.061         | 0.0107        | 0.648      | 0.681      | 0.715      |

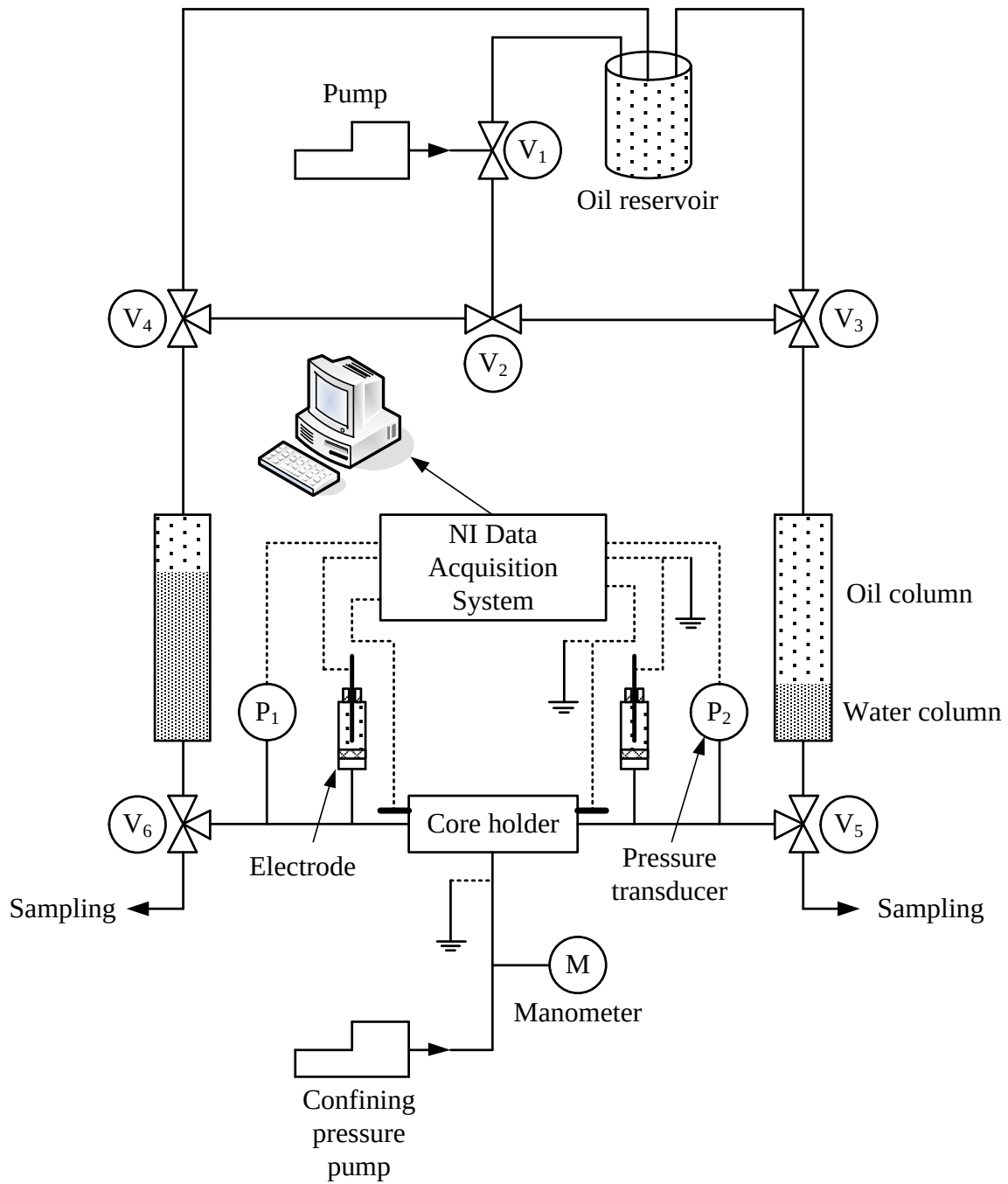
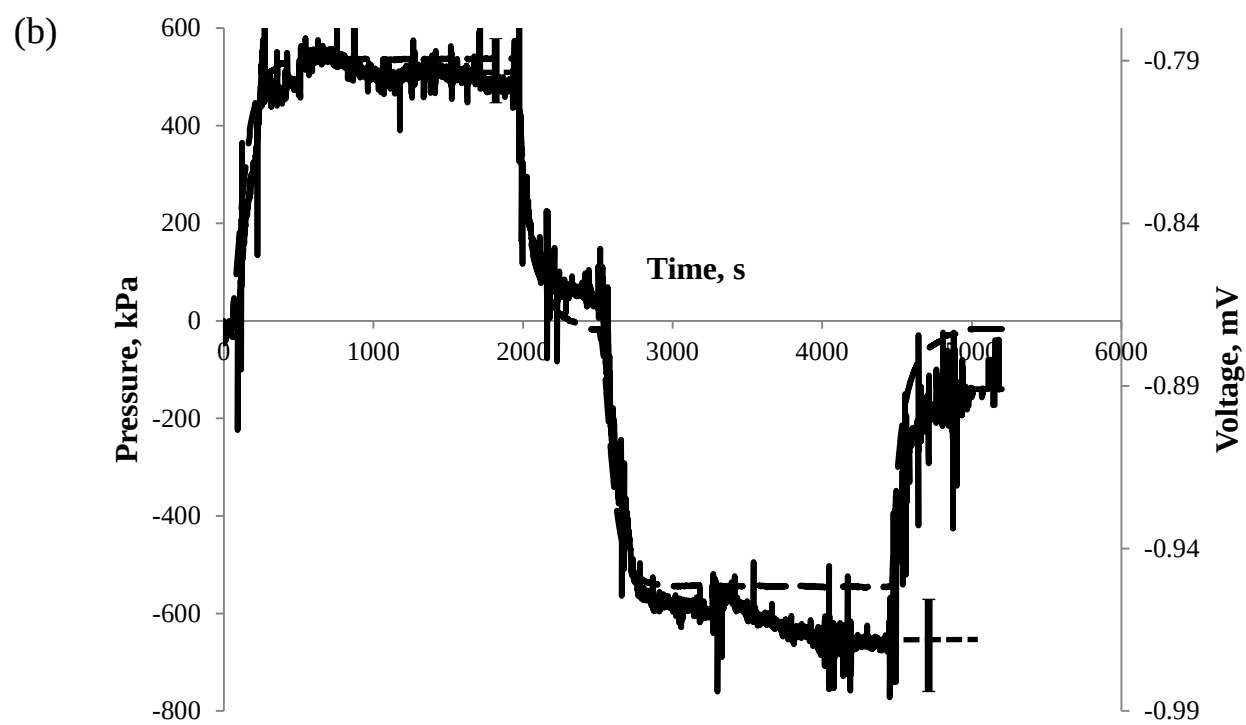
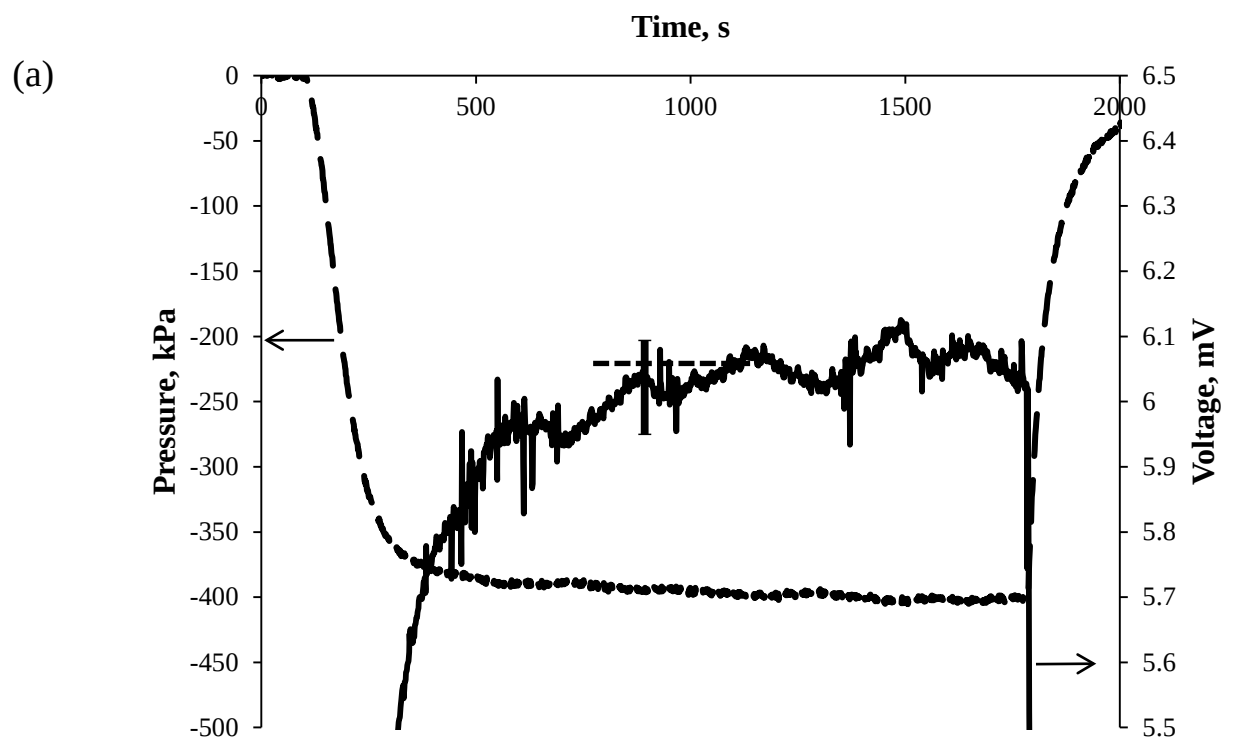


Figure 3. Experimental setup for measuring the streaming potential, which consists of a pressure vessel (core holder), electrolyte reservoirs, pump, flow lines (solid lines) and electrical connections (dashed lines). The oil column in the electrolyte reservoirs serves to isolate the electrolyte from the atmosphere (closed-system). The flow valves  $V_1 - V_6$  allow the pump the flow electrolyte through the sample in opposing directions. Modified from Jaafar et al. (2009).



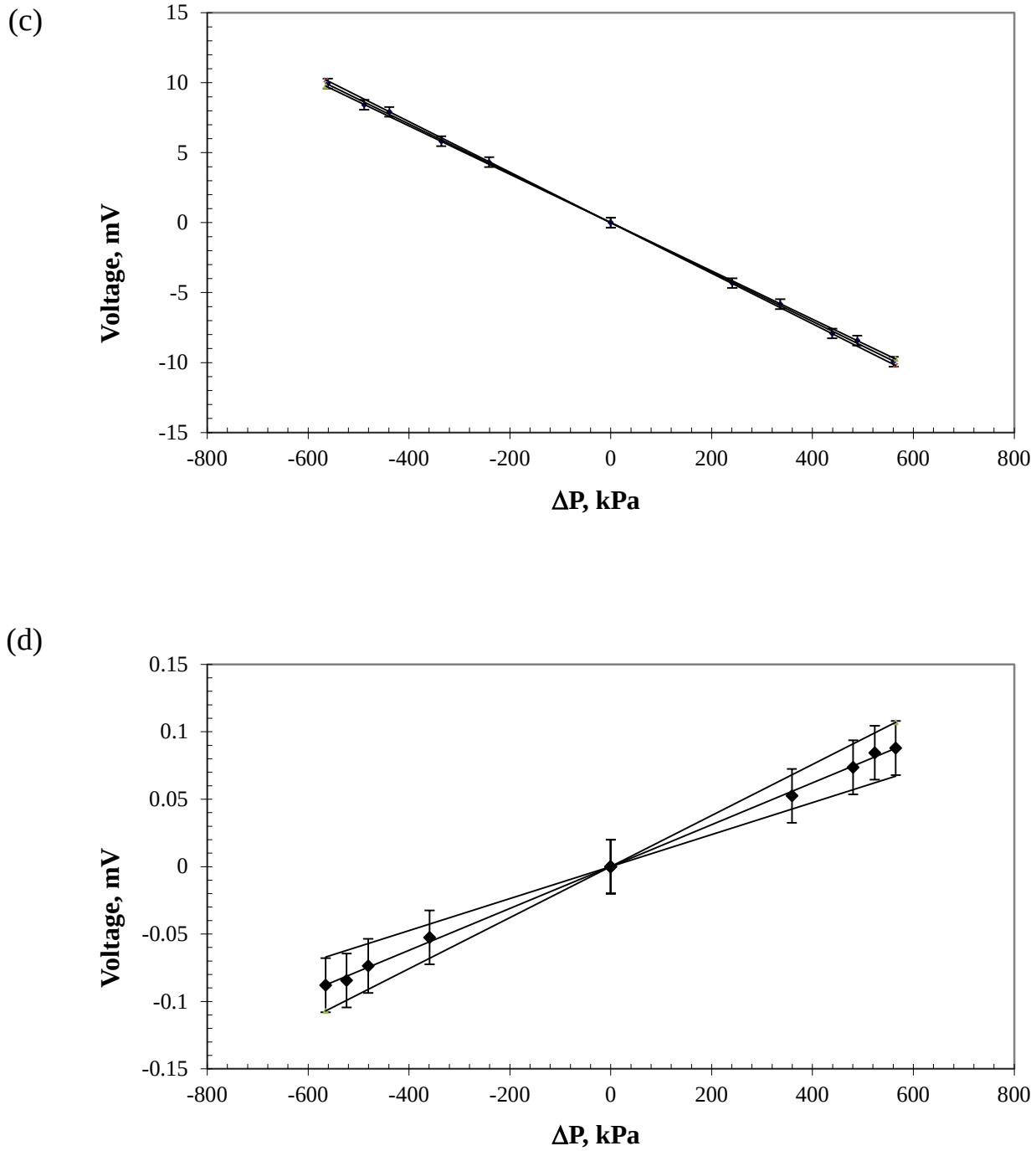


Figure 4. Typical experimental results used to determine the streaming potential coupling coefficient. Plots (a) and (b) show the voltage and pressure variation in experiments at a given flowrate using (a) low ionic strength 0.05M NaCl-EQ electrolyte and (b) high ionic strength synthetic formation brine (FMB). The horizontal dashed lines show the stabilized voltage and

pressure, and the error bar denotes the spread in these values. Plots (c) and (d) show the stabilized voltage plotted against stabilized pressure for the same electrolytes shown in (a) and (b). The gradient of a linear regression through these data yields the streaming potential coupling coefficient.

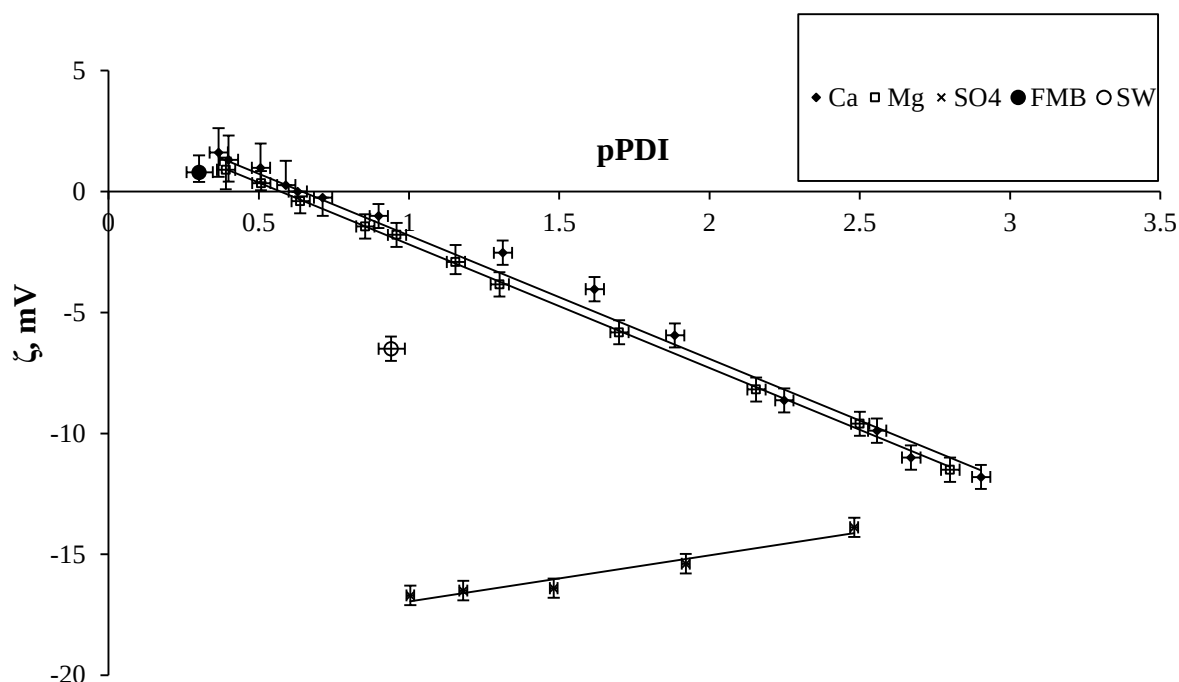
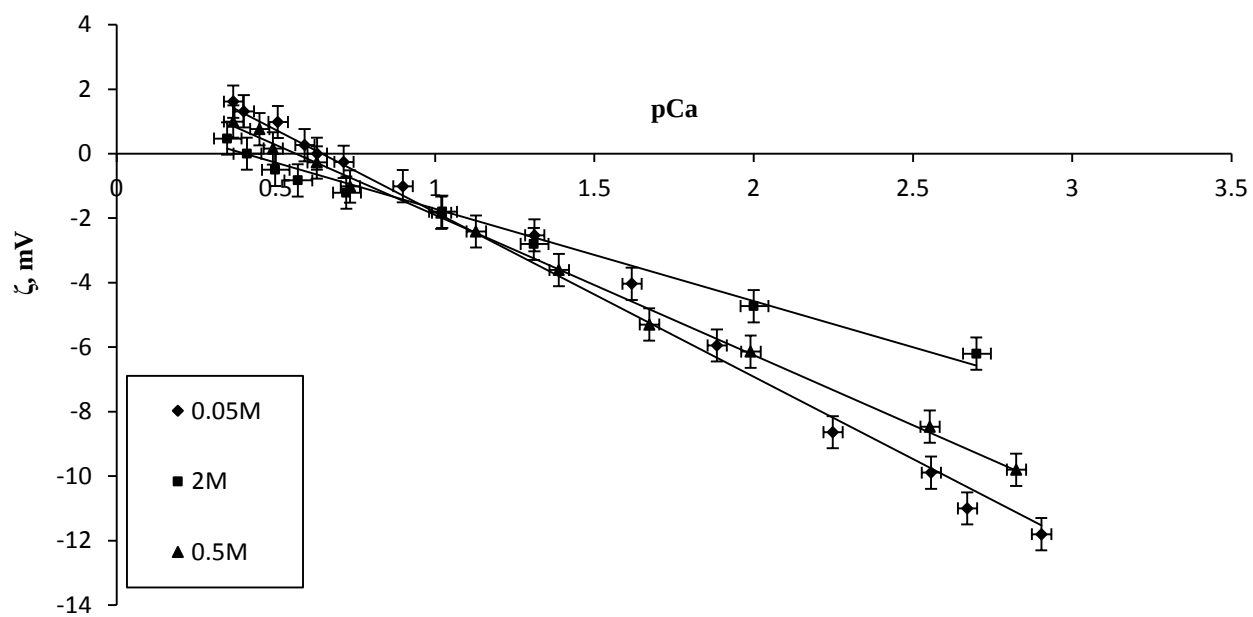
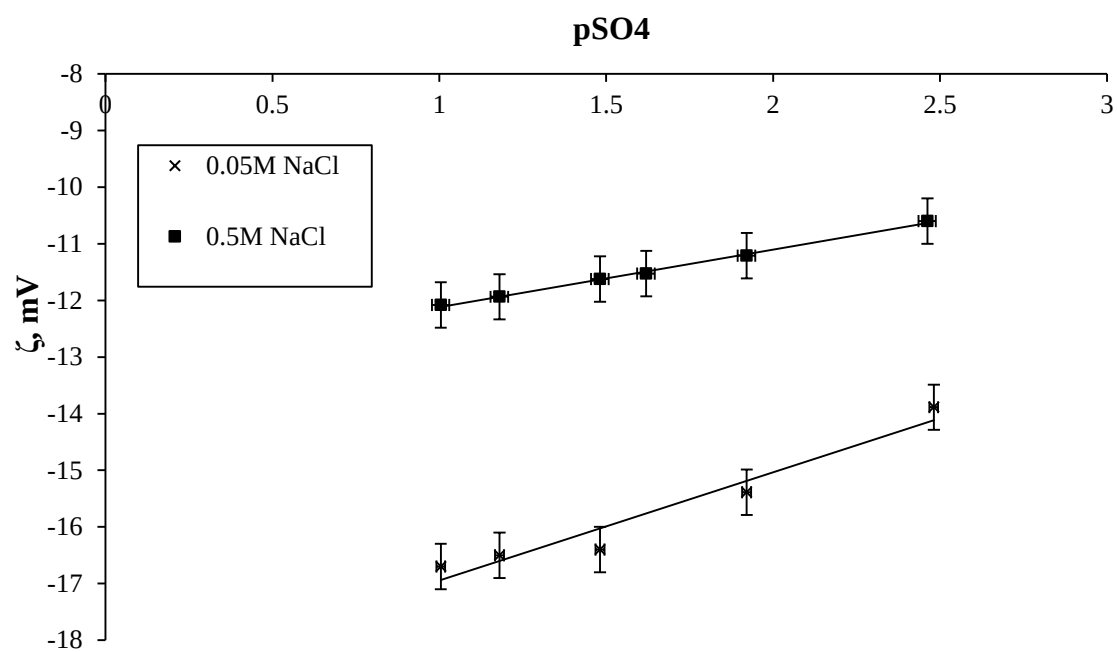


Figure 5. Effect of Ca, Mg and  $\text{SO}_4$  concentration (expressed as pPDI) in 0.05M NaCl electrolyte on the zeta potential of Portland limestone. The pH varied in the range  $7.2-8 \pm 0.3$  for the Ca and Mg effluent electrolytes, and  $7.9-8 \pm 0.3$  for the  $\text{SO}_4$  effluent electrolytes. Also shown are the results for the synthetic formation brine (FMB) and natural seawater (SW) plotted as a function of  $\text{pCa} + \text{pMg}$ .

(a)



(b)



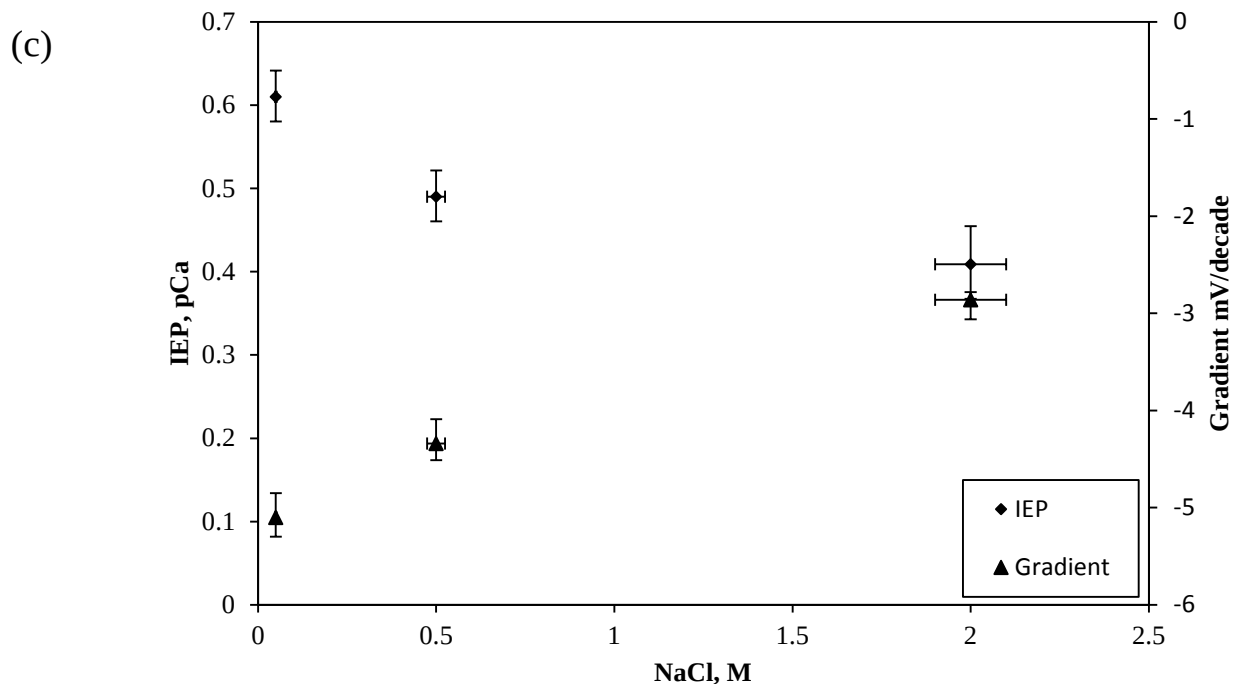


Figure 6. Effect of NaCl concentration on the relationship between PDI concentration and zeta potential of Portland limestone. (a) Effect of Ca concentration (expressed as pCa) in three different NaCl electrolytes (0.05M, 0.5M and 2M) on the zeta potential of Portland limestone, pH in the range  $7.2-8 \pm 0.3$ . (b) Effect of  $\text{SO}_4$  concentration (expressed as  $\text{pSO}_4$ ) in two different NaCl electrolytes (0.05M, 0.5M) on the zeta potential of Portland limestone, pH in the range  $7.9-8.1 \pm 0.3$ . (c) Effect of NaCl concentration on the IEP (expressed as pCa) and zeta potential sensitivity to pCa (expressed as the gradient of the linear regressions shown in (a)).

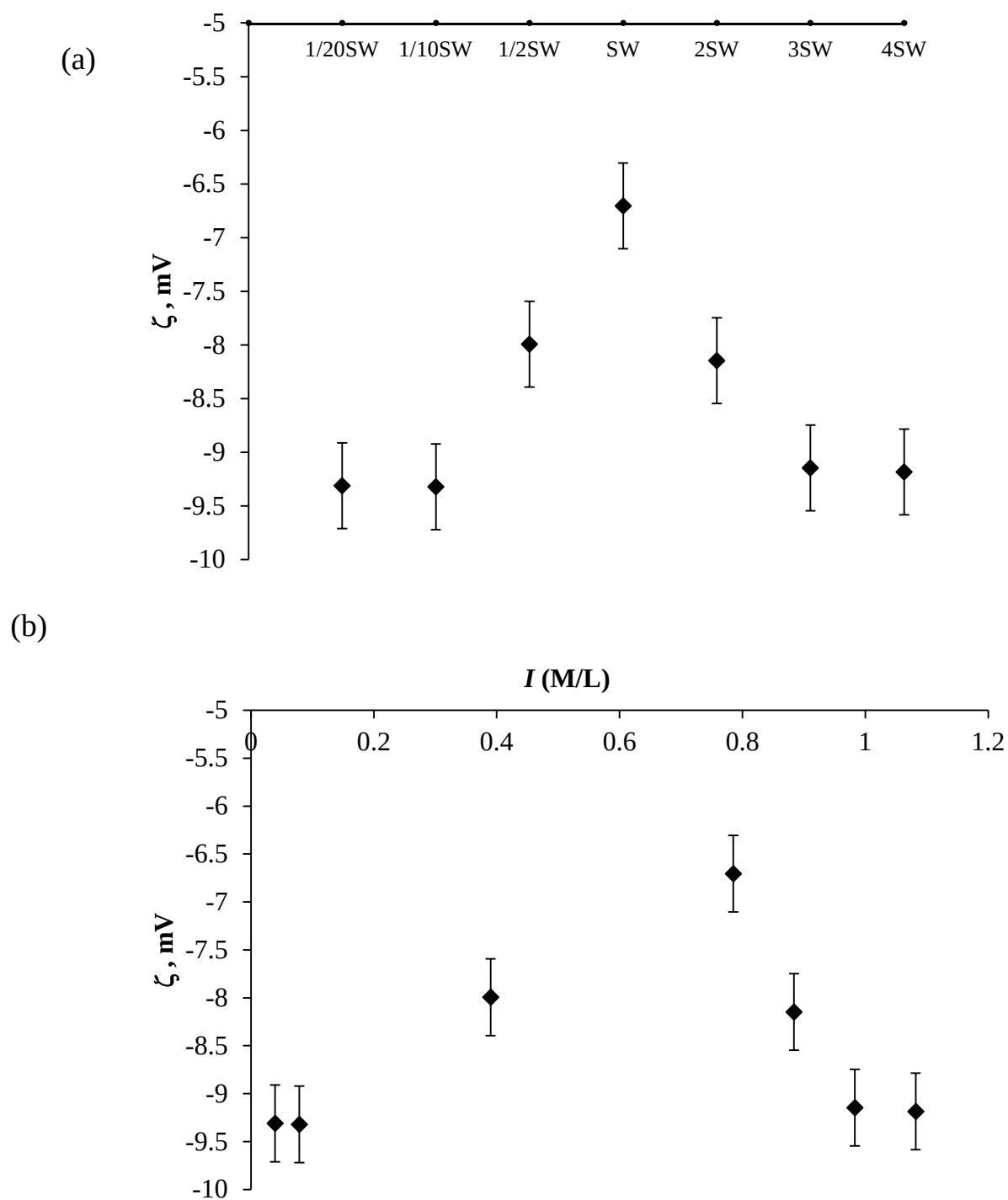


Figure 7. (a) Relationship between zeta potential and electrolyte compositions derived from seawater (SW),  $\text{pH}=8\pm0.3$ . (b) Zeta potential of the same compositions plotted as a function of ionic strength ( $I$ ).

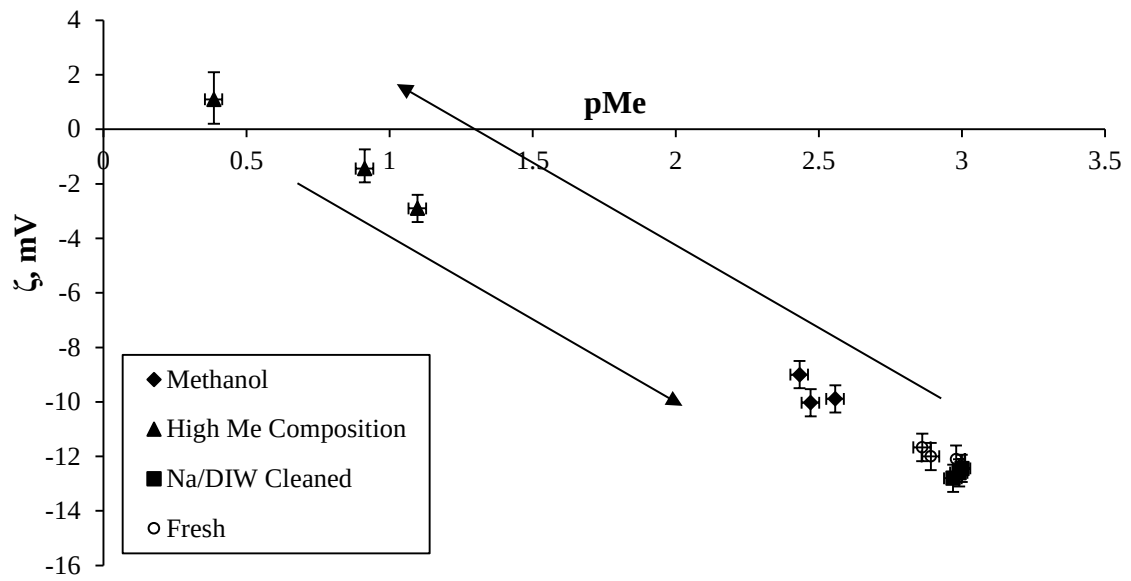


Figure 8. Zeta potential as a function of Ca + Mg concentration (expressed as pMe) for fresh samples (circles), pH =  $8.2 \pm 0.2$ , experiments at elevated Ca and Mg concentration (triangles), pH in the range  $7.2-7.5 \pm 0.3$ , after standard cleaning with methanol (diamonds), and after the enhanced cleaning with DIW used in this study (squares), pH =  $8.2 \pm 0.2$ .

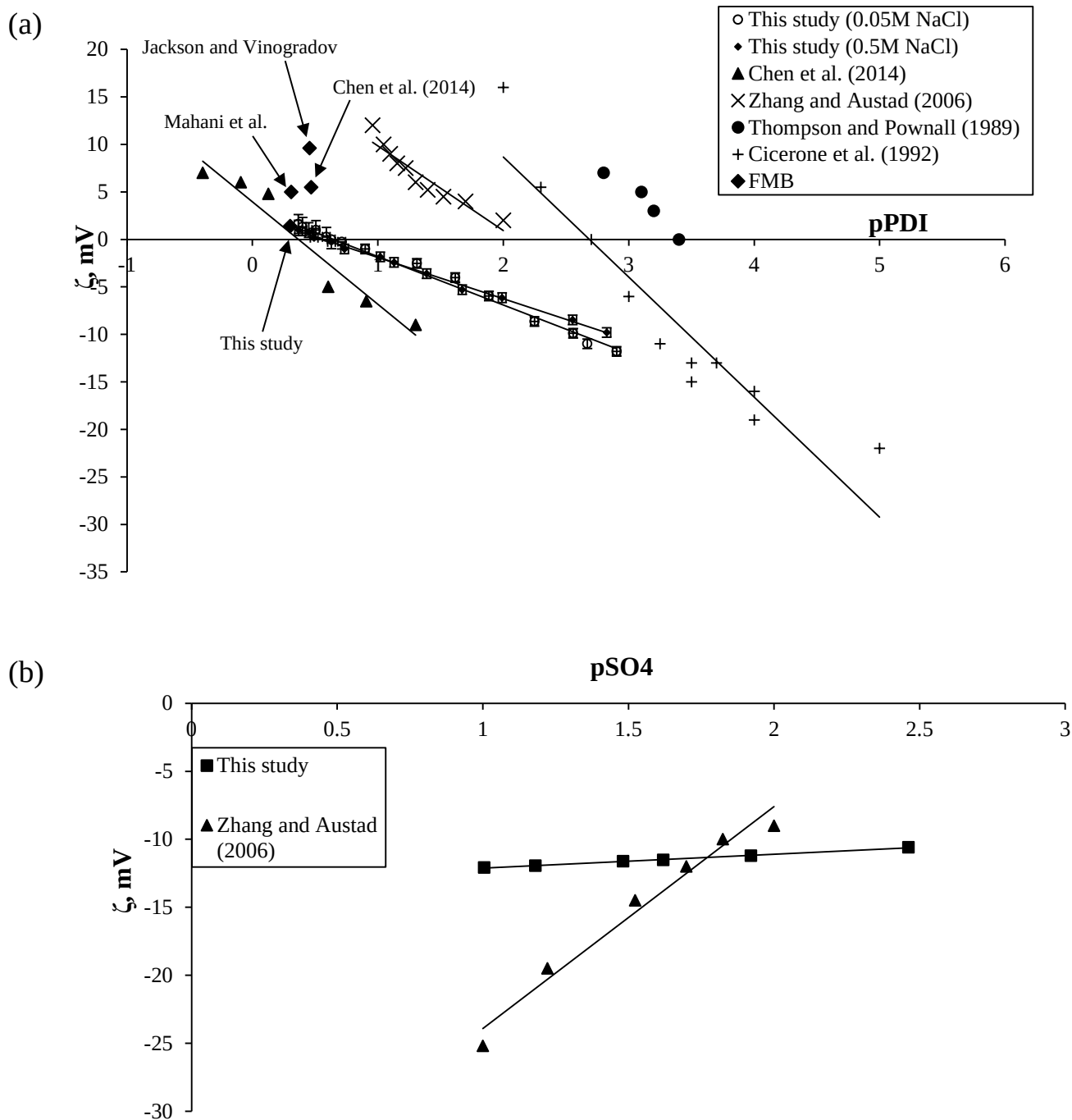


Figure 9. Comparison of the data obtained here and previously published measurements for the zeta potential sensitivity to (a) Ca and (b)  $\text{SO}_4$ . Errors are shown where reported by the authors. Thompson and Pownall (1989) used the SPM method, synthetic calcite and 0.002M NaCl electrolyte over the pH range 7-11. All other published studies used the EPM method. Cicerone et al. (1992) used synthetic calcite and 0.03M KCl electrolyte over the pH range 8.5-10.5 (error not reported). Zhang et al. (2006) used powered Stevns Klint chalk and 0.573M NaCl electrolyte at pH = 8.4 (error not reported). These

conditions are the most similar to those used here. Chen et al. (2014) used powdered natural limestone and DIW at pH = 8 (error not reported). The various labelled diamonds in (a) show data obtained using natural or synthetic formation brine (FMB).

Table 3. Values of the Stern layer capacitance and shear plane location used to match the experimental data using equation (5). The value of  $C_s$  was identified first for the EPM data using  $\Delta = 0$ , consistent with previous studies. The value of  $C_s$  was then fixed for the SPM data at the same NaCl concentration matched by adjusting  $\Delta$  to account for the complex pore-space. It was not possible to match the other NaCl concentrations tested without further adjusting  $C_s$ . The shear plane location is not expected to be significantly affected by the increase in ionic strength.

| Method | NaCl concentration (M) | Stern Layer capacitance $C_s$ (F/m <sup>2</sup> ) | Shear plane location $\Delta$ (nm) |
|--------|------------------------|---------------------------------------------------|------------------------------------|
| EPM    | 0.05                   | 1.13                                              | 0                                  |
| SPM    | 0.05                   | 1.13                                              | 0.245                              |
| SPM    | 0.5                    | 1.76                                              | 0.245                              |
| SPM    | 2                      | 2.75                                              | 0.245                              |

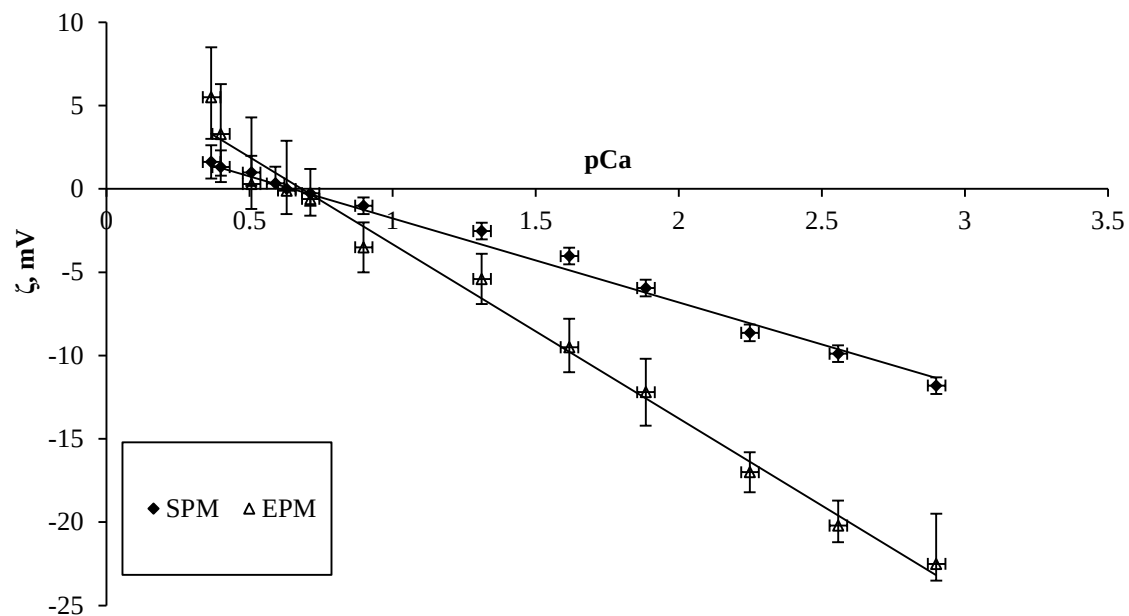


Figure 10. Comparison between zeta potential as a function of pCa obtained using the SPM and EPM method for the same natural Portland limestone and 0.05M NaCl electrolyte, pH =  $7.2-8 \pm 0.3$ .

Table 4. Compilation of reported IEP values, including the electrolyte, type of calcite, pCa and whether the IEP was directly measured or extrapolated.

| Reference                               | Background<br>Electrolyte                                                                                | Calcite   | IEP, pCa | Determination |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------|-----------|----------|---------------|
| Somasundaran<br>and Agar (1967)         | DIW                                                                                                      | Synthetic | 3.72     | extrapolated  |
| Fuerstenau et al.<br>(1968)             | $10^{-3}\text{M}$ ( $\text{SiO}_2/\text{Na}_2\text{O}$ )                                                 | Synthetic | 4.1      | extrapolated  |
| Mishra (1978)                           | $2 \times 10^{-3}\text{M}$ $\text{NaClO}_4$                                                              | Natural   | 3.09     | extrapolated  |
| Foxall et al. (1979)                    | 0.01-.15M $\text{NaCl}$                                                                                  | Synthetic | 4.4      | extrapolated  |
| Amankonah and<br>Somasundaran<br>(1985) | $2 \times 10^{-3}\text{M}$ $\text{KNO}_3$                                                                | Synthetic | 4.08     | extrapolated  |
| Thompson and<br>Pownall (1989)          | $2 \times 10^{-3}$ - $10^{-2}\text{M}$<br>( $\text{NaCl}/\text{HCl}/\text{NaOH}$ )                       | Synthetic | 2.02     | direct        |
| Thompson and<br>Pownall (1989)          | $2 \times 10^{-3}$ - $10^{-2}\text{M}$<br>( $\text{NaCl}/\text{NaHCO}_3/\text{HCl}$<br>/ $\text{NaOH}$ ) | Synthetic | 1.92     | direct        |
| Thompson and<br>Pownall (1989)          | $2 \times 10^{-3}$ - $10^{-2}\text{M}$<br>( $\text{NaCl}/\text{CaCl}_2/\text{HCl}/\text{NaOH}$ )         | Synthetic | 2.16     | direct        |
| Thompson and<br>Pownall (1989)          | $2 \times 10^{-3}$ - $10^{-2}\text{M}$<br>( $\text{NaCl}/\text{CaCl}_2/\text{HCl}/\text{NaOH}$ )         | Synthetic | 3.4      | direct        |

|                             |                                                                                                |           |          |              |
|-----------------------------|------------------------------------------------------------------------------------------------|-----------|----------|--------------|
| Thompson and Pownall (1989) | $2 \times 10^{-3}$ - $10^{-2}$ M<br>(NaCl/H <sub>2</sub> CO <sub>3</sub> )                     | Synthetic | 4        | direct       |
| Thompson and Pownall (1989) | $2 \times 10^{-3}$ - $10^{-2}$ M<br>(NaCl/NaHCO <sub>3</sub> /H <sub>2</sub> CO <sub>3</sub> ) | Synthetic | 3.8      | direct       |
| Thompson and Pownall (1989) | $2 \times 10^{-3}$ - $10^{-2}$ M<br>(NaCl/NaHCO <sub>3</sub> /Ca(OH) <sub>2</sub> )            | Synthetic | 3.8      | direct       |
| Pierre et al. (1990)        | $10^{-2}$ M NaCl                                                                               | Synthetic | 3.37     | direct       |
| Pierre et al. (1990)        | $10^{-3}$ - $10^{-1}$ M NaCl                                                                   | Natural   | 4        | direct       |
| Pierre et al. (1990)        | 0.03M NaCl<br>(constant pH=8.3)                                                                | Natural   | 2        | direct       |
| Pierre et al. (1990)        | $10^{-2}$ M NaCl<br>(constant pH=8.5)                                                          | Synthetic | 3.9      | direct       |
| Huang et al. (1991)         | DIW                                                                                            | Synthetic | 4.35     | extrapolated |
| Cicerone et al. (1992)      | 0.03M KCl                                                                                      | Synthetic | 2.7      | direct       |
| Chen et al. (2014)          | DIW                                                                                            | Natural   | 0.2-0.48 | extrapolated |

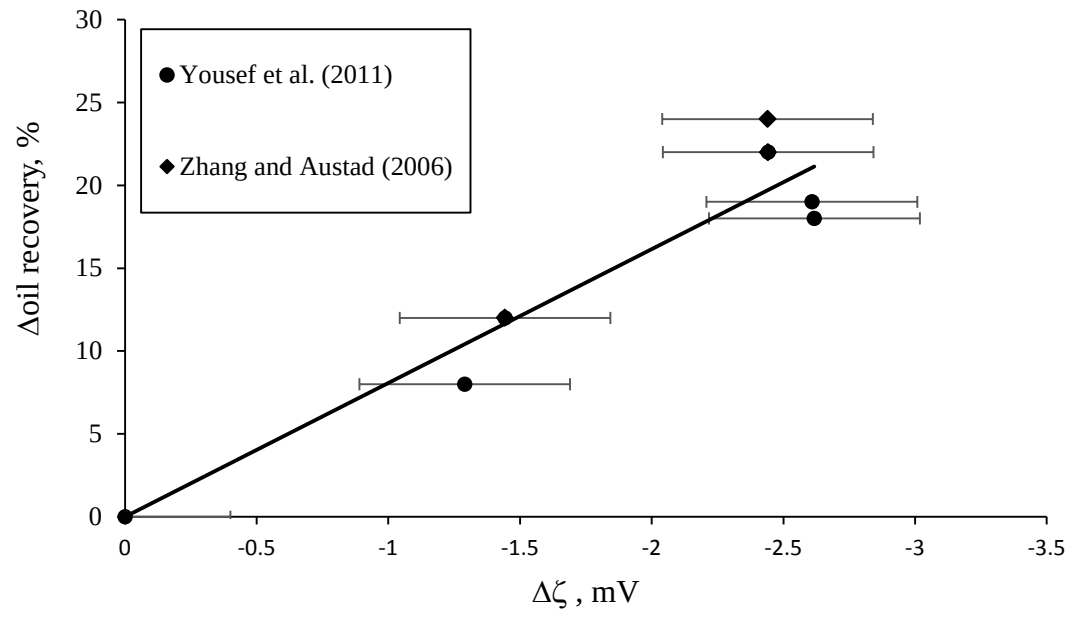


Figure 11. Comparison of the change in incremental oil recovery and zeta potential referenced to that of seawater for both controlled salinity (CSW) approaches: seawater dilution (Yousef et al, 2011) and sulfate addition to seawater (Zhang and Austad, 2006).